



# Essential Histology



## Question Bank for PGME



AREMS-HY

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**AREMS-HY**

Primary Reference: Junqueira's Basic Histology 17th Edition

Official Syllabus: 16 chapters 935 high-quality questions

## **Reference Statement**

This Question Bank is an educational synthesis aligned primarily with Junqueira's Basic Histology: Text and Atlas, 17th Edition. It is generated FROM the final textbook Essential Histology. Every question is traceable to a learning objective and content unit. No question tests untaught fact.

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# Chapter 1 Histology & Its Methods of Study

## Part A Recall

**\*\*Q1. In H&E staining, nuclei appear blue because:**

- A) Nucleic acids are negatively charged and bind the basic dye hematoxylin
- B) Proteins bind hematoxylin
- C) Eosin is a basic dye
- D) DNA binds eosin

**Answer: A**

Explanation: DNA/RNA phosphate groups (negative) attract positively charged hematoxylin blue. Eosin (acidic) binds proteins (D, C reversed; B wrong).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: basophilic = nucleic acid = binds a basic dye.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to basophilic = nucleic acid = binds a basic dye.

Difficulty: Recall

**\*\*Q2. Which stain requires a frozen section?**

- A) Oil Red O
- B) Prussian blue
- C) Von Kossa
- D) PAS

**Answer: A**

Explanation: Lipid dyes need frozen sections because routine processing dissolves lipid. PAS, Prussian blue, Von Kossa work on paraffin.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: lipid frozen.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to lipid frozen.

Difficulty: Recall

**\*\*Q3. TEM exceeds light microscopy in resolution because it uses:**

- A) Living cells
- B) Higher magnification
- C) A shorter wavelength (electrons)
- D) Fluorescent dyes

**Answer: C**

**Explanation:** Resolution depends on wavelength; electrons have far shorter wavelengths than light. Magnification (B) is not the mechanism.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: resolution = wavelength.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to resolution = wavelength.

Difficulty: Recall

**\*\*Q4. Freeze-fracture electron microscopy specifically reveals:**

- A) Whole-cell 3-D shape
- B) Internal organelles
- C) Membrane faces (E-face/P-face) and junctional strands
- D) Surface topography

**Answer: C**

**Explanation:** Fracturing splits the bilayer's hydrophobic interior, exposing membrane faces.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: freeze-fracture = the membrane plane.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to freeze-fracture = the membrane plane.

Difficulty: Recall

**\*\*Q5. The Papanicolaou (Pap) stain is designed to display:**

- A) Nuclear chromatin detail and cytoplasmic keratinization
- B) Bone mineral
- C) Iron deposits
- D) Lipid content

**Answer: A**

**Explanation:** Pap reveals nuclear detail (dysplasia) and cytoplasmic maturation/keratinization cervical cytology.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Pap = nuclear + cytoplasmic maturation.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Pap = nuclear + cytoplasmic maturation.

Difficulty: Recall

**\*\*Q6. The correct order of tissue processing is:**

- A) Fixation    dehydration    clearing    embedding    sectioning    staining
- B) Clearing    fixation    embedding    sectioning
- C) Fixation    staining    dehydration    embedding
- D) Embedding    fixation    dehydration    staining

**Answer: A**

Explanation: The canonical sequence; each step has a chemical purpose.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: know the pipeline order.

Learning Objective mapping: Chapter 1    Histology & Its Methods of Study    LO related to know the pipeline order.

Difficulty: Recall

**\*\*Q7. Which structure stains metachromatically with toluidine blue?**

- A) Mast-cell granules
- B) Erythrocytes
- C) Elastic fibers
- D) Collagen

**Answer: A**

Explanation: Mast-cell heparin (sulfated GAG) polymerizes the dye    purple/red.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: mast cells = metachromatic.

Learning Objective mapping: Chapter 1    Histology & Its Methods of Study    LO related to mast cells = metachromatic.

Difficulty: Recall

**\*\*Q8. A frozen section is superior to a paraffin section for:**

- A) Long-term storage
- B) Avoiding ice artifacts
- C) Best morphological detail
- D) Preserving enzyme activity and lipids

**Answer: D**

Explanation: Freezing bypasses the solvent/heat steps that destroy enzymes and dissolve lipids. Morphology is inferior (C, B wrong).

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B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: frozen = enzymes + lipids + speed.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to frozen = enzymes + lipids + speed.

Difficulty: Recall

**\*\*Q9. A researcher must count HER2 gene copies in breast cancer. The best method is:**

- A) PAS
- B) FISH
- C) Oil Red O
- D) Phase-contrast

**Answer: B**

Explanation: FISH counts gene copies via fluorescent probes. PAS/lipid/phase are irrelevant to DNA copy number.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: gene copy number FISH.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to gene copy number FISH.

Difficulty: Recall

**\*\*Q10. The "clearing" agent is classically:**

- A) Ethanol
- B) Xylene
- C) Paraffin
- D) Formalin

**Answer: B**

Explanation: Xylene clears alcohol and makes tissue paraffin-miscible.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: clearing = xylene.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to clearing = xylene.

Difficulty: Recall

**\*\*Q11. A lymph-node biopsy shows a delicate black branching mesh on a silver-stained section. The structure is:**

- A) Type I collagen
- B) Reticular (type III) fibers
- C) Elastic fibers
- D) Amyloid

**Answer: B**



**Explanation:** Reticular (type III) fibers form delicate black branching mesh on silver impregnation, seen in lymphoid stroma, liver, endocrine.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

**Key point:** Reticular (type III) fibers form delicate black branching mesh on silver impregnation, seen in lymph

**Learning Objective mapping:** Chapter 1 Histology & Its Methods of Study LO related to Reticular (type III) fibers form delicate black branching mesh on silver impregn

**Difficulty:** Recall

**\*\*Q12. The resolution of a light microscope is fundamentally limited by:**

- A) The wavelength of visible light
- B) Specimen thickness
- C) Staining intensity
- D) Lens quality only

**Answer: A**

**Explanation:** Resolution  $\sim 0.2 \mu\text{m}$  limited by wavelength of visible light; electrons shorter wavelength  $\sim 0.1 \text{ nm EM}$ .

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

**Key point:** Resolution  $\sim 0.2 \mu\text{m}$  limited by wavelength of visible light; electrons shorter wavelength  $\sim 0.1 \text{ nm EM}$

**Learning Objective mapping:** Chapter 1 Histology & Its Methods of Study LO related to Resolution  $\sim 0.2 \mu\text{m}$  limited by wavelength of visible light; electrons shorter wav

**Difficulty:** Recall

## **Part B Understanding**

**\*\*Q13. A structure that binds eosin and appears pink is best described as:**

- A) Basophilic
- B) Acidophilic
- C) Metachromatic
- D) Argyrophilic

**Answer: B**

**Explanation:** Eosinophilic/acidophilic structures (proteins, collagen) bind the acidic dye eosin pink. Basophilic binds hematoxylin (blue); metachromatic shifts color; argyrophilic reduces silver.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

**Key point:** acidophilic = pink = protein.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to acidophilic = pink = protein.

Difficulty: Understanding

**\*\*Q14. Amyloid is identified by:**

- A) Congo red with apple-green birefringence
- B) Metachromasia with toluidine blue
- C) Oil Red O positivity
- D) Prussian blue positivity

**Answer: A**

Explanation: Congo red binds amyloid -sheets and shows apple-green birefringence under polarized light. The others detect iron, GAGs, lipid.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: amyloid = Congo red + polarized light.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to amyloid = Congo red + polarized light.

Difficulty: Understanding

**\*\*Q15. Which technique images living, unstained cells?**

- A) TEM
- B) Polarizing microscopy
- C) H&E bright-field
- D) Phase-contrast microscopy

**Answer: D**

Explanation: Phase-contrast converts refractive-index differences into contrast without fixation/staining. TEM and H&E require fixed/stained specimens.

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B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: phase-contrast = live + unstained.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to phase-contrast = live + unstained.

Difficulty: Understanding

**\*\*Q16. Prussian blue (Perls') detects:**

- A) Amyloid
- B) Glycogen
- C) Calcium
- D) Iron (hemosiderin)

**Answer: D**

Explanation: Perls' reaction produces blue ferric ferrocyanide from tissue iron. Calcium = Von Kossa.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Prussian blue = iron.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Prussian blue = iron.

Difficulty: Understanding

**\*\*Q17. In immunohistochemistry, the primary antibody binds:**

- A) A chromogen
- B) The nucleus nonspecifically
- C) The secondary antibody only
- D) The specific antigen of interest

**Answer: D**

Explanation: The primary antibody recognizes the target antigen; a labeled secondary antibody binds the primary (indirect method).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: IHC = antibody-antigen.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to IHC = antibody-antigen.

Difficulty: Understanding

**\*\*Q18. Xylene is used in the clearing step because it:**

- A) Cross-links proteins
- B) Removes water
- C) Is miscible with both alcohol and paraffin
- D) Stains nuclei

**Answer: C**

Explanation: Xylene bridges alcohol (previous step) and paraffin (next step). Water removal = alcohol.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: clearing = the alcohol paraffin bridge.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to clearing = the alcohol paraffin bridge.

Difficulty: Understanding

**\*\*Q19. The most common fixative for routine light microscopy is:**

- A) Bouin solution

- B) Glutaraldehyde
- C) Formalin
- D) Osmium tetroxide

**Answer: C**

Explanation: Formalin (formaldehyde) is the standard LM fixative. Glutaraldehyde/osmium = EM.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

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D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: LM = formalin.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to LM = formalin.

Difficulty: Understanding

**\*\*Q20. Which modality shows a bright object against a black background using oblique illumination?**

- A) Bright-field
- B) Dark-field
- C) Phase-contrast
- D) Confocal

**Answer: B**

Explanation: Dark-field collects only scattered (oblique) light; the field stays dark. Classic use: spirochetes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

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D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: dark-field = bright on black.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to dark-field = bright on black.

Difficulty: Understanding

**\*\*Q21. Which is an artifact rather than a real structure?**

- A) PAS-positive basement membrane
- B) Metachromatic granules
- C) Basophilic nucleus
- D) Perinuclear halo from shrinkage

**Answer: D**

Explanation: Shrinkage halos are processing artifacts; the others are genuine staining patterns.

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B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: halos/gaps = artifact.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to halos/gaps = artifact.

Difficulty: Understanding

**\*\*Q22. Which statement about metachromasia is TRUE?**

- A) It occurs only with eosin
- B) It requires polarized light
- C) The dye stains tissue its own color
- D) It results from dye polymerization on densely charged structures

**Answer: D**

Explanation: Polymerization on dense polyanions shifts absorption (toluidine blue purple/red on heparin).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: metachromasia = polymerization color shift.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to metachromasia = polymerization color shift.

Difficulty: Understanding

**\*\*Q23. A hepatocyte's cytoplasm is uniformly pink on H&E and contains no visible blue granules. This indicates abundance of:**

- A) Glycogen
- B) Lipid
- C) Ribosomes
- D) Structural proteins

**Answer: D**

Explanation: Eosinophilic pink indicates structural proteins/mitochondria; basophilic blue indicates RER/RNA; plasma cell basophilic due to RER.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

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Key point: Eosinophilic pink indicates structural proteins/mitochondria; basophilic blue indicates RER/RNA; plasma cell basophilic due to RER.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Eosinophilic pink indicates structural proteins/mitochondria; basophilic blue in

Difficulty: Understanding

**\*\*Q24. Apple-green birefringence after Congo red indicates:**

- A) Amyloid
- B) Calcium
- C) Elastic fibers
- D) Iron

**Answer: A**

Explanation: Amyloid -pleated sheet binds Congo red apple-green birefringence polarized; diagnostic Congo red + green.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Amyloid -pleated sheet binds Congo red apple-green birefringence polarized; diagnostic Congo red

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Amyloid -pleated sheet binds Congo red apple-green birefringence polarized; d

Difficulty: Understanding

## **Part C Application**

**\*\*Q25. The PAS reaction detects:**

- A) Elastic fibers
- B) Type I collagen
- C) Carbohydrate-rich structures
- D) Reticular fibers

**Answer: C**

Explanation: PAS oxidizes hexose glycols to aldehydes that bind Schiff magenta (glycogen, mucin, BM, fungi). Collagen = trichrome; reticular = silver; elastic = Orcein.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: PAS = carbohydrate.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to PAS = carbohydrate.

Difficulty: Application

**\*\*Q26. The EM fixative combination is:**

- A) Xylene and paraffin
- B) Formalin and alcohol
- C) Glutaraldehyde and osmium tetroxide
- D) Hematoxylin and eosin

**Answer: C**

Explanation: Glutaraldehyde cross-links proteins; osmium fixes membrane lipids and adds electron contrast. Formalin is the LM fixative.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: EM = glutaraldehyde + osmium.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to EM = glutaraldehyde + osmium.

Difficulty: Application

**\*\*Q27. Confocal microscopy produces sharp images of thick specimens chiefly because of:**

- A) Electron illumination
- B) Staining with heavy metals
- C) Higher magnification
- D) A pinhole aperture that rejects out-of-focus light

**Answer: D**

Explanation: The pinhole admits only in-focus light optical sectioning. It still uses photons, not electrons.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: confocal = pinhole.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to confocal = pinhole.

Difficulty: Application

**\*\*Q28. Von Kossa stains:**

- A) Mucin
- B) Iron
- C) Calcium
- D) Lipid

**Answer: C**

Explanation: Von Kossa stains calcium black (silver substitution). Iron = Prussian blue; mucin = PAS/Alcian.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Von Kossa = calcium.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Von Kossa = calcium.

Difficulty: Application

**\*\*Q29. Immunofluorescence labels antibodies with:**

- A) A heavy metal
- B) An enzyme-chromogen only
- C) A radioactive isotope
- D) A fluorochrome

**Answer: D**

**Explanation:** Fluorescence uses fluorochromes excited by UV. Enzyme-chromogen = conventional IHC; isotope = autoradiography.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: immunofluorescence = fluorochrome.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to immunofluorescence = fluorochrome.

Difficulty: Application

**\*\*Q30. Delayed fixation produces a characteristic artifact of:**

- A) Over-staining
- B) Autolysis with shrunken, dark nuclei
- C) Lipid preservation
- D) Tissue expansion

**Answer: B**

**Explanation:** Autolysis before fixation shrinks and clumps nuclei mimicking necrosis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: delayed fixation = pseudo-necrosis.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to delayed fixation = pseudo-necrosis.

Difficulty: Application

**\*\*Q31. Which fixative preserves fine ultrastructure for electron microscopy?**

- A) Ethanol
- B) Bouin solution
- C) Formalin
- D) Glutaraldehyde

**Answer: D**

**Explanation:** Glutaraldehyde cross-links proteins strongly EM-grade detail.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: EM primary fix = glutaraldehyde.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to EM primary fix = glutaraldehyde.

Difficulty: Application



**\*\*Q32. Goodpasture/Alport relevance to histology rests on which structure best shown by PAS?**

- A) Reticular fibers
- B) Mast cells
- C) Elastic lamellae
- D) Basement membrane

**Answer: D**

Explanation: PAS demonstrates basement membrane (type IV collagen-containing glycoproteins) thickened in diabetic nephropathy, targeted in Goodpasture/Alport.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: PAS basement membrane.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to PAS basement membrane.

Difficulty: Application

**\*\*Q33. Congo red birefringence must be assessed with:**

- A) Bright-field
- B) Polarizing light
- C) Phase-contrast
- D) Dark-field

**Answer: B**

Explanation: Birefringence is only visible under polarized light.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: birefringence polarizer.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to birefringence polarizer.

Difficulty: Application

**\*\*Q34. Which technique demonstrates sites of DNA synthesis directly?**

- A) Autoradiography with <sup>3</sup>H-thymidine
- B) IHC for Ki-67
- C) FISH for a single gene
- D) Feulgen stain

**Answer: A**

Explanation: <sup>3</sup>H-thymidine is incorporated during synthesis and localized by silver grains. Feulgen = DNA amount; Ki-67 = cycling marker; FISH = copy number.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: synthesis H-thymidine.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to synthesis H-thymidine.

Difficulty: Application

**\*\*Q35. A pathologist must see the 3-D surface of a ciliated cell. The method is:**

- A) Freeze-fracture
- B) Phase-contrast
- C) TEM
- D) SEM

**Answer: D**

Explanation: SEM shows 3-D surface topography via secondary electrons; TEM shows internal ultrastructure 2-D.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: SEM shows 3-D surface topography via secondary electrons; TEM shows internal ultrastructure 2-D.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to SEM shows 3-D surface topography via secondary electrons; TEM shows internal ult

Difficulty: Application

**\*\*Q36. Which fixative preserves membranes AND adds electron contrast?**

- A) Osmium tetroxide
- B) Xylene
- C) Formalin
- D) Glutaraldehyde

**Answer: A**

Explanation: Osmium tetroxide binds phospholipid head groups, fixes lipids and scatters electrons membrane contrast TEM.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Osmium tetroxide binds phospholipid head groups, fixes lipids and scatters electrons membrane cont

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Osmium tetroxide binds phospholipid head groups, fixes lipids and scatters elect

Difficulty: Application

## **Part D Integration**

**\*\*Q37. Reticular fibers are best demonstrated by:**

- A) H&E alone
- B) PAS
- C) Silver impregnation
- D) Masson trichrome

**Answer: C**

Explanation: Reticular (type III) collagen is argyrophilic black with silver. PAS/trichrome/H&E do not selectively show reticular fibers.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: reticular = silver.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to reticular = silver.

Difficulty: Integration

**\*\*Q38. Osmium tetroxide is used in EM primarily because it:**

- A) Fixes membrane lipids and adds electron density
- B) Cross-links collagen
- C) Removes water
- D) Stains DNA

**Answer: A**

Explanation: Osmium binds phospholipids and, as a heavy metal, scatters electrons (contrast).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: osmium = membranes + contrast.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to osmium = membranes + contrast.

Difficulty: Integration

**\*\*Q39. Collagen birefringence is best visualized with:**

- A) Confocal
- B) Phase-contrast
- C) Polarizing microscopy
- D) Dark-field

**Answer: C**

Explanation: Birefringent structures split polarized light and appear bright against a dark background.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: birefringence = polarized light.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to birefringence = polarized light.

Difficulty: Integration

**\*\*Q40. Alcian blue selectively stains:**

- A) DNA
- B) Acidic mucins
- C) Neutral lipid
- D) Collagen

**Answer: B**

Explanation: Alcian blue (basic) binds acidic mucins/GAGs. Lipid = Oil Red O; DNA = Feulgen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Alcian blue = acidic mucin.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Alcian blue = acidic mucin.

Difficulty: Integration

**\*\*Q41. FISH detects:**

- A) Protein antigens
- B) Specific nucleic-acid sequences
- C) Enzyme activity
- D) Lipid content

**Answer: B**

Explanation: Fluorescent DNA probes hybridize to specific sequences (e.g., HER2 amplification). Protein = IHC.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: FISH = DNA sequence.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to FISH = DNA sequence.

Difficulty: Integration

**\*\*Q42. Empty, clear vacuoles in hepatocytes on a routine H&E section most likely represent:**

- A) Lipid that was dissolved during processing
- B) Calcification
- C) Iron deposition

D) Glycogen always

**Answer: A**

Explanation: Alcohol/xylene extract lipid, leaving clear spaces. Glycogen/water can also be clear, so confirmation (Oil Red O on frozen) is required but the classic cause is dissolved fat.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: clear vacuoles = dissolved lipid (confirm on frozen).

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to clear vacuoles = dissolved lipid (confirm on frozen).

Difficulty: Integration

**\*\*Q43. The routine thickness of a paraffin section is approximately:**

- A) 50 m
- B) 50 nm
- C) 5 m
- D) 0.5 mm

**Answer: C**

Explanation: ~5 m (range 1 10 m). EM sections are ~50 100 nm.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: paraffin 5 m.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to paraffin 5 m.

Difficulty: Integration

**\*\*Q44. Which statement about resolution is correct?**

- A) Resolution is the smallest distance two points can be separated
- B) Resolution is unlimited in light microscopy
- C) Longer wavelength improves resolution
- D) Magnification improves resolution

**Answer: A**

Explanation: Resolution is the minimum resolvable separation. Magnification cannot improve it (D); shorter wavelength improves it (C).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: resolution magnification.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to resolution magnification.

Difficulty: Integration

**\*\*Q45. Which pair is correctly matched regarding stains and their targets?**

- A) Collagen    Orcein
- B) Calcium    Prussian blue
- C) Elastic fibers    silver impregnation
- D) Reticular fibers    silver impregnation

**Answer: D**

Explanation: Reticular = silver. Elastic = Orcein; collagen = trichrome; calcium = Von Kossa (C, A, B mismatched).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: reticular    silver.

Learning Objective mapping: Chapter 1    Histology & Its Methods of Study    LO related to reticular silver.

Difficulty: Integration

**\*\*Q46. A slide shows a "signet-ring" cell with a single large clear space and a flattened peripheral nucleus. On H&E alone, the space is most safely interpreted as:**

- A) Amyloid
- B) Iron
- C) Glycogen
- D) Dissolved lipid (requires frozen/Oil Red O to confirm)

**Answer: D**

Explanation: The signet-ring pattern is classic for adipocyte/fatty change; but H&E only shows where lipid was    confirmation needs a lipid stain on frozen tissue.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: signet ring = fat (confirm on frozen).

Learning Objective mapping: Chapter 1    Histology & Its Methods of Study    LO related to signet ring = fat (confirm on frozen).

Difficulty: Integration

**\*\*Q47. A liver biopsy for suspected fatty change must be processed by:**

- A) Paraffin + Von Kossa
- B) Routine paraffin + H&E only
- C) Frozen section + Oil Red O
- D) Paraffin + Prussian blue

**Answer: C**

Explanation: Lipid dissolves in ethanol/xylene, routine paraffin shows empty vacuoles; need frozen + Oil Red O for lipid.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Lipid dissolves in ethanol/xylene, routine paraffin shows empty vacuoles; need frozen + Oil Red O fo

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Lipid dissolves in ethanol/xylene, routine paraffin shows empty vacuoles; need f

Difficulty: Integration

**\*\*Q48. H-thymidine autoradiography localizes:**

- A) Antigen expression
- B) Protein synthesis
- C) DNA synthesis
- D) Lipid synthesis

**Answer: C**

Explanation: H-thymidine incorporation marks DNA synthesis S-phase; Feulgen measures DNA amount; Ki-67 marks cycling cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: H-thymidine incorporation marks DNA synthesis S-phase; Feulgen measures DNA amount; Ki-67 marks cyc

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to H-thymidine incorporation marks DNA synthesis S-phase; Feulgen measures DNA amo

Difficulty: Integration

**Part E High-Difficulty**

**\*\*Q49. Masson trichrome stains:**

- A) Iron blue
- B) Collagen blue/green and muscle red
- C) Reticular black
- D) Amyloid green

**Answer: B**

Explanation: Trichrome differentiates collagen (blue/green) from muscle (red) fibrosis assessment. The others are different stains.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: trichrome = collagen vs muscle.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to trichrome = collagen vs muscle.

Difficulty: High-Difficulty

**\*\*Q50. The resolution limit of the light microscope is approximately:**

- A) 20  $\mu\text{m}$
- B) 0.2 nm
- C)  $\sim 0.2 \mu\text{m}$
- D) 2  $\mu\text{m}$

**Answer: C**

Explanation:  $\sim 0.2 \mu\text{m}$  (200 nm), set by visible-light wavelength. EM reaches  $\sim 0.1 \text{ nm}$ .

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: light =  $0.2 \mu\text{m}$ .

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to light =  $0.2 \mu\text{m}$ .

Difficulty: High-Difficulty

**\*\*Q51. TEM differs from SEM in that TEM:**

- A) Images surfaces in 3-D
- B) Images internal ultrastructure from thin sections
- C) Requires gold coating of the surface
- D) Uses secondary electrons only

**Answer: B**

Explanation: TEM passes electrons through ultrathin sections internal structure. SEM images surfaces (A, C, D describe SEM).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: TEM inside SEM surface.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to TEM inside SEM surface.

Difficulty: High-Difficulty

**\*\*Q52. The Feulgen reaction specifically demonstrates:**

- A) Glycogen
- B) DNA
- C) Lipid
- D) Reticular fibers

**Answer: B**

Explanation: Acid hydrolysis exposes deoxyribose aldehydes that react with Schiff specific for DNA and quantifiable.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Feulgen = DNA.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Feulgen = DNA.

Difficulty: High-Difficulty

**\*\*Q53. H-thymidine autoradiography demonstrates:**

- A) Antibody secretion
- B) Protein synthesis
- C) DNA synthesis
- D) Lipid synthesis

**Answer: C**

Explanation: Thymidine is incorporated into DNA; silver grains mark cells synthesizing DNA during the pulse.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: H-thymidine = DNA synthesis.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to H-thymidine = DNA synthesis.

Difficulty: High-Difficulty

**\*\*Q54. Metachromasia refers to:**

- A) Staining with two dyes
- B) Heavy-metal staining
- C) A dye staining tissue its own color
- D) A dye staining tissue a different color (dye polymerization)

**Answer: D**

Explanation: Polymerized dye on dense polyanions (heparin) shifts color (toluidine blue purple/red).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: metachromasia = color shift.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to metachromasia = color shift.

Difficulty: High-Difficulty

**\*\*Q55. Semi-thin (~1 μm) sections for EM orientation are stained with:**

- A) Toluidine blue
- B) Congo red
- C) Prussian blue
- D) H&E

**Answer: A**

Explanation: Resin-embedded semi-thin sections are stained with toluidine blue to select regions for ultrathin sectioning.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: semi-thin    toluidine blue.

Learning Objective mapping: Chapter 1    Histology & Its Methods of Study    LO related to semi-thin toluidine blue.

Difficulty: High-Difficulty

**\*\*Q56. Why is TEM required for podocyte foot-process evaluation?**

- A) Foot processes are acidophilic
- B) The alterations are below light resolution (~nm)
- C) TEM stains basement membrane
- D) Light microscopy cannot cut glomeruli

**Answer: B**

Explanation: Foot-process effacement is nanometer-scale, below the 0.2  $\mu$ m light limit.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: below 0.2  $\mu$ m    EM.

Learning Objective mapping: Chapter 1    Histology & Its Methods of Study    LO related to below 0.2  $\mu$ m    EM.

Difficulty: High-Difficulty

**\*\*Q57. A biopsy must show whether fibrosis (collagen) or muscle dominates a scar. The best stain is:**

- A) PAS
- B) Masson trichrome
- C) Oil Red O
- D) Feulgen

**Answer: B**

Explanation: Trichrome colors collagen (blue/green) vs muscle (red)    fibrosis assessment.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: fibrosis    trichrome.

Learning Objective mapping: Chapter 1    Histology & Its Methods of Study    LO related to fibrosis trichrome.

Difficulty: High-Difficulty

**\*\*Q58. Which sequence correctly ranks resolution from best to worst?**

- A) TEM > SEM > light
- B) Light > SEM > TEM
- C) TEM light > SEM
- D) Light > TEM > SEM

**Answer: A**

Explanation: TEM has the best resolution (~0.1 nm), then SEM (nm tens of nm), then light (0.2 m). Practice Test

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: TEM resolution > SEM > light.

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to TEM resolution > SEM > light.

Difficulty: High-Difficulty

**\*\*Q59. A slide shows purple granules in a cell stained with blue toluidine dye. This is:**

- A) Basophilia
- B) Metachromasia
- C) Acidophilia
- D) Birefringence

**Answer: B**

Explanation: Metachromasia = color shift blue purple/red due to dye polymerization on dense polyanions like heparin in mast cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Metachromasia = color shift blue purple/red due to dye polymerization on dense polyanions like hepar

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Metachromasia = color shift blue purple/red due to dye polymerization on dense p

Difficulty: High-Difficulty

**\*\*Q60. An intraoperative margin must be reported in 15 minutes. The processing is:**

- A) Paraffin embedding
- B) Frozen section
- C) Plastic embedding
- D) Decalcification

**Answer: B**

Explanation: Frozen section for intraoperative diagnosis, preserves enzymes/lipids, fast but ice crystal artifacts, inferior morphology.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point: Frozen section for intraoperative diagnosis, preserves enzymes/lipids, fast but ice crystal artifact

Learning Objective mapping: Chapter 1 Histology & Its Methods of Study LO related to Frozen section for intraoperative diagnosis, preserves enzymes/lipids, fast but

Difficulty: High-Difficulty

# Chapter 2      The Cytoplasm

## Part A      Recall

**\*\*Q1. The cytoskeletal element used as a cell-type marker in tumor diagnosis is:**

- A) Actin
- B) Microfilaments
- C) Intermediate filaments
- D) Microtubules

**Answer: C**

Explanation: Intermediate filaments are cell-type specific (keratin, vimentin, desmin ). Actin and tubulin are identical across cell types (B, D, A wrong).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: identity = intermediate filaments.

Learning Objective mapping: Chapter 2      The Cytoplasm      LO related to identity = intermediate filaments.

Difficulty: Recall

**\*\*Q2. Neurofilaments are the marker of:**

- A) Epithelium
- B) Glia
- C) Neurons
- D) Muscle

**Answer: C**

Explanation: Neurofilaments are the neuronal intermediate filament.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: neurofilament = neuron.

Learning Objective mapping: Chapter 2      The Cytoplasm      LO related to neurofilament = neuron.

Difficulty: Recall

**\*\*Q3. Zellweger syndrome results from defective biogenesis of:**

- A) Lysosomes
- B) Mitochondria
- C) Golgi apparatus
- D) Peroxisomes

**Answer: D**

Explanation: Peroxisome biogenesis failure      VLCFA accumulation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Zellweger = peroxisome.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Zellweger = peroxisome.

Difficulty: Recall

**\*\*Q4. Mitochondrial DNA is inherited:**

- A) From both equally
- B) Only in males
- C) From the father
- D) From the mother

**Answer: D**

Explanation: All mitochondria come from the oocyte maternal inheritance.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: mtDNA = maternal.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to mtDNA = maternal.

Difficulty: Recall

**\*\*Q5. Centrioles are composed of:**

- A) Six doublets
- B) Thirteen protofilaments only
- C) 9+2 doublets
- D) Nine triplets with no central pair

**Answer: D**

Explanation: Centrioles = 9 triplets, no central pair the classic contrast with cilia.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: centriole = 9 triplets.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to centriole = 9 triplets.

Difficulty: Recall

**\*\*Q6. The signal recognition particle (SRP) functions to:**

- A) Dock ribosomes translating secretory proteins onto the RER
- B) Package proteins into secretory vesicles
- C) Import proteins into peroxisomes
- D) Tag proteins for degradation

**Answer: A**

**Explanation:** SRP binds the signal peptide and docks the ribosome mRNA complex onto the RER.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: SRP RER docking.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to SRP RER docking.

Difficulty: Recall

**\*\*Q7. Catalase, which detoxifies hydrogen peroxide, is found in:**

- A) Lysosomes
- B) Peroxisomes
- C) Mitochondria
- D) Golgi

**Answer: B**

**Explanation:** Peroxisomal catalase converts  $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$ .

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: catalase = peroxisome.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to catalase = peroxisome.

Difficulty: Recall

**\*\*Q8. LDL uptake occurs by:**

- A) Simple diffusion
- B) Receptor-mediated endocytosis via clathrin-coated pits
- C) Phagocytosis
- D) Pinocytosis only

**Answer: B**

**Explanation:** LDL receptors cluster in clathrin-coated pits internalization.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: LDL = receptor-mediated endocytosis.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to LDL = receptor-mediated endocytosis.

Difficulty: Recall

**\*\*Q9. Lipofuscin is best described as:**

- A) A benign wear-and-tear pigment
- B) Stored iron

- C) A form of glycogen
- D) A harmful cause of cell death

**Answer: A**

Explanation: Lipofuscin is indigestible lysosomal residue, benign, increasing with age.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: lipofuscin = benign aging pigment.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to lipofuscin = benign aging pigment.

Difficulty: Recall

**\*\*Q10. Which cell performs phagocytosis of large particles?**

- A) Adipocyte
- B) Fibroblast
- C) Macrophage
- D) Erythrocyte

**Answer: C**

Explanation: Macrophages (and neutrophils) phagocytose large particles.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: phagocytosis = macrophages/neutrophils.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to phagocytosis = macrophages/neutrophils.

Difficulty: Recall

**\*\*Q11. A malignant tumor stains positively for desmin. The most likely diagnosis is:**

- A) Astrocytoma
- B) Lymphoma
- C) Carcinoma
- D) Rhabdomyosarcoma

**Answer: D**

Explanation: Desmin intermediate filament marks muscle differentiation rhabdomyosarcoma; GFAP astrocyte, cytokeratin carcinoma.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Desmin intermediate filament marks muscle differentiation rhabdomyosarcoma; GFAP astrocyte, cytoke



Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Desmin intermediate filament marks muscle differentiation rhabdomyosarcoma; GF

Difficulty: Recall

**\*\*Q12. A vesicle moves a protein from the ER to the Golgi. Its coat is:**

- A) COPI
- B) COPII
- C) Clathrin
- D) Caveolin

**Answer: B**

Explanation: COPII ER Golgi anterograde, COPI Golgi ER retrograde, clathrin Golgi membrane and membrane endosome.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: COPII ER Golgi anterograde, COPI Golgi ER retrograde, clathrin Golgi membrane and membrane endosome.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to COPII ER Golgi anterograde, COPI Golgi ER retrograde, clathrin Golgi membrane an

Difficulty: Recall

## Part B Understanding

**\*\*Q13. Cytokeratin is the marker of:**

- A) Muscle cells
- B) Epithelial cells
- C) Neurons
- D) Astrocytes

**Answer: B**

Explanation: Keratin epithelial cells carcinoma. Desmin = muscle; neurofilament = neuron; GFAP = astrocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: keratin = epithelium.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to keratin = epithelium.

Difficulty: Understanding

**\*\*Q14. The core of microvilli contains:**

- A) Microtubules
- B) Actin filaments
- C) Intermediate filaments
- D) Dynein arms

**Answer: B**

**Explanation:** Microvilli have an actin core (with villin/fimbrin). Microtubules are in cilia.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: microvilli = actin.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to microvilli = actin.

Difficulty: Understanding

**\*\*Q15. I-cell disease results from failure to tag lysosomal enzymes with:**

- A) Ubiquitin
- B) Mannose-6-phosphate
- C) A GPI anchor
- D) Signal peptide

**Answer: B**

**Explanation:** Without M6P, enzymes are secreted instead of delivered to lysosomes.

Ubiquitin proteasome.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: I-cell = M6P defect.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to I-cell = M6P defect.

Difficulty: Understanding

**\*\*Q16. Microtubules are polymers of:**

- A) Actin
- B) Tubulin
- C) Keratin
- D) Vimentin

**Answer: B**

**Explanation:** -tubulin dimers 13 protofilaments hollow tube.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: microtubule = tubulin.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to microtubule = tubulin.

Difficulty: Understanding

**\*\*Q17. Keratin intermediate filaments mark:**

- A) Neurons
- B) Muscle
- C) Epithelial cells

D) Astrocytes

**Answer: C**

Explanation: Keratin   epithelial cells   carcinoma.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: keratin = epithelium.

Learning Objective mapping: Chapter 2   The Cytoplasm   LO related to keratin = epithelium.

Difficulty: Understanding

**\*\*Q18. COPII-coated vesicles mediate transport from the:**

- A) Rough ER to the Golgi
- B) Lysosome to the Golgi
- C) Nucleus to the cytoplasm
- D) Golgi to the plasma membrane

**Answer: A**

Explanation: COPII = ER   Golgi (anterograde). Clathrin = Golgi   membrane.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: COPII = ER   Golgi.

Learning Objective mapping: Chapter 2   The Cytoplasm   LO related to COPII = ER   Golgi.

Difficulty: Understanding

**\*\*Q19. The Golgi apparatus receives proteins at the \_\_\_\_ face and exits at the \_\_\_\_ face:**

- A) Both identical
- B) No polarity
- C) Trans; cis
- D) Cis; trans

**Answer: D**

Explanation: Entry = cis (facing RER); exit = trans.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: cis   trans.

Learning Objective mapping: Chapter 2   The Cytoplasm   LO related to cis   trans.

Difficulty: Understanding

**\*\*Q20. Cytosolic proteins such as hemoglobin are synthesized on:**

- A) The smooth ER
- B) Bound ribosomes (RER)
- C) Free ribosomes
- D) The Golgi

**Answer: C**

Explanation: Free polyribosomes make cytosolic (and nuclear/mitochondrial) proteins.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: free ribosomes = cytosolic proteins.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to free ribosomes = cytosolic proteins.

Difficulty: Understanding

**\*\*Q21. Which is a NON-membranous organelle?**

- A) Ribosome
- B) Mitochondrion
- C) Lysosome
- D) Peroxisome

**Answer: A**

Explanation: Ribosomes (and proteasomes, centrioles) lack membranes.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: ribosome = non-membranous.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to ribosome = non-membranous.

Difficulty: Understanding

**\*\*Q22. The glycocalyx functions chiefly in:**

- A) Cell recognition, adhesion, and protection
- B) Protein synthesis
- C) Chromosome separation
- D) ATP production

**Answer: A**

Explanation: The carbohydrate coat mediates recognition/adhesion and protects the surface.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: glycocalyx = recognition.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to glycocalyx = recognition.  
Difficulty: Understanding

**\*\*Q23. A drug causes mitotic arrest by preventing microtubule disassembly. The drug is:**

- A) Paclitaxel
- B) Phalloidin
- C) Colchicine
- D) Vincristine

**Answer: A**

Explanation: Paclitaxel stabilizes microtubules preventing disassembly mitotic arrest; colchicine/vincristine prevent assembly.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Paclitaxel stabilizes microtubules preventing disassembly mitotic arrest; colchicine/vincristine p

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Paclitaxel stabilizes microtubules preventing disassembly mitotic arrest; colc

Difficulty: Understanding

**\*\*Q24. Chronic sinusitis, bronchiectasis, and situs inversus point to a defect in:**

- A) Lamin
- B) Keratin
- C) Dynein arms
- D) Actin

**Answer: C**

Explanation: Kartagener primary ciliary dyskinesia dynein arm defect immotile cilia sinusitis bronchiectasis situs inversus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Kartagener primary ciliary dyskinesia dynein arm defect immotile cilia sinusitis bronchiectasis si

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Kartagener primary ciliary dyskinesia dynein arm defect immotile cilia sinusit

Difficulty: Understanding

## Part C Application

**\*\*Q25. Vimentin marks:**

- A) Muscle
- B) Glia
- C) Epithelial cells

D) Mesenchymal / connective-tissue cells

**Answer: D**

Explanation: Vimentin    fibroblasts/endothelium    sarcoma. Keratin = epithelium; desmin = muscle.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: vimentin = mesenchyme.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to vimentin = mesenchyme.

Difficulty: Application

**\*\*Q26. Cilia and flagella contain:**

- A) Keratin filaments
- B) Collagen
- C) Actin bundles
- D) A 9+2 microtubule axoneme

**Answer: D**

Explanation: 9 peripheral doublets + central pair, with dynein arms.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: cilia = 9+2 tubulin.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to cilia = 9+2 tubulin.

Difficulty: Application

**\*\*Q27. Proteins destined for secretion are synthesized on:**

- A) The Golgi
- B) Free ribosomes
- C) Ribosomes bound to the rough ER
- D) The smooth ER

**Answer: C**

Explanation: Secretory/membrane/lysosomal proteins are made on RER-bound ribosomes after SRP docking.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: secretion = bound ribosomes (RER).

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to secretion = bound ribosomes (RER).

Difficulty: Application

**\*\*Q28. Retrograde axonal transport is mediated by:**

- A) Dynein
- B) Myosin
- C) Dynamin
- D) Kinesin

**Answer: A**

Explanation: Dynein moves cargo toward the cell body (minus end).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: dynein = retrograde.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to dynein = retrograde.

Difficulty: Application

**\*\*Q29. Desmin is most characteristic of:**

- A) Neurons
- B) Epithelium
- C) Muscle
- D) Astrocytes

**Answer: C**

Explanation: Desmin = muscle (rhabdo/leiomyosarcoma).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: desmin = muscle.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to desmin = muscle.

Difficulty: Application

**\*\*Q30. Tay-Sachs disease is caused by dysfunction of the:**

- A) Mitochondrion
- B) Golgi
- C) Peroxisome
- D) Lysosome

**Answer: D**

Explanation: Hexosaminidase A deficiency    GM2 ganglioside storage in lysosomes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Tay-Sachs = lysosome.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to Tay-Sachs = lysosome.

Difficulty: Application

**\*\*Q31. Paclitaxel (taxol) acts by:**

- A) Blocking actin
- B) Inhibiting kinesin
- C) Depolymerizing microtubules
- D) Stabilizing microtubules

**Answer: D**

Explanation: Taxol stabilizes microtubules, freezing the spindle still mitotic arrest.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: taxol = stabilize.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to taxol = stabilize.

Difficulty: Application

**\*\*Q32. Cells active in detoxification (hepatocytes) are rich in smooth ER containing:**

- A) Catalase
- B) Cytochrome P450 enzymes
- C) Acid hydrolases
- D) Dynein

**Answer: B**

Explanation: SER carries CYP450 oxidation of drugs/toxins.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: SER = CYP450 detox.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to SER = CYP450 detox.

Difficulty: Application

**\*\*Q33. Steroid-secreting cells are rich in:**

- A) Lysosomes
- B) Glycogen
- C) Rough ER
- D) Smooth ER

**Answer: D**

Explanation: SER synthesizes steroids (adrenal cortex, Leydig cells).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.



C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: steroid cells = SER.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to steroid cells = SER.

Difficulty: Application

**\*\*Q34. A protein lacking a signal peptide will be synthesized on:**

- A) Free ribosomes
- B) The Golgi
- C) The smooth ER
- D) RER-bound ribosomes

**Answer: A**

Explanation: Without a signal peptide, SRP cannot dock the ribosome free-ribosome (cytosolic) synthesis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: no signal peptide free ribosomes.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to no signal peptide free ribosomes.

Difficulty: Application

**\*\*Q35. A cell shows intensely blue cytoplasm and a pale perinuclear zone. The dominant organelles are:**

- A) SER and mitochondria
- B) RER and Golgi
- C) Lysosomes and peroxisomes
- D) Actin and myosin

**Answer: B**

Explanation: Plasma cell: basophilic cytoplasm = RER, perinuclear halo = Golgi, eccentric clock-face nucleus, Russell bodies immunoglobulin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Plasma cell: basophilic cytoplasm = RER, perinuclear halo = Golgi, eccentric clock-face nucleus, Rus

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Plasma cell: basophilic cytoplasm = RER, perinuclear halo = Golgi, eccentric clo

Difficulty: Application

**\*\*Q36. A disease is transmitted only through females but affects both sexes. The genetic lesion is in:**

- A) The X chromosome only
- B) The Y chromosome

- C) Nuclear DNA
- D) Mitochondrial DNA

**Answer: D**

Explanation: Mitochondrial DNA maternal inheritance, affects both sexes but transmitted only through females.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Mitochondrial DNA maternal inheritance, affects both sexes but transmitted only through females.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Mitochondrial DNA maternal inheritance, affects both sexes but transmitted only

Difficulty: Application

## Part D Integration

**\*\*Q37. Desmin is the marker of:**

- A) Fibroblasts
- B) Neurons
- C) Muscle cells
- D) Epithelium

**Answer: C**

Explanation: Desmin is the muscle intermediate filament (rhabdomyosarcoma, leiomyosarcoma).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: desmin = muscle.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to desmin = muscle.

Difficulty: Integration

**\*\*Q38. The mitotic spindle is composed of:**

- A) Actin
- B) Microtubules
- C) Intermediate filaments
- D) Myosin

**Answer: B**

Explanation: Tubulin microtubules separate chromosomes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: spindle = microtubules.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to spindle = microtubules.

Difficulty: Integration

**\*\*Q39. The proteasome degrades proteins tagged with:**

- A) GPI anchor
- B) Mannose-6-phosphate
- C) Ubiquitin
- D) Signal peptide

**Answer: C**

Explanation: Ubiquitin tags proteins for proteasomal degradation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: ubiquitin    proteasome.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to ubiquitin    proteasome.

Difficulty: Integration

**\*\*Q40. Anterograde transport away from the cell body is mediated by:**

- A) Kinesin
- B) Myosin II
- C) Tubulin alone
- D) Dynein

**Answer: A**

Explanation: Kinesin walks toward the plus end (away from the soma).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: kinesin = anterograde.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to kinesin = anterograde.

Difficulty: Integration

**\*\*Q41. Lamins form a meshwork supporting the:**

- A) Inner nuclear membrane
- B) Golgi apparatus
- C) Mitochondrial outer membrane
- D) Plasma membrane

**Answer: A**

Explanation: Lamins (nuclear intermediate filaments) underlie the inner nuclear membrane.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: lamin = nuclear lamina.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to lamin = nuclear lamina.

Difficulty: Integration

**\*\*Q42. Mitochondria contain their own:**

- A) Lysosomal enzymes
- B) Centrioles
- C) Ribosomes and DNA
- D) Golgi stacks

**Answer: C**

Explanation: Circular mtDNA + mitochondrial ribosomes (semi-autonomous).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: mitochondria = own DNA + ribosomes.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to mitochondria = own DNA + ribosomes.

Difficulty: Integration

**\*\*Q43. Kartagener syndrome is caused by a defect in:**

- A) Dynein arms of the ciliary axoneme
- B) Intermediate filaments
- C) Centriolar triplets
- D) Actin filaments

**Answer: A**

Explanation: Dynein-arm defect immotile cilia situs inversus, sinusitis, bronchiectasis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Kartagener = dynein arms.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Kartagener = dynein arms.

Difficulty: Integration

**\*\*Q44. The diameter of intermediate filaments is approximately:**

- A) 25 nm
- B) ~0.2  $\mu$ m
- C) 7 nm
- D) 10 nm

**Answer: D**

Explanation: Intermediate = 10 nm (between actin 7 nm and tubulin 25 nm).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: intermediate = 10 nm.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to intermediate = 10 nm.

Difficulty: Integration

**\*\*Q45. Release of cytochrome c from mitochondria triggers:**

- A) Apoptosis
- B) Autophagy
- C) Mitosis
- D) Protein synthesis

**Answer: A**

Explanation: Cytochrome c release activates the caspase cascade apoptosis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: cytochrome c apoptosis.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to cytochrome c apoptosis.

Difficulty: Integration

**\*\*Q46. The lysosomal lumen is acidic because of:**

- A) Diffusion of HCl
- B) CO<sub>2</sub> accumulation
- C) Acid hydrolase activity
- D) A proton pump (V-ATPase) in the lysosomal membrane

**Answer: D**

Explanation: A membrane H<sup>+</sup>-ATPase acidifies the lysosome so hydrolases are active.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: lysosome pH = proton pump.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to lysosome pH = proton pump.

Difficulty: Integration

**\*\*Q47. A neonate has elevated VLCFA, hypotonia, and dysmorphic facies. The defective organelle is:**

- A) Peroxisome
- B) Golgi
- C) Lysosome

D) Mitochondrion

**Answer: A**

Explanation: Zellweger peroxisome biogenesis defect    elevated VLCFA, hypotonia, dysmorphism, liver dysfunction.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Zellweger peroxisome biogenesis defect    elevated VLCFA, hypotonia, dysmorphism, liver dysfunction.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to Zellweger peroxisome biogenesis defect    elevated VLCFA, hypotonia, dysmorphism,

Difficulty: Integration

**\*\*Q48. A protein tagged with ubiquitin is destined for the:**

- A) Golgi
- B) Secretory vesicle
- C) Lysosome
- D) Proteasome

**Answer: D**

Explanation: Ubiquitin tags proteins for proteasomal degradation; M6P receptor tags for lysosome.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Ubiquitin tags proteins for proteasomal degradation; M6P receptor tags for lysosome.

Learning Objective mapping: Chapter 2    The Cytoplasm    LO related to Ubiquitin tags proteins for proteasomal degradation; M6P receptor tags for lysos

Difficulty: Integration

## **Part E    High-Difficulty**

**\*\*Q49. GFAP identifies:**

- A) Astrocytes
- B) Oligodendrocytes
- C) Microglia
- D) Neurons

**Answer: A**

Explanation: GFAP    astrocytes    astrocytoma/glioblastoma. Neurofilaments mark neurons.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: GFAP = glia.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to GFAP = glia.

Difficulty: High-Difficulty

**\*\*Q50. Colchicine and vincristine act by:**

- A) Depolymerizing microtubules
- B) Blocking keratin assembly
- C) Inhibiting myosin
- D) Stabilizing actin

**Answer: A**

Explanation: These drugs depolymerize microtubules spindle collapse mitotic arrest.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: colchicine/vincristine = depolymerize.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to colchicine/vincristine = depolymerize.

Difficulty: High-Difficulty

**\*\*Q51. Autophagy degrades a cell's own organelles by fusion of the autophagosome with a:**

- A) Peroxisome
- B) Lysosome
- C) Mitochondrion
- D) Golgi cisterna

**Answer: B**

Explanation: Autophagosome + lysosome = autolysosome.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: autophagy lysosome.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to autophagy lysosome.

Difficulty: High-Difficulty

**\*\*Q52. The 9+2 arrangement is found in the axoneme of:**

- A) Centrioles
- B) Microvilli
- C) Cilia and flagella
- D) Stereocilia

**Answer: C**

Explanation: Motile cilia/flagella have 9+2. Centrioles have 9 triplets.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point:  $9+2 = \text{cilium}$ .

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to  $9+2 = \text{cilium}$ .

Difficulty: High-Difficulty

**\*\*Q53. N-linked glycosylation begins in the:**

- A) Lysosome
- B) Golgi apparatus
- C) Rough endoplasmic reticulum
- D) Cytosol

**Answer: C**

Explanation: The core oligosaccharide is added in the RER; the Golgi then modifies it.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: N-glycosylation starts in the RER.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to N-glycosylation starts in the RER.

Difficulty: High-Difficulty

**\*\*Q54. The diameter of actin microfilaments is approximately:**

- A) 100 nm
- B) 6-8 nm
- C) 10 nm
- D) 25 nm

**Answer: B**

Explanation: Actin (7 nm) < intermediate (10 nm) < microtubules (25 nm).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: actin = thinnest (7 nm).

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to actin = thinnest (7 nm).

Difficulty: High-Difficulty

**\*\*Q55. The sarcoplasmic reticulum of muscle is a specialized:**

- A) Golgi
- B) Lysosome
- C) Rough ER
- D) Smooth ER (Ca<sup>2+</sup> store)

**Answer: D**



**Explanation:** The SR is SER specialized for Ca<sup>2+</sup> sequestration/release.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: SR = SER (Ca<sup>2+</sup>).

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to SR = SER (Ca<sup>2+</sup>).

Difficulty: High-Difficulty

**\*\*Q56. A single microtubule contains how many protofilaments?**

- A) 9
- B) 13
- C) 20
- D) 2

**Answer: B**

**Explanation:** 13 protofilaments of  $\alpha$ -tubulin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: 13 protofilaments.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to 13 protofilaments.

Difficulty: High-Difficulty

**\*\*Q57. Chediak-Higashi syndrome is best understood as a defect of:**

- A) Peroxisomal  $\beta$ -oxidation
- B) Lysosomal/granule trafficking and fusion
- C) Proteasome function
- D) Mitochondrial DNA

**Answer: B**

**Explanation:** Defective granule/lysosome trafficking giant granules, albinism, immune deficiency.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Chediak-Higashi = trafficking.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Chediak-Higashi = trafficking.

Difficulty: High-Difficulty

**\*\*Q58. Which is correctly paired regarding vesicle coats and trafficking?**

- A) Clathrin ER to Golgi
- B) COPI Golgi to plasma membrane

- C) Clathrin Golgi to membrane / membrane to endosome
- D) COPII Golgi to ER

**Answer: C**

Explanation: Clathrin mediates Golgi membrane and membrane endosome. COPII = ER Golgi; COPI = retrograde. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: clathrin = secretion + endocytosis.

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to clathrin = secretion + endocytosis.

Difficulty: High-Difficulty

**\*\*Q59. The structure with a 9+2 arrangement is the:**

- A) Nuclear pore
- B) Centriole
- C) Cilium
- D) Microvillus

**Answer: C**

Explanation: Motile cilium 9+2 doublet + central pair + dynein arms; microvillus actin core, centriole 9+0 triplet.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: Motile cilium 9+2 doublet + central pair + dynein arms; microvillus actin core, centriole 9+0 triple

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to Motile cilium 9+2 doublet + central pair + dynein arms; microvillus actin core,

Difficulty: High-Difficulty

**\*\*Q60. A clear cytoplasmic vacuole on H&E could be glycogen or lipid. The definitive discriminator is:**

- A) Prussian blue
- B) H&E with higher magnification
- C) PAS diastase vs Oil Red O on a frozen section
- D) Congo red

**Answer: C**

Explanation: PAS diastase vs Oil Red O frozen distinguishes glycogen (PAS positive diastase sensitive) vs lipid (Oil Red O frozen positive).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Cytoplasm section for decisive features.

Key point: PAS + diastase vs Oil Red O frozen distinguishes glycogen (PAS positive diastase sensitive) vs lipid

Learning Objective mapping: Chapter 2 The Cytoplasm LO related to PAS + diastase vs Oil Red O frozen distinguishes glycogen (PAS positive diastase

Difficulty: High-Difficulty

# Chapter 3      The Nucleus

## Part A      Recall

**\*\*Q1. Euchromatin is characterized by being:**

- A) Composed of RNA only
- B) Transcriptionally inactive and dark
- C) Transcriptionally active and lightly stained
- D) Located only at the periphery

**Answer: C**

Explanation: Decondensed euchromatin allows transcription      pale. Heterochromatin is inactive/dark (B); peripheral location is heterochromatin (D).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: euchromatin = active + pale.

Learning Objective mapping: Chapter 3      The Nucleus      LO related to euchromatin = active + pale.

Difficulty: Recall

**\*\*Q2. A cell with a large, prominent nucleolus is most likely:**

- A) A mature RBC
- B) Inactive
- C) Actively synthesizing protein/ribosomes
- D) Undergoing apoptosis

**Answer: C**

Explanation: High rRNA demand = high protein synthesis or rapid proliferation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: prominent nucleolus = active/proliferating cell.

Learning Objective mapping: Chapter 3      The Nucleus      LO related to prominent nucleolus = active/proliferating cell.

Difficulty: Recall

**\*\*Q3. The nucleolus is the principal site of synthesis and assembly of:**

- A) Ribosomal subunits
- B) tRNA only
- C) DNA
- D) mRNA

**Answer: A**

Explanation: Ribosomal subunits; mRNA is made in the nucleoplasm.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: nucleolus = ribosomal subunits.

Learning Objective mapping: Chapter 3 The Nucleus LO related to nucleolus = ribosomal subunits.

Difficulty: Recall

**\*\*Q4. The outer nuclear membrane is continuous with the:**

- A) Rough endoplasmic reticulum
- B) Golgi apparatus
- C) Lysosomal membrane
- D) Plasma membrane

**Answer: A**

Explanation: The outer membrane bears ribosomes and is continuous with the RER.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: outer membrane = RER.

Learning Objective mapping: Chapter 3 The Nucleus LO related to outer membrane = RER.

Difficulty: Recall

**\*\*Q5. The nuclear envelope breaks down during:**

- A) Interphase
- B) Prophase
- C) Telophase only
- D) It never breaks down

**Answer: B**

Explanation: Prophase (lamin phosphorylation).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: breakdown = prophase.

Learning Objective mapping: Chapter 3 The Nucleus LO related to breakdown = prophase.

Difficulty: Recall

**\*\*Q6. The Barr body, a dense mass at the nuclear periphery, represents:**

- A) The nucleolus
- B) Condensed rRNA genes
- C) An extra Y chromosome
- D) An inactivated X chromosome

**Answer: D**

**Explanation: The Lyonized X.**

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Barr = inactivated X.

Learning Objective mapping: Chapter 3 The Nucleus LO related to Barr = inactivated X.

Difficulty: Recall

**\*\*Q7. Ribosomal RNA genes are transcribed within the:**

- A) Nucleolus
- B) Cytoplasm
- C) Mitochondria only
- D) Nucleoplasm generally

**Answer: A**

**Explanation: The nucleolus (NORs).**

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: rRNA = nucleolus.

Learning Objective mapping: Chapter 3 The Nucleus LO related to rRNA = nucleolus.

Difficulty: Recall

**\*\*Q8. A patient with Turner syndrome (45,X) has how many Barr bodies?**

- A) 1
- B) 2
- C) 3
- D) 0

**Answer: D**

**Explanation:  $X - 1 = 0$  (only one X, which stays active).**

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point:  $X0 = 0$  Barr bodies.

Learning Objective mapping: Chapter 3 The Nucleus LO related to  $X0 = 0$  Barr bodies.

Difficulty: Recall

**\*\*Q9. DNA replication is best described as:**

- A) Conservative
- B) Semiconservative
- C) Dispersive
- D) Random

**Answer: B**

Explanation: Each daughter helix retains one parental strand.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: semiconservative.

Learning Objective mapping: Chapter 3 The Nucleus LO related to semiconservative.

Difficulty: Recall

**\*\*Q10. A key difference between necrosis and apoptosis is that necrosis:**

- A) Never involves the nucleus
- B) Triggers inflammation via release of cell contents
- C) Is always programmed
- D) Produces apoptotic bodies

**Answer: B**

Explanation: Necrosis ruptures the membrane inflammation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: necrosis = inflammatory.

Learning Objective mapping: Chapter 3 The Nucleus LO related to necrosis = inflammatory.

Difficulty: Recall

**\*\*Q11. A buccal smear from a patient shows two Barr bodies. The most likely karyotype is:**

- A) 47,XXX
- B) 47,XXY
- C) 45,X
- D) 46,XX

**Answer: A**

Explanation: Barr body count = X chromosomes -1; 47,XXX has 2 Barr bodies, 47,XXY has 1, 45,X has 0, 46,XX has 1, 46,XY has 0.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Barr body count = X chromosomes -1; 47,XXX has 2 Barr bodies, 47,XXY has 1, 45,X has 0, 46,XX has 1,

Learning Objective mapping: Chapter 3 The Nucleus LO related to Barr body count = X chromosomes -1; 47,XXX has 2 Barr bodies, 47,XXY has 1, 45,X

Difficulty: Recall

**\*\*Q12. The "guardian of the genome" acting at the G1/S checkpoint is:**

- A) Telomerase
- B) Lamin A
- C) Cyclin D
- D) p53

**Answer: D**

Explanation: p53 guardian of genome, G1/S checkpoint, DNA damage response; mutated in many cancers.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: p53 guardian of genome, G1/S checkpoint, DNA damage response; mutated in many cancers.

Learning Objective mapping: Chapter 3 The Nucleus LO related to p53 guardian of genome, G1/S checkpoint, DNA damage response; mutated in many ca

Difficulty: Recall

## **Part B Understanding**

**\*\*Q13. Heterochromatin is:**

- A) Active and pale
- B) Inactive and densely stained
- C) Absent in neurons
- D) The site of rRNA synthesis

**Answer: B**

Explanation: Condensed, transcriptionally silent, dark. rRNA synthesis = nucleolus (D).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: heterochromatin = inactive + dark.

Learning Objective mapping: Chapter 3 The Nucleus LO related to heterochromatin = inactive + dark.

Difficulty: Understanding

**\*\*Q14. The nuclear envelope consists of:**

- A) No membrane
- B) A single membrane
- C) A double membrane with nuclear pores
- D) Three membranes

**Answer: C**

Explanation: Two membranes + perinuclear cisterna + pores.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.



B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: envelope = double membrane.

Learning Objective mapping: Chapter 3 The Nucleus LO related to envelope = double membrane.

Difficulty: Understanding

**\*\*Q15. Which best describes euchromatin?**

- A) Transcriptionally active, loosely packed
- B) Found only in Barr bodies
- C) Absent in interphase nuclei
- D) Transcriptionally inactive, densely packed

**Answer: A**

Explanation: Active and loose.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: euchromatin = loose + active.

Learning Objective mapping: Chapter 3 The Nucleus LO related to euchromatin = loose + active.

Difficulty: Understanding

**\*\*Q16. A nucleosome consists of DNA wrapped around an octamer of:**

- A) Tubulin
- B) Histone proteins
- C) Lamin
- D) Actin

**Answer: B**

Explanation: H2A, H2B, H3, H4 ( 2).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: nucleosome = histone octamer.

Learning Objective mapping: Chapter 3 The Nucleus LO related to nucleosome = histone octamer.

Difficulty: Understanding

**\*\*Q17. Pyknosis, karyorrhexis, and karyolysis characterize:**

- A) Cell necrosis
- B) Meiosis
- C) DNA repair
- D) Normal mitosis

**Answer: A**

Explanation: The necrotic nuclear sequence.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: pyknosis lysis = necrosis.

Learning Objective mapping: Chapter 3 The Nucleus LO related to pyknosis lysis = necrosis.

Difficulty: Understanding

**\*\*Q18. Heterochromatin is characteristically located:**

- A) At the nuclear periphery and around the nucleolus
- B) Exclusively in the cytoplasm
- C) Only during mitosis
- D) In the center of the nucleus only

**Answer: A**

Explanation: Peripheral + perinucleolar.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: heterochromatin = periphery.

Learning Objective mapping: Chapter 3 The Nucleus LO related to heterochromatin = periphery.

Difficulty: Understanding

**\*\*Q19. A cell in G0 is best described as:**

- A) Actively dividing
- B) Quiescent and non-dividing
- C) Undergoing DNA synthesis
- D) In mitotic arrest

**Answer: B**

Explanation: G0 = out of the cycle (neurons, cardiac muscle).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: G0 = quiescent.

Learning Objective mapping: Chapter 3 The Nucleus LO related to G0 = quiescent.

Difficulty: Understanding

**\*\*Q20. A 47,XXX female has how many Barr bodies?**

- A) 3
- B) 0
- C) 1
- D) 2

**Answer: D**

Explanation: X 1 = 2.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: XXX = 2.

Learning Objective mapping: Chapter 3 The Nucleus LO related to XXX = 2.

Difficulty: Understanding

**\*\*Q21. Progression through the cell cycle is driven by:**

- A) Ribosomal RNA
- B) Hemoglobin
- C) Cyclins and cyclin-dependent kinases
- D) Histones alone

**Answer: C**

Explanation: Cyclins/CDKs drive phase transitions.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: cyclins + CDKs.

Learning Objective mapping: Chapter 3 The Nucleus LO related to cyclins + CDKs.

Difficulty: Understanding

**\*\*Q22. Telomeres function to:**

- A) Are absent in cancer cells
- B) Hold sister chromatids together
- C) Code for ribosomal RNA
- D) Protect chromosome ends from degradation and fusion

**Answer: D**

Explanation: TTAGGG repeats protect ends.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: telomere = end cap.

Learning Objective mapping: Chapter 3 The Nucleus LO related to telomere = end cap.

Difficulty: Understanding

**\*\*Q23. A cell with an intensely dark, shrunken, irregular nucleus amid inflamed tissue is undergoing:**

- A) Apoptosis
- B) Necrosis
- C) Mitosis

## D) Transcription

### Answer: B

Explanation: Pyknosis dark shrunken irregular nucleus with inflammation = necrosis; apoptosis condensed but no inflammation, caspase-mediated.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Pyknosis dark shrunken irregular nucleus with inflammation = necrosis; apoptosis condensed but no in

Learning Objective mapping: Chapter 3 The Nucleus LO related to Pyknosis dark shrunken irregular nucleus with inflammation = necrosis; apoptosis

Difficulty: Understanding

### **\*\*Q24. A cell in permanent G0 is characteristic of:**

- A) Intestinal epithelium
- B) Cardiac muscle
- C) Bone marrow stem cells
- D) Epidermis

### Answer: B

Explanation: Cardiac muscle and neurons permanent G0 post-mitotic no centrioles no regeneration; intestinal epithelium labile, stem cells quiescent G0 but can re-enter.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Cardiac muscle and neurons permanent G0 post-mitotic no centrioles no regeneration; intestinal epith

Learning Objective mapping: Chapter 3 The Nucleus LO related to Cardiac muscle and neurons permanent G0 post-mitotic no centrioles no regenerati

Difficulty: Understanding

## Part C Application

### **\*\*Q25. The nucleolus is the site of:**

- A) DNA replication
- B) rRNA synthesis and ribosome assembly
- C) Protein translation
- D) Lipid synthesis

### Answer: B

Explanation: rRNA genes (NORs) are transcribed here and ribosomal subunits assembled. Replication is nucleoplasmic; translation is cytoplasmic.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: nucleolus = rRNA + ribosomes.

Learning Objective mapping: Chapter 3 The Nucleus LO related to nucleolus = rRNA + ribosomes.

Difficulty: Application

**\*\*Q26. Chromatin is composed of:**

- A) Collagen and elastin
- B) Actin and tubulin
- C) RNA and lipids
- D) DNA and histones

**Answer: D**

Explanation: DNA wrapped around histone octamers nucleosomes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: chromatin = DNA + histones.

Learning Objective mapping: Chapter 3 The Nucleus LO related to chromatin = DNA + histones.

Difficulty: Application

**\*\*Q27. A prominent, enlarged nucleolus on cytology is most suggestive of:**

- A) A malignant or rapidly proliferating cell
- B) A senescent cell
- C) A quiescent cell
- D) An apoptotic cell

**Answer: A**

Explanation: Nucleolar enlargement tracks proliferative activity a malignancy clue.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: enlarged nucleolus = malignancy concern.

Learning Objective mapping: Chapter 3 The Nucleus LO related to enlarged nucleolus = malignancy concern.

Difficulty: Application

**\*\*Q28. Which phase of the cell cycle is dedicated to DNA replication?**

- A) M
- B) G1
- C) S
- D) G2

**Answer: C**

Explanation: S = synthesis of DNA (and histones).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: S phase = replication.

Learning Objective mapping: Chapter 3 The Nucleus LO related to S phase = replication.

Difficulty: Application

**\*\*Q29. Apoptosis, in contrast to necrosis, is characterized by:**

- A) Nuclear swelling
- B) Chromatin condensation and fragmentation into apoptotic bodies
- C) Karyolysis with dissolution
- D) No nuclear change

**Answer: B**

Explanation: Clean condensation + fragmentation, no inflammation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: apoptosis = condensation + bodies.

Learning Objective mapping: Chapter 3 The Nucleus LO related to apoptosis = condensation + bodies.

Difficulty: Application

**\*\*Q30. The G1/S checkpoint, preventing cells with damaged DNA from replicating, is regulated chiefly by:**

- A) Actin
- B) Lamin
- C) Cyclin D
- D) p53

**Answer: D**

Explanation: p53 = guardian of the genome.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: p53 = G1/S brake.

Learning Objective mapping: Chapter 3 The Nucleus LO related to p53 = G1/S brake.

Difficulty: Application

**\*\*Q31. Down syndrome is most commonly caused by:**

- A) A balanced translocation only
- B) Loss of a Barr body
- C) Monosomy of chromosome 21

D) Trisomy of chromosome 21

**Answer: D**

Explanation: Trisomy 21 (meiotic nondisjunction).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Down = trisomy 21.

Learning Objective mapping: Chapter 3 The Nucleus LO related to Down = trisomy 21.

Difficulty: Application

**\*\*Q32. Histones are synthesized primarily during which phase?**

A) M

B) G1

C) S

D) G2

**Answer: C**

Explanation: Histones are made alongside DNA replication in S phase.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: histones = S phase.

Learning Objective mapping: Chapter 3 The Nucleus LO related to histones = S phase.

Difficulty: Application

**\*\*Q33. Which cells are permanent (post-mitotic)?**

A) Intestinal epithelial cells

B) Bone marrow stem cells

C) Skin keratinocytes

D) Neurons and cardiac muscle cells

**Answer: D**

Explanation: Neurons and cardiac muscle remain in G0.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: permanent = neuron + cardiac muscle.

Learning Objective mapping: Chapter 3 The Nucleus LO related to permanent = neuron + cardiac muscle.

Difficulty: Application

**\*\*Q34. Nuclear pores permit the exit of:**

- A) DNA
- B) mRNA and ribosomal subunits
- C) Histones only
- D) Nucleosomes

**Answer: B**

Explanation: mRNA and ribosomal subunits exit to the cytoplasm; DNA remains nuclear.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: RNA exits, DNA stays.

Learning Objective mapping: Chapter 3 The Nucleus LO related to RNA exits, DNA stays.

Difficulty: Application

**\*\*Q35. The structure that disassembles when lamins are phosphorylated is the:**

- A) Nucleolus
- B) Nuclear envelope
- C) Centriole
- D) Barr body

**Answer: B**

Explanation: Lamin phosphorylation disassembles nuclear lamina nuclear envelope breakdown in mitosis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Lamin phosphorylation disassembles nuclear lamina nuclear envelope breakdown in mitosis.

Learning Objective mapping: Chapter 3 The Nucleus LO related to Lamin phosphorylation disassembles nuclear lamina nuclear envelope breakdown i

Difficulty: Application

**\*\*Q36. The pale, open nucleus of an activated cell indicates:**

- A) Euchromatin predominance (active transcription)
- B) Apoptosis
- C) Polyploidy
- D) Heterochromatin predominance

**Answer: A**

Explanation: Pale open euchromatic nucleus = active transcription, dark heterochromatic = inactive.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Pale open euchromatic nucleus = active transcription, dark heterochromatic = inactive.

Learning Objective mapping: Chapter 3 The Nucleus LO related to Pale open euchromatic nucleus = active transcription, dark heterochromatic = inactive

Difficulty: Application

## Part D Integration

### **\*\*Q37. A Barr body represents:**

- A) The Y chromosome
- B) A nucleolus
- C) An active X chromosome
- D) An inactivated X chromosome

#### **Answer: D**

Explanation: Lyonization condenses one X into peripheral heterochromatin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Barr body = inactivated X.

Learning Objective mapping: Chapter 3 The Nucleus LO related to Barr body = inactivated X.

Difficulty: Integration

### **\*\*Q38. In Klinefelter syndrome (47,XXY), the number of Barr bodies is:**

- A) 2
- B) 3
- C) 0
- D) 1

#### **Answer: D**

Explanation:  $X - 1 = 1$  (the Y contributes none).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point:  $XXY = 1$  Barr body.

Learning Objective mapping: Chapter 3 The Nucleus LO related to  $XXY = 1$  Barr body.

Difficulty: Integration

### **\*\*Q39. The nuclear envelope is composed of how many concentric membranes?**

- A) Two
- B) Three
- C) Four
- D) One

**Answer: A**

Explanation: Two, separated by the perinuclear cisterna.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: two membranes.

Learning Objective mapping: Chapter 3 The Nucleus LO related to two membranes.

Difficulty: Integration

**\*\*Q40. Chromosomes align at the metaphase plate during:**

- A) Telophase
- B) Prophase
- C) Metaphase
- D) Anaphase

**Answer: C**

Explanation: Metaphase = equatorial alignment.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: align = metaphase.

Learning Objective mapping: Chapter 3 The Nucleus LO related to align = metaphase.

Difficulty: Integration

**\*\*Q41. A normal human somatic cell contains how many chromosomes?**

- A) 48
- B) 23
- C) 46
- D) 44

**Answer: C**

Explanation: 46 (23 pairs).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point:  $2n = 46$ .

Learning Objective mapping: Chapter 3 The Nucleus LO related to  $2n = 46$ .

Difficulty: Integration

**\*\*Q42. Polyploidy is a normal finding in:**

- A) Neurons
- B) Fibroblasts

- C) Lymphocytes
- D) Hepatocytes and megakaryocytes

**Answer: D**

Explanation: Hepatocytes (endoreduplication) and megakaryocytes (endomitosis).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: normal polyploidy = hepatocyte + megakaryocyte.

Learning Objective mapping: Chapter 3 The Nucleus LO related to normal polyploidy = hepatocyte + megakaryocyte.

Difficulty: Integration

**\*\*Q43. Nucleolar organizer regions are chromosomal segments that:**

- A) Bear rRNA genes and form the nucleolus
- B) Are found only on the Y chromosome
- C) Contain no genetic material
- D) Are visible only in mitosis

**Answer: A**

Explanation: NORs on acrocentric chromosomes 13, 14, 15, 21, 22.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: NOR = rRNA genes.

Learning Objective mapping: Chapter 3 The Nucleus LO related to NOR = rRNA genes.

Difficulty: Integration

**\*\*Q44. Meiosis differs from mitosis in that meiosis:**

- A) Has a single division
- B) Produces two identical diploid cells
- C) Halves the chromosome number to produce haploid gametes
- D) Occurs in all somatic cells

**Answer: C**

Explanation: Two divisions four haploid gametes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: meiosis = haploid gametes.

Learning Objective mapping: Chapter 3 The Nucleus LO related to meiosis = haploid gametes.

Difficulty: Integration

**\*\*Q45. Sister chromatids are held together at the:**

- A) Telomere
- B) Centromere
- C) Nucleolus
- D) Nuclear pore

**Answer: B**

Explanation: Cohesin at the centromere until anaphase.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: centromere = hold.

Learning Objective mapping: Chapter 3 The Nucleus LO related to centromere = hold.

Difficulty: Integration

**\*\*Q46. During telophase, which structure re-forms around each chromosome set?**

- A) The nuclear envelope
- B) The spindle
- C) The Barr body
- D) The nucleolus only

**Answer: A**

Explanation: ER-derived membrane re-forms the envelope.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: telophase = envelope re-forms.

Learning Objective mapping: Chapter 3 The Nucleus LO related to telophase = envelope re-forms.

Difficulty: Integration

**\*\*Q47. A cytology specimen shows large nuclei with very prominent nucleoli. The primary concern is:**

- A) Apoptosis
- B) Senescence
- C) Malignancy
- D) Quiescence

**Answer: C**

Explanation: Large nuclei with prominent nucleoli indicate high ribosome demand malignancy concern; small dark nucleus = quiescence.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Large nuclei with prominent nucleoli indicate high ribosome demand    malignancy concern; small dark

Learning Objective mapping: Chapter 3    The Nucleus    LO related to Large nuclei with prominent nucleoli indicate high ribosome demand    malignancy

Difficulty: Integration

**\*\*Q48. In 46,XY males, the Barr-body count is:**

- A) 2
- B) 3
- C) 0
- D) 1

**Answer: C**

Explanation: 46,XY male has 0 Barr bodies; 46,XX female has 1; 47,XXY has 1; 47,XXX has 2.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: 46,XY male has 0 Barr bodies; 46,XX female has 1; 47,XXY has 1; 47,XXX has 2.

Learning Objective mapping: Chapter 3    The Nucleus    LO related to 46,XY male has 0 Barr bodies; 46,XX female has 1; 47,XXY has 1; 47,XXX has 2.

Difficulty: Integration

## Part E    High-Difficulty

**\*\*Q49. The number of Barr bodies in a normal female (46,XX) is:**

- A) 0
- B) 1
- C) 2
- D) 3

**Answer: B**

Explanation:  $X - 1 = 1$ .

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: female = 1 Barr body.

Learning Objective mapping: Chapter 3    The Nucleus    LO related to female = 1 Barr body.

Difficulty: High-Difficulty

**\*\*Q50. Heterochromatin is typically located:**

- A) Evenly dispersed as active genes
- B) Outside the nucleus
- C) Only in the nucleolus
- D) At the nuclear periphery and around the nucleolus

**Answer: D**

**Explanation:** Condensed chromatin concentrates at the periphery and around the nucleolus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: heterochromatin = periphery.

Learning Objective mapping: Chapter 3 The Nucleus LO related to heterochromatin = periphery.

Difficulty: High-Difficulty

**\*\*Q51. Nuclear pores are largely impermeable to passively diffusing molecules larger than approximately:**

A) 500 kDa

B) 5 kDa

C) There is no size limit

D) 40 kDa

**Answer: D**

**Explanation:** ~40-60 kDa; larger molecules use active transport.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: pore passive limit ~40 kDa.

Learning Objective mapping: Chapter 3 The Nucleus LO related to pore passive limit ~40 kDa.

Difficulty: High-Difficulty

**\*\*Q52. Sister chromatids separate during:**

A) Anaphase

B) Telophase

C) Prophase

D) Metaphase

**Answer: A**

**Explanation:** Anaphase = separation to opposite poles.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: separate = anaphase.

Learning Objective mapping: Chapter 3 The Nucleus LO related to separate = anaphase.

Difficulty: High-Difficulty

**\*\*Q53. Klinefelter syndrome (47,XXY) shows how many Barr bodies?**

A) Three

B) Zero

- C) One
- D) Two

**Answer: C**

Explanation:  $X - 1 = 1$ .

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point:  $XXY = 1$ .

Learning Objective mapping: Chapter 3 The Nucleus LO related to  $XXY = 1$ .

Difficulty: High-Difficulty

**\*\*Q54. The nuclear lamina is composed of:**

- A) Keratin
- B) Actin filaments
- C) Lamin intermediate filaments
- D) Microtubules

**Answer: C**

Explanation: Lamins (nuclear intermediate filaments).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: lamina = lamin.

Learning Objective mapping: Chapter 3 The Nucleus LO related to lamina = lamin.

Difficulty: High-Difficulty

**\*\*Q55. Telomerase is characteristically reactivated in:**

- A) Normal somatic cells throughout life
- B) The majority of malignant tumors
- C) Only red blood cells
- D) It is never active in humans

**Answer: B**

Explanation: Telomerase confers replicative immortality to most cancers.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: cancer = telomerase ON.

Learning Objective mapping: Chapter 3 The Nucleus LO related to cancer = telomerase ON.

Difficulty: High-Difficulty

**\*\*Q56. The space between the two nuclear membranes is the:**

- A) Nucleoplasm
- B) Perinuclear cisterna
- C) Nucleolus
- D) Nuclear lamina

**Answer: B**

Explanation: Perinuclear cisterna.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: perinuclear cisterna.

Learning Objective mapping: Chapter 3 The Nucleus LO related to perinuclear cisterna.

Difficulty: High-Difficulty

**\*\*Q57. Which is an example of physiological apoptosis?**

- A) Removal of interdigital webbing in the embryo
- B) Burn injury
- C) Frostbite
- D) Ischemic infarction

**Answer: A**

Explanation: Programmed, silent cell death shapes the digits.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: embryonic webbing = apoptosis.

Learning Objective mapping: Chapter 3 The Nucleus LO related to embryonic webbing = apoptosis.

Difficulty: High-Difficulty

**\*\*Q58. "Beads on a string" represents:**

- A) Microtubule assembly
- B) The nuclear lamina
- C) DNA wrapped around histone octamers
- D) rRNA folding

**Answer: C**

Explanation: Nucleosomes = beads; linker DNA = string. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: nucleosome = bead on string.

Learning Objective mapping: Chapter 3 The Nucleus LO related to nucleosome = bead on string.

Difficulty: High-Difficulty



**\*\*Q59. Which phase is defined by separation of sister chromatids?**

- A) Telophase
- B) Prophase
- C) Metaphase
- D) Anaphase

**Answer: D**

Explanation: Anaphase = sister chromatids separate; metaphase = alignment, prophase = condensation, telophase = decondensation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Anaphase = sister chromatids separate; metaphase = alignment, prophase = condensation, telophase = d

Learning Objective mapping: Chapter 3 The Nucleus LO related to Anaphase = sister chromatids separate; metaphase = alignment, prophase = condens

Difficulty: High-Difficulty

**\*\*Q60. Karyolysis is a feature of:**

- A) Necrosis
- B) Mitosis
- C) S phase
- D) Apoptosis

**Answer: A**

Explanation: Karyolysis fading nucleus is necrosis feature, along with pyknosis, karyorrhexis; apoptosis has karyorrhexis but no inflammation.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Nucleus section for decisive features.

Key point: Karyolysis fading nucleus is necrosis feature, along with pyknosis, karyorrhexis; apoptosis has kary

Learning Objective mapping: Chapter 3 The Nucleus LO related to Karyolysis fading nucleus is necrosis feature, along with pyknosis, karyorrhexis

Difficulty: High-Difficulty

# Chapter 4      Epithelial Tissue

## Part A      Recall

**\*\*Q1. Which feature definitively distinguishes epithelium from connective tissue?**

- A) Rich vascularity
- B) Avascularity + a basement membrane
- C) Abundant fibers
- D) A large extracellular matrix

**Answer: B**

Explanation: Epithelium is avascular and rests on a basement membrane. Connective tissue is vascular with abundant ECM (A, C, D).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: avascular + basement membrane = epithelium.

Learning Objective mapping: Chapter 4      Epithelial Tissue      LO related to avascular + basement membrane = epithelium.

Difficulty: Recall

**\*\*Q2. Hemidesmosomes anchor cells to the basement membrane via:**

- A) Claudins
- B) Cadherins
- C) Integrins
- D) Connexins

**Answer: C**

Explanation: Integrin  $\alpha 6 \beta 4$  binds laminin. Cadherins = desmosomes/adherens; connexins = gap; claudins = tight.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: hemidesmosome = integrin      laminin.

Learning Objective mapping: Chapter 4      Epithelial Tissue      LO related to hemidesmosome = integrin      laminin.

Difficulty: Recall

**\*\*Q3. Barrett esophagus is an example of:**

- A) Hyperplasia
- B) Anaplasia
- C) Dysplasia
- D) Metaplasia

**Answer: D**

**Explanation:** One mature epithelium (squamous) replaced by another (intestinal columnar).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Barrett = metaplasia.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to Barrett = metaplasia.

Difficulty: Recall

**\*\*Q4. The primary cilium functions mainly as a:**

- A) Locomotor organelle
- B) Sensory/signaling organelle
- C) Secretory duct
- D) Site of ATP production

**Answer: B**

**Explanation:** The single 9+0 primary cilium is a signaling antenna (e.g., Hedgehog).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: primary cilium = 9+0, sensory.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to primary cilium = 9+0, sensory.

Difficulty: Recall

**\*\*Q5. Goblet cells are unicellular exocrine glands that secrete:**

- A) Digestive enzymes
- B) Mucin
- C) Hydrochloric acid
- D) Intrinsic factor

**Answer: B**

**Explanation:** Goblet cells synthesize and secrete mucin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: goblet = mucin.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to goblet = mucin.

Difficulty: Recall

**\*\*Q6. A gland with a single unbranched duct is:**

- A) Acinar
- B) Compound

- C) Simple
- D) Endocrine

**Answer: C**

Explanation: Simple = one unbranched duct; compound = branched.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: simple vs compound = duct.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to simple vs compound = duct.

Difficulty: Recall

**\*\*Q7. Which stain best demonstrates the basement membrane?**

- A) Silver for reticular only
- B) H&E
- C) PAS
- D) Sudan black

**Answer: C**

Explanation: PAS stains the glycoprotein-rich basement membrane magenta.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: PAS basement membrane.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to PAS basement membrane.

Difficulty: Recall

**\*\*Q8. Alport syndrome results from a mutation in:**

- A) Elastin
- B) Fibrillin
- C) Type I collagen
- D) Type IV collagen ( 5 chain)

**Answer: D**

Explanation: Type IV 5 hereditary nephritis + deafness.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Alport = type IV 5.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Alport = type IV 5.

Difficulty: Recall

**\*\*Q9. The reticular lamina beneath the basal lamina is composed of:**

- A) Type II collagen
- B) Elastin
- C) Keratin
- D) Type III collagen

**Answer: D**

Explanation: Type III (reticular) collagen, made by fibroblasts.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: reticular lamina = type III.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to reticular lamina = type III.

Difficulty: Recall

**\*\*Q10. A blister splitting the epidermis at the desmosome level is most characteristic of:**

- A) Dermatitis herpetiformis
- B) Erythema multiforme
- C) Bullous pemphigoid
- D) Pemphigus vulgaris

**Answer: D**

Explanation: Anti-desmoglein   intraepidermal acantholysis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: pemphigus = intraepidermal.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to pemphigus = intraepidermal.

Difficulty: Recall

**\*\*Q11. A biopsy of the distal esophagus shows columnar epithelium with goblet cells. The diagnosis is:**

- A) Squamous cell carcinoma
- B) Barrett esophagus (metaplasia)
- C) Achalasia
- D) Eosinophilic esophagitis

**Answer: B**

Explanation: Barrett esophagus = metaplasia stratified squamous   columnar with goblet cells due to chronic GERD, premalignant.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Barrett esophagus = metaplasia stratified squamous columnar with goblet cells due to chronic GERD,

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Barrett esophagus = metaplasia stratified squamous columnar with goblet cells

Difficulty: Recall

**\*\*Q12. Which molecule is the autoantigen in Goodpasture syndrome?**

- A) Laminin
- B) Type IV collagen
- C) Desmoglein
- D) Integrin 6 4

**Answer: B**

Explanation: Goodpasture anti-type IV collagen 3 chain in basement membrane lung hemorrhage + glomerulonephritis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Goodpasture anti-type IV collagen 3 chain in basement membrane lung hemorrhage + glomerulonephrit

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Goodpasture anti-type IV collagen 3 chain in basement membrane lung hemorrhag

Difficulty: Recall

## **Part B Understanding**

**\*\*Q13. The epithelium of the urinary bladder is:**

- A) Transitional (urothelium)
- B) Simple columnar
- C) Pseudostratified
- D) Stratified squamous

**Answer: A**

Explanation: Urothelium with umbrella cells permits distension without leaking. Squamous resists abrasion, not stretch.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: bladder = transitional.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to bladder = transitional.

Difficulty: Understanding

**\*\*Q14. Pemphigus vulgaris involves autoantibodies against:**

- A) Tight junctions
- B) Hemidesmosomes

- C) Desmoglein (desmosomes)
- D) Type IV collagen

**Answer: C**

Explanation: Anti-desmoglein    desmosome loss    intraepidermal acantholysis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: pemphigus = desmoglein.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to pemphigus = desmoglein.

Difficulty: Understanding

**\*\*Q15. Stereocilia of the epididymis are:**

- A) Flagella
- B) Motile cilia
- C) Long, immotile microvilli (actin)
- D) Keratin projections

**Answer: C**

Explanation: Stereocilia are elongated actin microvilli    immotile.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: stereocilia = long microvilli.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to stereocilia = long microvilli.

Difficulty: Understanding

**\*\*Q16. The junctional order from apical to basal is:**

- A) Tight    adherens    desmosome
- B) Adherens    desmosome    tight
- C) Gap    tight    desmosome
- D) Desmosome    tight    adherens

**Answer: A**

Explanation: Tight (most apical)    adherens    desmosome(s)    gap/hemidesmosomes.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: tight    adherens    desmosome.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to tight    adherens desmosome.

Difficulty: Understanding

**\*\*Q17. Which epithelium lines large gland ducts near the surface?**

- A) Stratified cuboidal/columnar
- B) Transitional only
- C) Pseudostratified only
- D) Simple squamous throughout

**Answer: A**

Explanation: Ducts strengthen (stratify) as they approach the surface.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: ducts stratify near the surface.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to ducts stratify near the surface.

Difficulty: Understanding

**\*\*Q18. Epithelial mesenchymal transition (EMT) is characterized by:**

- A) Losing basement-membrane attachment only
- B) Losing nuclear material
- C) Gaining more E-cadherin
- D) Losing junctions/polarity and gaining motility

**Answer: D**

Explanation: EMT = loss of E-cadherin/polarity + gain of mesenchymal motility.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: EMT = adhesion loss + motility.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to EMT = adhesion loss + motility.

Difficulty: Understanding

**\*\*Q19. The junction permitting ion/small-molecule passage between cells is the:**

- A) Desmosome
- B) Hemidesmosome
- C) Tight junction
- D) Gap junction

**Answer: D**

Explanation: Gap junctions (connexins) allow communication; tight junctions block it.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.



Key point: gap = connexin = communicate.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to gap = connexin = communicate.

Difficulty: Understanding

**\*\*Q20. The functional loss in bronchial squamous metaplasia is:**

- A) Mucociliary clearance
- B) Absorption
- C) Distensibility
- D) Acid secretion

**Answer: A**

Explanation: Replacing ciliated/mucous cells removes the mucociliary escalator.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: squamous metaplasia sacrifices clearance.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to squamous metaplasia sacrifices clearance.

Difficulty: Understanding

**\*\*Q21. Which cells produce basal-lamina components?**

- A) Mast cells
- B) Fibroblasts alone
- C) Epithelial cells and connective-tissue cells jointly
- D) Adipocytes

**Answer: C**

Explanation: Epithelium secretes laminin/type IV; fibroblasts contribute reticular lamina.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: basal lamina = joint product.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to basal lamina = joint product.

Difficulty: Understanding

**\*\*Q22. Which is a unicellular exocrine gland?**

- A) Goblet cell
- B) Sweat gland
- C) Salivary acinus
- D) Sebaceous gland

**Answer: A**

Explanation: Goblet cells are single cells secreting mucin.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: goblet = unicellular gland.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to goblet = unicellular gland.

Difficulty: Understanding

**\*\*Q23. Which apical specialization is built from actin and is non-motile?**

- A) Flagella
- B) Axoneme
- C) Cilia
- D) Microvilli

**Answer: D**

Explanation: Microvilli actin core non-motile absorption; cilia 9+2 tubulin motile; stereocilia long branched microvilli.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Microvilli actin core non-motile absorption; cilia 9+2 tubulin motile; stereocilia long branched mic

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Microvilli actin core non-motile absorption; cilia 9+2 tubulin motile; stereocil

Difficulty: Understanding

**\*\*Q24. Chronic smoking converts bronchial ciliated columnar epithelium into:**

- A) Simple squamous epithelium
- B) Intestinal columnar epithelium
- C) Transitional epithelium
- D) Stratified squamous epithelium

**Answer: D**

Explanation: Chronic smoking squamous metaplasia pseudostratified ciliated stratified squamous, premalignant, goblet loss.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Chronic smoking squamous metaplasia pseudostratified ciliated stratified squamous, premalignant,

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Chronic smoking squamous metaplasia pseudostratified ciliated stratified squ

Difficulty: Understanding

## **Part C Application**

**\*\*Q25. The respiratory tract is lined by:**

- A) Transitional
- B) Simple squamous
- C) Pseudostratified ciliated columnar
- D) Stratified squamous keratinized

**Answer: C**

Explanation: Pseudostratified ciliated columnar + goblet cells = mucociliary escalator.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: airway = pseudostratified ciliated.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to airway = pseudostratified ciliated.

Difficulty: Application

**\*\*Q26. The basement membrane contains predominantly:**

- A) Type III collagen
- B) Elastin
- C) Type I collagen
- D) Type IV collagen

**Answer: D**

Explanation: Type IV forms the lamina densa sheet. Type III is the reticular lamina; type I is fibrillar.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: basement membrane = type IV.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to basement membrane = type IV.

Difficulty: Application

**\*\*Q27. Simple squamous epithelium lining blood vessels is termed:**

- A) Mesothelium
- B) Endothelium
- C) Urothelium
- D) Mesenchyme

**Answer: B**

Explanation: Endothelium = vessels; mesothelium = cavities.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: endothelium = vessels.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to endothelium = vessels.

Difficulty: Application

**\*\*Q28. The basal lamina contains type IV collagen, laminin, and:**

- A) Perlecan (heparan sulfate proteoglycan)
- B) Type I collagen
- C) Fibrillin
- D) Elastin

**Answer: A**

Explanation: Perlecan provides the anionic charge for filtration.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: basal lamina = type IV + laminin + perlecan.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to basal lamina = type IV + laminin + perlecan.

Difficulty: Application

**\*\*Q29. Cell polarity refers to the difference between:**

- A) Left and right borders
- B) Apical and basolateral domains
- C) Anterior and posterior poles
- D) Nuclear and cytoplasmic domains

**Answer: B**

Explanation: Apical vs lateral vs basal domains with distinct proteins.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: polarity = apical vs basolateral.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to polarity = apical vs basolateral.

Difficulty: Application

**\*\*Q30. Epithelial regeneration depends on:**

- A) Multinucleation
- B) Its own vasculature
- C) A proliferative basal/stem-cell compartment
- D) Absence of a basement membrane

**Answer: C**

Explanation: Renewal proceeds from basal/crypt stem cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: basal stem cells drive renewal.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to basal stem cells drive renewal.

Difficulty: Application

**\*\*Q31. Transitional epithelium is characterized by:**

- A) Flat surface cells with keratin
- B) Dome-shaped umbrella cells that flatten with distension
- C) Ciliated columnar surface cells
- D) A single squamous layer

**Answer: B**

Explanation: Umbrella cells + distensibility define urothelium.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: umbrella cells = transitional.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to umbrella cells = transitional.

Difficulty: Application

**\*\*Q32. Which epithelium is optimal for gas exchange?**

- A) Simple columnar
- B) Transitional
- C) Stratified squamous
- D) Simple squamous

**Answer: D**

Explanation: Minimal diffusion distance (one flat cell).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: simple squamous = diffusion surfaces.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to simple squamous = diffusion surfaces.

Difficulty: Application

**\*\*Q33. Stratified squamous keratinized epithelium is found in the:**

- A) Oral cavity
- B) Esophagus
- C) Epidermis of skin
- D) Vagina

**Answer: C**

Explanation: Only the dry epidermis is keratinized.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: keratinized = skin.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to keratinized = skin.

Difficulty: Application

**\*\*Q34. The terminal web of microvilli is composed of:**

- A) Microtubules
- B) Actin and associated proteins at the cell apex
- C) Collagen
- D) Dynein arms

**Answer: B**

Explanation: The terminal web anchors actin bundles of microvilli at the apical cortex.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: terminal web = actin meshwork.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to terminal web = actin meshwork.

Difficulty: Application

**\*\*Q35. Autoantibodies against desmoglein produce:**

- A) Intraepidermal acantholysis
- B) Basement-membrane thickening
- C) Fibrosis
- D) Subepidermal blisters

**Answer: A**

Explanation: Anti-desmoglein desmosome loss intraepidermal acantholysis pemphigus vulgaris; anti-hemidesmosome subepidermal bullous pemphigoid.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Anti-desmoglein desmosome loss intraepidermal acantholysis pemphigus vulgaris; anti-hemidesmosome subepidermal bullous pemphigoid.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Anti-desmoglein desmosome loss intraepidermal acantholysis pemphigus vulgaris

Difficulty: Application

**\*\*Q36. Which statement about pseudostratified epithelium is correct?**

- A) All cells touch the basal lamina
- B) It lacks goblet cells
- C) It lines the bladder
- D) It is a true stratified epithelium

**Answer: A**

Explanation: Pseudostratified: all cells touch basal lamina but nuclei at different heights, appears stratified but simple.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Pseudostratified: all cells touch basal lamina but nuclei at different heights, appears stratified b

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Pseudostratified: all cells touch basal lamina but nuclei at different heights,

Difficulty: Application

## **Part D Integration**

**\*\*Q37. Microvilli contain a core of:**

- A) Actin filaments
- B) Intermediate filaments
- C) Dynein arms
- D) Microtubules

**Answer: A**

Explanation: Microvilli are actin-based (villin/fimbrin + terminal web). Cilia contain microtubules.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: microvilli = actin.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to microvilli = actin.

Difficulty: Integration

**\*\*Q38. Sebaceous glands secrete by:**

- A) Merocrine
- B) Holocrine
- C) Apocrine
- D) Endocrine

**Answer: B**

Explanation: The whole cell disintegrates (holocrine).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: sebaceous = holocrine.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to sebaceous = holocrine.

Difficulty: Integration

**\*\*Q39. Simple squamous lining the pleura/peritoneum/pericardium is:**

- A) Mesothelium
- B) Urothelium
- C) Endocardium
- D) Endothelium

**Answer: A**

Explanation: Mesothelium lines serous cavities.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: mesothelium = cavities.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to mesothelium = cavities.

Difficulty: Integration

**\*\*Q40. Squamous metaplasia of the bronchus is an adaptive response to:**

- A) Viral infection only
- B) Cold exposure
- C) Vitamin A excess
- D) Chronic cigarette smoke exposure

**Answer: D**

Explanation: Smoking (and vitamin A deficiency) drive ciliated columnar   squamous.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: smoking   bronchial squamous metaplasia.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to smoking   bronchial squamous metaplasia.

Difficulty: Integration

**\*\*Q41. An exocrine gland releasing product without cell loss is:**

- A) Endocrine
- B) Apocrine
- C) Merocrine
- D) Holocrine

**Answer: C**



**Explanation:** Merocrine = exocytosis, no cell loss.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: merocrine = exocytosis.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to merocrine = exocytosis.

Difficulty: Integration

**\*\*Q42. A compound acinar gland with serous acini is the:**

- A) Thyroid
- B) Sweat gland
- C) Parotid gland
- D) Liver

**Answer: C**

**Explanation:** Parotid = compound acinar, purely serous.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: parotid = serous acini.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to parotid = serous acini.

Difficulty: Integration

**\*\*Q43. Which is TRUE of pseudostratified epithelium?**

- A) Only surface cells touch the basal lamina
- B) All cells touch the basal lamina, but not all reach the surface
- C) It contains no goblet cells
- D) It is always keratinized

**Answer: B**

**Explanation:** Nuclei at different heights; every cell reaches the basal lamina.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: pseudostratified = all cells reach base.

Learning Objective mapping: Chapter 4   Epithelial Tissue   LO related to pseudostratified = all cells reach base.

Difficulty: Integration

**\*\*Q44. Myoepithelial cells are absent in:**

- A) Salivary glands
- B) Mammary glands
- C) The pancreas

D) Sweat glands

**Answer: C**

Explanation: The pancreas lacks myoepithelial cells a key discriminator.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: pancreas = no myoepithelial cells.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to pancreas = no myoepithelial cells.

Difficulty: Integration

**\*\*Q45. The glycoprotein that binds epithelial integrins in the basal lamina is:**

- A) Keratin
- B) Laminin
- C) Fibrillin
- D) Elastin

**Answer: B**

Explanation: Laminin is the integrin ligand.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: integrin laminin.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to integrin laminin.

Difficulty: Integration

**\*\*Q46. Which mechanism underlies bladder urothelial impermeability?**

- A) Tight junctions between umbrella cells + apical plaques
- B) A keratin layer
- C) Goblet-cell mucus
- D) Cilia

**Answer: A**

Explanation: Umbrella-cell tight junctions and thickened apical plaques prevent leakage.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: urothelium barrier = umbrella cells.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to urothelium barrier = umbrella cells.

Difficulty: Integration

**\*\*Q47. The epithelium that can stretch without leaking contains:**

- A) Ciliated cells
- B) Keratinocytes
- C) Umbrella cells
- D) Goblet cells

**Answer: C**

Explanation: Transitional epithelium umbrella cells dome-shaped binucleate stretch without leaking, line bladder ureter.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Transitional epithelium umbrella cells dome-shaped binucleate stretch without leaking, line bladder

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Transitional epithelium umbrella cells dome-shaped binucleate stretch without le

Difficulty: Integration

**\*\*Q48. Hemidesmosomes anchor epithelial cells to the basement membrane by binding:**

- A) Connexin
- B) Type I collagen
- C) Laminin
- D) Elastin

**Answer: C**

Explanation: Hemidesmosome integrin  $\alpha 6 \beta 4$  binds laminin-332 in basal lamina, BP180/230 + plectin + keratin intermediate filament.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Hemidesmosome integrin  $\alpha 6 \beta 4$  binds laminin-332 in basal lamina, BP180/230 + plectin + keratin interme

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Hemidesmosome integrin  $\alpha 6 \beta 4$  binds laminin-332 in basal lamina, BP180/230 + plect

Difficulty: Integration

**Part E High-Difficulty**

**\*\*Q49. The junction forming a belt-like seal is the:**

- A) Desmosome
- B) Tight junction
- C) Gap junction
- D) Hemidesmosome

**Answer: B**

Explanation: Tight junctions (claudin/occludin) seal the paracellular space.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: tight = seal (claudin).

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to tight = seal (claudin).

Difficulty: High-Difficulty

**\*\*Q50. The esophagus is lined by:**

- A) Stratified squamous non-keratinized
- B) Pseudostratified ciliated
- C) Transitional
- D) Simple columnar

**Answer: A**

Explanation: Non-keratinized squamous resists abrasion while staying moist.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: esophagus = non-keratinized squamous.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to esophagus = non-keratinized squamous.

Difficulty: High-Difficulty

**\*\*Q51. Which junction couples the actin cytoskeletons of adjacent cells?**

- A) Zonula adherens
- B) Gap junction
- C) Hemidesmosome
- D) Tight junction

**Answer: A**

Explanation: The zonula adherens uses E-cadherin linked to actin.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: adherens = E-cadherin actin.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to adherens = E-cadherin actin.

Difficulty: High-Difficulty

**\*\*Q52. Barrett esophagus refers to metaplasia into:**

- A) Simple squamous epithelium
- B) Keratinized squamous epithelium
- C) Intestinal-type columnar epithelium with goblet cells

D) Transitional epithelium

**Answer: C**

Explanation: Goblet-cell intestinal metaplasia is diagnostic.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Barrett = goblet cells in the esophagus.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Barrett = goblet cells in the esophagus.

Difficulty: High-Difficulty

**\*\*Q53. Mammary-gland secretion (apical cytoplasm released) is:**

- A) Holocrine
- B) Endocrine
- C) Merocrine
- D) Apocrine

**Answer: D**

Explanation: Apocrine = apical pinch-off with product.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: apocrine = mammary.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to apocrine = mammary.

Difficulty: High-Difficulty

**\*\*Q54. Cells specialized to contract and expel glandular secretion are:**

- A) Parietal cells
- B) Paneth cells
- C) Goblet cells
- D) Myoepithelial cells

**Answer: D**

Explanation: Myoepithelial cells (actin/myosin) wrap acini and ducts.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: myoepithelial = contractile.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to myoepithelial = contractile.

Difficulty: High-Difficulty

**\*\*Q55. Goodpasture syndrome targets which basement-membrane component?**

- A) Type IV collagen
- B) Fibronectin
- C) Perlecan
- D) Laminin

**Answer: A**

Explanation: Anti-type IV collagen    glomerular + alveolar basement-membrane damage.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Goodpasture = anti-type IV.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to Goodpasture = anti-type IV.

Difficulty: High-Difficulty

**\*\*Q56. Which junction is a spot-weld linking intermediate filaments?**

- A) Desmosome
- B) Gap junction
- C) Zonula adherens
- D) Tight junction

**Answer: A**

Explanation: Desmosome = desmoglein/desmocollin    keratin filaments.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: desmosome = intermediate filaments.

Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to desmosome = intermediate filaments.

Difficulty: High-Difficulty

**\*\*Q57. Which apical structure is motile and moves mucus?**

- A) Stereocilia
- B) Terminal web
- C) Microvilli
- D) Cilia

**Answer: D**

Explanation: Cilia (9+2 + dynein) beat to move mucus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: cilia = motile.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to cilia = motile.

Difficulty: High-Difficulty

**\*\*Q58. Stratified squamous epithelial stem cells reside in the:**

- A) Superficial layer
- B) Basal layer
- C) Middle spinous layer
- D) Dermal papillae

**Answer: B**

Explanation: The stratum basale contains the proliferating stem cells. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: basal layer = stem-cell compartment.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to basal layer = stem-cell compartment.

Difficulty: High-Difficulty

**\*\*Q59. A gland whose cells disintegrate entirely to release their product is:**

- A) Endocrine
- B) Merocrine
- C) Apocrine
- D) Holocrine

**Answer: D**

Explanation: Holocrine = whole cell disintegrates sebaceous; merocrine = exocytosis, apocrine = decapitation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

Key point: Holocrine = whole cell disintegrates sebaceous; merocrine = exocytosis, apocrine = decapitation.

Learning Objective mapping: Chapter 4 Epithelial Tissue LO related to Holocrine = whole cell disintegrates sebaceous; merocrine = exocytosis, apocrine

Difficulty: High-Difficulty

**\*\*Q60. A third-degree burn fails to re-epithelialize without grafting because it destroys:**

- A) Only the stratum corneum
- B) The dermal collagen
- C) The epidermal stem-cell compartment (basal layer + appendages)
- D) The subcutaneous fat

**Answer: C**

**Explanation: Third-degree burn destroys epidermis + dermis including basal stem cells and appendages    needs graft; superficial burn heals from appendages.**

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Epithelial Tissue section for decisive features.

**Key point: Third-degree burn destroys epidermis + dermis including basal stem cells and appendages needs graf**

**Learning Objective mapping: Chapter 4    Epithelial Tissue    LO related to Third-degree burn destroys epidermis + dermis including basal stem cells and app**

**Difficulty: High-Difficulty**



# Chapter 5      Connective Tissue

## Part A      Recall

**\*\*Q1. The most abundant protein in the human body is:**

- A) Elastin
- B) Collagen
- C) Keratin
- D) Fibrillin

**Answer: B**

Explanation: Collagen is the most abundant body protein (~25-30% of protein mass).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: collagen = most abundant protein.

Learning Objective mapping: Chapter 5      Connective Tissue      LO related to collagen = most abundant protein.

Difficulty: Recall

**\*\*Q2. The principal resident cell of connective tissue is the:**

- A) Mast cell
- B) Plasma cell
- C) Macrophage
- D) Fibroblast

**Answer: D**

Explanation: The fibroblast is the resident matrix-producing cell.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: resident = fibroblast.

Learning Objective mapping: Chapter 5      Connective Tissue      LO related to resident = fibroblast.

Difficulty: Recall

**\*\*Q3. Osteogenesis imperfecta is due to a defect in:**

- A) Fibrillin
- B) Elastin
- C) Type II collagen
- D) Type I collagen

**Answer: D**

Explanation: Type I collagen mutation      brittle bones.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: OI = type I.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to OI = type I.

Difficulty: Recall

**\*\*Q4. Mast-cell histamine is stored preformed within:**

- A) Cytoplasmic granules
- B) The nucleus
- C) Mitochondria
- D) Lysosomes

**Answer: A**

Explanation: Histamine (with heparin) is stored in secretory granules.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: histamine = granules.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to histamine = granules.

Difficulty: Recall

**\*\*Q5. Hyaluronic acid is unique among GAGs because it:**

- A) Forms collagen fibrils
- B) Is found only in bone
- C) Is sulfated
- D) Is not sulfated and not covalently bound to a core protein

**Answer: D**

Explanation: HA is non-sulfated and free (not protein-bound)    the GAG exception.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: HA = non-sulfated, free.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to HA = non-sulfated, free.

Difficulty: Recall

**\*\*Q6. Granulation tissue is characterized by fibroblasts and:**

- A) New capillary formation (angiogenesis)
- B) Calcified matrix
- C) Adipose tissue
- D) Mature dense collagen

**Answer: A**

Explanation: New vessels + fibroblasts + type III = granulation.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: granulation = new vessels.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to granulation = new vessels.

Difficulty: Recall

**\*\*Q7. Adult CT proper is classified as loose or dense based on:**

- A) Presence of vessels only
- B) The type of GAG present
- C) Cell number alone
- D) The ratio/density of collagen fibers to ground substance

**Answer: D**

Explanation: Fiber-to-ground-substance ratio defines loose vs dense.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: classification = fiber density.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to classification = fiber density.

Difficulty: Recall

**\*\*Q8. The resident tissue macrophage of the liver is the:**

- A) Langerhans cell
- B) Microglia
- C) Kupffer cell
- D) Osteoclast

**Answer: C**

Explanation: Kupffer cell = liver macrophage.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Kupffer = liver.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Kupffer = liver.

Difficulty: Recall

**\*\*Q9. Which is a glycoprotein of the ECM?**

- A) Hyaluronic acid
- B) Fibronectin
- C) Chondroitin sulfate

D) Elastin

**Answer: B**

Explanation: Fibronectin (protein + short sugars) = glycoprotein; HA/chondroitin = GAGs; elastin = fiber protein.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: fibronectin = glycoprotein.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to fibronectin = glycoprotein.

Difficulty: Recall

**\*\*Q10. Desmosine cross-links are characteristic of:**

- A) Fibrillin
- B) Collagen
- C) Elastin
- D) Keratin

**Answer: C**

Explanation: Desmosine/isodesmosine cross-links give elastin its recoil.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: desmosine = elastin.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to desmosine = elastin.

Difficulty: Recall

**\*\*Q11. A patient has aortic root dilation, lens dislocation, and tall stature. The defective molecule is:**

- A) Fibrillin-1
- B) Elastin
- C) Type IV collagen
- D) Type I collagen

**Answer: A**

Explanation: Marfan fibrillin-1 defect    elastic fiber scaffold weak    aortic root dilation, lens dislocation, tall arachnodactyly.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Marfan fibrillin-1 defect    elastic fiber scaffold weak    aortic root dilation, lens dislocation, tall stature

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Marfan fibrillin-1 defect    elastic fiber scaffold weak    aortic root dilation,

Difficulty: Recall

**\*\*Q12. The cell with an eccentric clock-face nucleus and perinuclear hof is:**

- A) Mast cell
- B) Plasma cell
- C) Fibroblast
- D) Macrophage

**Answer: B**

Explanation: Plasma cell eccentric clock-face nucleus basophilic cytoplasm perinuclear hof Golgi Russell bodies antibody.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Plasma cell eccentric clock-face nucleus basophilic cytoplasm perinuclear hof Golgi Russell bodies a

Learning Objective mapping: Chapter 5 Connective Tissue LO related to Plasma cell eccentric clock-face nucleus basophilic cytoplasm perinuclear hof Go

Difficulty: Recall

## Part B Understanding

**\*\*Q13. Type II collagen is characteristic of:**

- A) Basement membrane
- B) Tendon
- C) Bone
- D) Cartilage

**Answer: D**

Explanation: Type II = cartilage (and vitreous). Bone/tendon = type I; basement membrane = type IV.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: II = cartilage.

Learning Objective mapping: Chapter 5 Connective Tissue LO related to II = cartilage.

Difficulty: Understanding

**\*\*Q14. Mast cells mediate which hypersensitivity?**

- A) Type II
- B) Type I (IgE-mediated)
- C) Type III
- D) Type IV

**Answer: B**

Explanation: Mast cells release histamine upon IgE cross-linking type I.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: mast cell = type I.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to mast cell = type I.

Difficulty: Understanding

**\*\*Q15. Early granulation tissue contains predominantly:**

- A) Type I collagen
- B) Type III collagen
- C) Type II collagen
- D) Type IV collagen

**Answer: B**

Explanation: Early granulation = type III; remodeling replaces it with type I.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: granulation = type III.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to granulation = type III.

Difficulty: Understanding

**\*\*Q16. Tissue macrophages are derived from circulating:**

- A) Monocytes
- B) Neutrophils
- C) Eosinophils
- D) Lymphocytes

**Answer: A**

Explanation: Monocytes    tissue macrophages.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: macrophage    monocyte.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to macrophage    monocyte.

Difficulty: Understanding

**\*\*Q17. Elastic fibers consist of an elastin core surrounded by:**

- A) Reticulin
- B) Type I collagen
- C) Fibrillin microfibrils
- D) Laminin

**Answer: C**

**Explanation:** Fibrillin microfibrils scaffold the elastin core.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: elastic = elastin + fibrillin.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to elastic = elastin + fibrillin.

Difficulty: Understanding

**\*\*Q18. During remodeling, type III collagen is replaced by stronger:**

- A) Type IV collagen
- B) Elastin
- C) Type I collagen
- D) Type II collagen

**Answer: C**

**Explanation:** Type I replaces type III as the scar matures.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: III    I with maturity.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to III    I with maturity.

Difficulty: Understanding

**\*\*Q19. Fibronectin functions primarily to:**

- A) Store lipid
- B) Transmit nerve impulses
- C) Provide tensile strength
- D) Mediate cell adhesion to the extracellular matrix

**Answer: D**

**Explanation:** Fibronectin binds integrins    adhesion/migration.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: fibronectin = adhesion.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to fibronectin = adhesion.

Difficulty: Understanding

**\*\*Q20. Which stain best demonstrates elastic fibers?**

- A) Orcein or Verhoeff Van Gieson
- B) Silver
- C) PAS

D) H&E

**Answer: A**

Explanation: Orcein/Verhoeff are elastic-fiber stains.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: elastic = Orcein.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to elastic = Orcein.

Difficulty: Understanding

**\*\*Q21. The perinuclear hof of a plasma cell corresponds to:**

A) The Golgi apparatus

B) A lysosome

C) Lipid droplet

D) The nucleus

**Answer: A**

Explanation: The pale zone is the Golgi (no ribosomes    unstained).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: hof = Golgi.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to hof = Golgi.

Difficulty: Understanding

**\*\*Q22. Which statement about reticular fibers is TRUE?**

A) They are type I collagen

B) They form the stroma of lymphoid and hematopoietic organs

C) They stain pink on H&E

D) They provide elastic recoil

**Answer: B**

Explanation: Reticular (type III) fibers form the delicate stroma of lymphoid organs/bone marrow.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: reticular = lymphoid stroma.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to reticular = lymphoid stroma.

Difficulty: Understanding

**\*\*Q23. Which collagen type forms the basement membrane sheet?**



- A) III
- B) IV
- C) I
- D) II

**Answer: B**

Explanation: Type IV collagen sheet-forming network via NC1 domains forms basal lamina, filtration, Goodpasture, Alport.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Type IV collagen sheet-forming network via NC1 domains forms basal lamina, filtration, Goodpasture,

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Type IV collagen sheet-forming network via NC1 domains forms basal lamina, filtr

Difficulty: Understanding

**\*\*Q24. A scar that extends beyond the original wound margins is a:**

- A) Keloid
- B) Granulation tissue
- C) Contracture
- D) Hypertrophic scar

**Answer: A**

Explanation: Keloid extends beyond original wound margins and does not regress; hypertrophic scar stays within margins.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Keloid beyond margins.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Keloid beyond margins.

Difficulty: Understanding

## **Part C    Application**

**\*\*Q25. Reticular fibers are composed of:**

- A) Type III collagen
- B) Elastin
- C) Type IV collagen
- D) Type I collagen

**Answer: A**

Explanation: Reticular = type III collagen (argyrophilic).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: reticular = type III.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to reticular = type III.

Difficulty: Application

**\*\*Q26. Plasma cells show microscopically:**

- A) A multilobed nucleus
- B) An eccentric clock-face nucleus + perinuclear hof
- C) A spindle shape
- D) Granular eosinophilic cytoplasm only

**Answer: B**

Explanation: Clock-face chromatin + pale Golgi hof = plasma cell.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: plasma cell = clock-face + hof.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to plasma cell = clock-face + hof.

Difficulty: Application

**\*\*Q27. Type II collagen is the predominant type in:**

- A) Basement membrane
- B) Tendon
- C) Skin
- D) Hyaline cartilage

**Answer: D**

Explanation: Hyaline cartilage = type II.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: cartilage = type II.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to cartilage = type II.

Difficulty: Application

**\*\*Q28. Dense regular CT is characteristic of:**

- A) Bone marrow stroma
- B) Dermis
- C) Tendons and ligaments
- D) Lamina propria

**Answer: C**

Explanation: Parallel collagen = tendon/ligament.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: regular = tendon.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to regular = tendon.

Difficulty: Application

**\*\*Q29. Plasma cells are recognized by an eccentric clock-face nucleus and:**

- A) Multiple nuclei
- B) Cytoplasmic lipid droplets
- C) Absence of a nucleus
- D) Abundant RER with a perinuclear pale Golgi zone

**Answer: D**

Explanation: The pale hof is the Golgi; the blue cytoplasm is RER.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: plasma cell = RER + Golgi hof.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to plasma cell = RER + Golgi hof.

Difficulty: Application

**\*\*Q30. Muroid (mucous) CT rich in hyaluronic acid is characteristic of:**

- A) Cartilage
- B) Adult dermis
- C) The umbilical cord (Wharton's jelly)
- D) Tendon

**Answer: C**

Explanation: Wharton's jelly = muroid CT.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Wharton = umbilical cord.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Wharton = umbilical cord.

Difficulty: Application

**\*\*Q31. Marfan syndrome results from a mutation in the gene encoding:**

- A) Type I collagen
- B) Fibrillin-1
- C) Elastin

D) Laminin

**Answer: B**

Explanation: FBN1 gene    fibrillin-1.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Marfan = FBN1.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Marfan = FBN1.

Difficulty: Application

**\*\*Q32. The negative charge of GAGs is responsible for:**

A) Hydration and compression resistance

B) Elastic recoil

C) Basement-membrane anchoring

D) Tensile strength

**Answer: A**

Explanation: Polyanionic GAGs trap water    compression resistance.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: GAG charge = water = cushion.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to GAG charge = water = cushion.

Difficulty: Application

**\*\*Q33. Which collagen type is found in the vitreous body and hyaline cartilage?**

A) Type IV

B) Type I

C) Type II

D) Type III

**Answer: C**

Explanation: Type II = cartilage + vitreous.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: II = cartilage + vitreous.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to II = cartilage + vitreous.

Difficulty: Application

**\*\*Q34. Blue sclerae in osteogenesis imperfecta result from:**

- A) Thin sclera revealing the underlying choroid
- B) Iron deposition
- C) Elastin loss
- D) Excess collagen

**Answer: A**

Explanation: Defective type I collagen    thin sclera    choroid shows through (blue).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: OI = blue sclera.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to OI = blue sclera.

Difficulty: Application

**\*\*Q35. Early granulation tissue is rich in:**

- A) Type III collagen and new capillaries
- B) Elastin
- C) Type IV collagen only
- D) Type I collagen

**Answer: A**

Explanation: Early granulation tissue rich in type III collagen reticular and new capillaries, later remodeling to type I.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Early granulation tissue rich in type III collagen reticular and new capillaries, later remodeling t

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Early granulation tissue rich in type III collagen reticular and new capillaries

Difficulty: Application

**\*\*Q36. Tendon is an example of:**

- A) Reticular CT
- B) Dense irregular CT
- C) Dense regular CT
- D) Loose CT

**Answer: C**

Explanation: Tendon dense regular parallel type I collagen tensile one direction; dense irregular interwoven many directions.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Tendon dense regular parallel type I collagen tensile one direction; dense irregular interwoven many

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Tendon dense regular parallel type I collagen tensile one direction; dense irreg

Difficulty: Application

## Part D    Integration

**\*\*Q37. Marfan syndrome is due to a defect in:**

- A) Elastin only
- B) Type IV collagen
- C) Type I collagen
- D) Fibrillin-1

**Answer: D**

Explanation: Fibrillin-1 mutation    elastic-fiber scaffold defect.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Marfan = fibrillin.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Marfan = fibrillin.

Difficulty: Integration

**\*\*Q38. Tendons are composed of:**

- A) Loose CT
- B) Dense regular CT
- C) Dense irregular CT
- D) Elastic CT

**Answer: B**

Explanation: Parallel collagen bundles resist unidirectional tension.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: tendon = dense regular.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to tendon = dense regular.

Difficulty: Integration

**\*\*Q39. Which cell degrades and remodels collagen via matrix metalloproteinases?**

- A) Adipocyte
- B) Fibroblast
- C) Mast cell
- D) Eosinophil

**Answer: B**

**Explanation:** Fibroblasts both synthesize and degrade collagen (MMPs) during remodeling.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: fibroblast = build + remodel.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to fibroblast = build + remodel.

Difficulty: Integration

**\*\*Q40. Dense irregular CT is characteristic of:**

- A) Dermis of skin
- B) Nerve
- C) Cartilage
- D) Tendon

**Answer: A**

**Explanation:** Interwoven collagen = dermis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: irregular = dermis.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to irregular = dermis.

Difficulty: Integration

**\*\*Q41. The primary source of type I and III collagen in wound healing is the:**

- A) Macrophage
- B) Fibroblast
- C) Mast cell
- D) Eosinophil

**Answer: B**

**Explanation:** Fibroblasts synthesize the wound collagen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: fibroblast = wound collagen.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to fibroblast = wound collagen.

Difficulty: Integration

**\*\*Q42. Eosinophils in CT are associated with:**

- A) Fungal infection only
- B) Bacterial infection
- C) Parasitic infection and allergic reactions
- D) Viral infection only

**Answer: C**

Explanation: Eosinophils    parasites and allergy.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: eosinophil = parasite/allergy.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to eosinophil = parasite/allergy.

Difficulty: Integration

**\*\*Q43. A keloid represents:**

- A) Excessive collagen extending beyond the wound margins
- B) Failure of re-epithelialization
- C) Chronic ulceration
- D) Insufficient collagen deposition

**Answer: A**

Explanation: Keloid = excess collagen beyond wound margins.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: keloid = beyond margins.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to keloid = beyond margins.

Difficulty: Integration

**\*\*Q44. Which cell is the mast cell's blood counterpart with similar granules?**

- A) Eosinophil
- B) Monocyte
- C) Neutrophil
- D) Basophil

**Answer: D**

Explanation: Basophils share metachromatic, histamine/heparin granules.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: basophil = blood mast cell.



Learning Objective mapping: Chapter 5    Connective Tissue    LO related to basophil = blood mast cell.

Difficulty: Integration

**\*\*Q45. Hemostasis and neutrophil/macrophage influx describe which wound phase?**

- A) Remodeling
- B) Maturation
- C) Proliferative
- D) Inflammatory

**Answer: D**

Explanation: Inflammatory phase.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: inflammation first.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to inflammation first.

Difficulty: Integration

**\*\*Q46. Which cell produces the collagen of a mature scar?**

- A) Plasma cell
- B) Macrophage
- C) Fibroblast
- D) Mast cell

**Answer: C**

Explanation: Fibroblasts deposit the scar collagen (type I).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: scar = fibroblast collagen.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to scar = fibroblast collagen.

Difficulty: Integration

**\*\*Q47. A diet deficient in fresh fruit produces bleeding gums. The mechanism is:**

- A) Failed elastin cross-linking
- B) Failed collagen hydroxylation (vitamin C)
- C) Excess collagen
- D) Fibrillin loss

**Answer: B**

Explanation: Scurvy vitamin C deficiency    proline/lysine hydroxylation fails    weak collagen    bleeding gums poor wound healing.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Scurvy vitamin C deficiency    proline/lysine hydroxylation fails    weak collagen    bleeding gums poo

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Scurvy vitamin C deficiency    proline/lysine hydroxylation fails    weak collagen

Difficulty: Integration

**\*\*Q48. Hyaluronic acid is unique because it is:**

- A) A glycoprotein
- B) Found only in bone
- C) Sulfated and protein-bound
- D) Non-sulfated and not covalently bound to a core protein

**Answer: D**

Explanation: Hyaluronan non-sulfated GAG not covalently bound to core protein, highly hydrated, compression resistance, CD44 receptor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Hyaluronan non-sulfated GAG not covalently bound to core protein, highly hydrated, compression resis

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Hyaluronan non-sulfated GAG not covalently bound to core protein, highly hydrate

Difficulty: Integration

## **Part E    High-Difficulty**

**\*\*Q49. Scurvy is due to deficiency of:**

- A) Vitamin K
- B) Vitamin D
- C) Vitamin C
- D) Vitamin A

**Answer: C**

Explanation: Vitamin C is the cofactor for prolyl/lysyl hydroxylase.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: scurvy = vitamin C.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to scurvy = vitamin C.

Difficulty: High-Difficulty

**\*\*Q50. The dermis of skin is:**

- A) Elastic CT only
- B) Dense regular CT
- C) Dense irregular CT
- D) Loose CT

**Answer: C**

Explanation: Interwoven collagen resists multidirectional stress.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: dermis = dense irregular.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to dermis = dense irregular.

Difficulty: High-Difficulty

**\*\*Q51. Ehlers-Danlos syndrome (classic) most commonly results from a defect in:**

- A) Fibrillin-1
- B) Laminin
- C) Elastin
- D) Type V collagen

**Answer: D**

Explanation: Classic Ehlers-Danlos syndrome is caused by type V collagen mutations (COL5A1/COL5A2); the vascular form involves type III collagen (COL3A1).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: classic EDS = type V collagen.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to classic EDS = type V collagen.

Difficulty: High-Difficulty

**\*\*Q52. Ground substance is composed mainly of water, glycosaminoglycans, and:**

- A) Keratin
- B) Collagen fibers
- C) Proteoglycans
- D) Elastin only

**Answer: C**

Explanation: Proteoglycans (core protein + GAGs) plus glycoproteins.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: ground substance = GAGs + proteoglycans.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to ground substance = GAGs + proteoglycans.

Difficulty: High-Difficulty

**\*\*Q53. The early phase of wound healing (neutrophils then macrophages) is termed:**

- A) Proliferative
- B) Inflammatory
- C) Remodeling
- D) Maturation

**Answer: B**

Explanation: Inflammatory phase precedes granulation tissue.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: first = inflammation.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to first = inflammation.

Difficulty: High-Difficulty

**\*\*Q54. Scurvy impairs collagen synthesis by preventing:**

- A) Hydroxylation of proline and lysine
- B) Triple-helix formation directly
- C) Collagen gene transcription
- D) Glycosylation of collagen

**Answer: A**

Explanation: Vitamin C is required for prolyl/lysyl hydroxylation.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: scurvy = failed hydroxylation.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to scurvy = failed hydroxylation.

Difficulty: High-Difficulty

**\*\*Q55. Type IV collagen characteristically forms a:**

- A) Cross-banded fiber on H&E
- B) Structure identical to elastin
- C) Rope-like fibril
- D) Sheet-like network in the basement membrane

**Answer: D**

Explanation: Type IV = 2-D sheet (lamina densa).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: IV = sheet.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to IV = sheet.

Difficulty: High-Difficulty

**\*\*Q56. A tendon heals slowly primarily because it is:**

- A) Highly cellular
- B) Poorly vascularized with parallel fibers
- C) Elastic
- D) Reticular

**Answer: B**

Explanation: Dense regular CT has few vessels; injury disrupts the sparse supply.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: tendon = slow healing.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to tendon = slow healing.

Difficulty: High-Difficulty

**\*\*Q57. Which tissue resists multidirectional tension?**

- A) Tendon
- B) Dense irregular CT (dermis)
- C) Ligament
- D) Reticular tissue

**Answer: B**

Explanation: Interwoven fibers resist tension in all planes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: multidirectional = irregular.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to multidirectional = irregular.

Difficulty: High-Difficulty

**\*\*Q58. A biopsy shows metachromatic granules in cells near dermal vessels. These cells are:**

- A) Mast cells
- B) Fibroblasts

- C) Adipocytes
- D) Plasma cells

**Answer: A**

Explanation: Metachromatic granules + perivascular = mast cells. Practice Test

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: metachromatic = mast cell.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to metachromatic = mast cell.

Difficulty: High-Difficulty

**\*\*Q59. Which stain demonstrates reticular fibers?**

- A) Oil Red O
- B) Orcein
- C) Silver impregnation
- D) H&E

**Answer: C**

Explanation: Reticular fibers type III collagen demonstrated by silver impregnation black argyrophilic.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Reticular fibers type III collagen demonstrated by silver impregnation black argyrophilic.

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Reticular fibers type III collagen demonstrated by silver impregnation black arg

Difficulty: High-Difficulty

**\*\*Q60. A child has recurrent fractures and blue sclerae. The defect is in:**

- A) Type II collagen
- B) Elastin
- C) Fibrillin-1
- D) Type I collagen

**Answer: D**

Explanation: Osteogenesis imperfecta type I collagen mutation brittle bones blue sclerae; EDS collagen hypermobile, Marfan fibrillin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Connective Tissue section for decisive features.

Key point: Osteogenesis imperfecta type I collagen mutation brittle bones blue sclerae; EDS collagen hypermobil

Learning Objective mapping: Chapter 5    Connective Tissue    LO related to Osteogenesis imperfecta  
type I collagen mutation brittle bones blue sclerae; EDS

Difficulty: High-Difficulty

# Chapter 6      Adipose Tissue

## Part A      Recall

**\*\*Q1. White adipocytes are characterized by:**

- A) Abundant mitochondria
- B) A central nucleus
- C) Multiple small droplets
- D) A single large droplet (unilocular)

**Answer: D**

Explanation: White fat = one large droplet, peripheral nucleus. Multiple droplets + central nucleus + many mitochondria = brown fat.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: white = unilocular.

Learning Objective mapping: Chapter 6      Adipose Tissue      LO related to white = unilocular.

Difficulty: Recall

**\*\*Q2. The "signet-ring" appearance is characteristic of:**

- A) White adipocytes
- B) Fibroblasts
- C) Macrophages
- D) Brown adipocytes

**Answer: A**

Explanation: One large droplet pushes the nucleus to the rim      signet ring.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: signet ring = white fat.

Learning Objective mapping: Chapter 6      Adipose Tissue      LO related to signet ring = white fat.

Difficulty: Recall

**\*\*Q3. Brown adipose tissue is characterized histologically by:**

- A) A thick fibrous capsule
- B) A single large droplet and few mitochondria
- C) Multiple small droplets and abundant mitochondria
- D) Absence of lipid

**Answer: C**

Explanation: Multilocular + mitochondrial-rich = brown fat.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.



B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: brown = multilocular.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to brown = multilocular.

Difficulty: Recall

**\*\*Q4. Insulin promotes triglyceride storage chiefly by stimulating:**

- A) Lipoprotein lipase
- B) Adenylate cyclase
- C) UCP1
- D) Hormone-sensitive lipase

**Answer: A**

Explanation: Insulin activates lipoprotein lipase triglyceride uptake/storage.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: insulin = LPL = storage.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to insulin = LPL = storage.

Difficulty: Recall

**\*\*Q5. "Beige" (brite) adipocytes resemble:**

- A) Brown adipocytes
- B) Osteoblasts
- C) Fibroblasts
- D) Chondrocytes

**Answer: A**

Explanation: Beige cells are UCP1-positive, brown-like cells induced within white depots by cold/ stimulation.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: beige = brown-like.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to beige = brown-like.

Difficulty: Recall

**\*\*Q6. The signet-ring appearance results from:**

- A) A single large droplet compressing nucleus and cytoplasm to the periphery
- B) A central nucleus with no lipid
- C) Mitochondria around a central droplet
- D) Multiple droplets compressing a round nucleus

**Answer: A**

**Explanation:** One giant droplet    peripheral nucleus/cytoplasm.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: signet ring = one droplet.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to signet ring = one droplet.

Difficulty: Recall

**\*\*Q7. The brown color of brown fat is due to its vascularity and:**

- A) Melanin
- B) Cytochromes in abundant mitochondria
- C) Hemosiderin
- D) Lipofuscin

**Answer: B**

**Explanation:** Mitochondrial cytochromes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: brown = cytochromes.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to brown = cytochromes.

Difficulty: Recall

**\*\*Q8. Congenital leptin deficiency produces:**

- A) Cachexia
- B) Diabetes insipidus
- C) Lipodystrophy
- D) Early-onset severe obesity with hyperphagia

**Answer: D**

**Explanation:** No satiety signal    hyperphagia    severe obesity (leptin-responsive).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: no leptin = hyperphagia.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to no leptin = hyperphagia.

Difficulty: Recall

**\*\*Q9. White adipose tissue's endocrine role includes secretion of:**

- A) Cortisol
- B) Insulin
- C) Leptin and adiponectin
- D) Glucagon

**Answer: C**

Explanation: Leptin + adiponectin (adipokines).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: white fat = endocrine.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to white fat = endocrine.

Difficulty: Recall

**\*\*Q10. Hibernoma is best described as:**

- A) Inflammatory fat necrosis
- B) Malignant brown-fat tumor
- C) Benign brown-fat tumor
- D) Malignant white-fat tumor

**Answer: C**

Explanation: Hibernoma = benign brown-fat tumor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: hibernoma = benign brown.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to hibernoma = benign brown.

Difficulty: Recall

**\*\*Q11. A premature infant becomes hypothermic in a cool room. The missing thermogenic tissue is:**

- A) White adipose tissue
- B) Brown adipose tissue
- C) Skeletal muscle
- D) Liver

**Answer: B**

Explanation: Brown adipose tissue thermogenesis via UCP1, abundant in neonate perirenal supraclavicular, sympathetic 3 mediated.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Brown adipose tissue thermogenesis via UCP1, abundant in neonate perirenal supraclavicular, sympathetic

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to Brown adipose tissue thermogenesis via UCP1, abundant in neonate perirenal supra

Difficulty: Recall

**\*\*Q12. In obesity, adiponectin levels:**

- A) Stay constant
- B) Are unaffected
- C) Rise
- D) Fall

**Answer: D**

Explanation: Adiponectin falls in obesity, insulin-sensitizing; leptin rises satiety signal.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Adiponectin falls in obesity, insulin-sensitizing; leptin rises satiety signal.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to Adiponectin falls in obesity, insulin-sensitizing; leptin rises satiety signal.

Difficulty: Recall

## **Part B Understanding**

**\*\*Q13. Brown fat produces heat via:**

- A) Glycolysis
- B) Shivering
- C) ATP synthase
- D) UCP1 (thermogenin) uncoupling

**Answer: D**

Explanation: UCP1 dissipates the proton gradient as heat. Shivering is muscle thermogenesis, not brown fat.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: brown heat = UCP1.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to brown heat = UCP1.

Difficulty: Understanding

**\*\*Q14. Brown-fat thermogenesis is stimulated by:**

- A) Parasympathetic
- B) Sympathetic ( ) stimulation
- C) Insulin
- D) Glucagon only

**Answer: B**

Explanation: Cold sympathetic norepinephrine lipolysis + UCP1 activation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: brown fat = sympathetic.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to brown fat = sympathetic.

Difficulty: Understanding

**\*\*Q15. Leptin, signaling satiety to the hypothalamus, is secreted primarily by:**

- A) White adipocytes
- B) Chief cells
- C) Hepatocytes
- D) Brown adipocytes

**Answer: A**

Explanation: Leptin is the white-adipocyte satiety hormone.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: leptin white fat.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to leptin white fat.

Difficulty: Understanding

**\*\*Q16. A benign neoplasm of mature white adipocytes is a:**

- A) Hibernoma
- B) Adenoma
- C) Liposarcoma
- D) Lipoma

**Answer: D**

Explanation: Lipoma = benign white-fat tumor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: lipoma = benign.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to lipoma = benign.

Difficulty: Understanding

**\*\*Q17. White adipose tissue is innervated by the:**

- A) Parasympathetic system
- B) Sympathetic system
- C) Somatic motor system
- D) Enteric system only

**Answer: B**

Explanation: Sympathetic fibers regulate lipolysis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: fat = sympathetic innervation.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to fat = sympathetic innervation.

Difficulty: Understanding

**\*\*Q18. Hormones secreted by adipose tissue are termed:**

- A) Chalone
- B) Eicosanoids exclusively
- C) Cytokines only
- D) Adipokines

**Answer: D**

Explanation: Adipokines (leptin, adiponectin, resistin ).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: adipokines = fat hormones.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to adipokines = fat hormones.

Difficulty: Understanding

**\*\*Q19. The most potent activator of brown-fat thermogenesis is:**

- A) Fasting alone
- B) Warm ambient temperature
- C) Cold exposure via sympathetic norepinephrine
- D) High-carbohydrate meals

**Answer: C**

Explanation: Cold    sympathetic norepinephrine    thermogenesis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: cold = BAT activation.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to cold = BAT activation.

Difficulty: Understanding

**\*\*Q20. Lipodystrophy (absence of fat) causes metabolic disease primarily because of:**

- A) Excess adiponectin
- B) Leptin deficiency
- C) Excess UCP1

D) Brown-fat overactivity

**Answer: B**

Explanation: No fat    no leptin    hyperphagia + insulin resistance.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: lipodystrophy = leptin deficiency.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to lipodystrophy = leptin deficiency.

Difficulty: Understanding

**\*\*Q21. A multivacuolated cell with a scalloped nucleus in an adipose tumor is a:**

- A) Lipoblast
- B) Mature adipocyte
- C) Hibernoma cell
- D) Macrophage

**Answer: A**

Explanation: Lipoblast (scalloped nucleus + vacuoles) = liposarcoma.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: lipoblast = malignancy.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to lipoblast = malignancy.

Difficulty: Understanding

**\*\*Q22. Which hormone's receptor is the    -adrenergic receptor on brown adipocytes?**

- A) Cortisol
- B) Insulin
- C) Norepinephrine (sympathetic)
- D) Glucagon

**Answer: C**

Explanation: Sympathetic norepinephrine acts via    on brown adipocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point:    = norepinephrine.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to    = norepinephrine.

Difficulty: Understanding

**\*\*Q23. UCP1 produces heat by:**

- A) Blocking the electron transport chain
- B) Activating ATP synthase
- C) Dissipating the proton gradient as heat
- D) Increasing glycolysis

**Answer: C**

Explanation: UCP1 uncouples oxidative phosphorylation dissipating proton gradient as heat in brown fat mitochondria.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: UCP1 uncouples oxidative phosphorylation dissipating proton gradient as heat in brown fat mitochondria

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to UCP1 uncouples oxidative phosphorylation dissipating proton gradient as heat in

Difficulty: Understanding

**\*\*Q24. The lipid droplet of an adipocyte is bounded by:**

- A) No boundary
- B) A phospholipid bilayer
- C) A phospholipid monolayer
- D) A cell wall

**Answer: C**

Explanation: Lipid droplet bounded by phospholipid monolayer with perilipin, not bilayer.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Lipid droplet bounded by phospholipid monolayer with perilipin, not bilayer.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to Lipid droplet bounded by phospholipid monolayer with perilipin, not bilayer.

Difficulty: Understanding

## **Part C Application**

**\*\*Q25. Brown fat appears brown because of:**

- A) Melanin
- B) Abundant mitochondria (cytochromes) and vasculature
- C) Hemoglobin only
- D) Lipofuscin

**Answer: B**

Explanation: Mitochondrial cytochromes + rich blood supply give the brown color.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.



C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: brown = mitochondria.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to brown = mitochondria.

Difficulty: Application

**\*\*Q26. A benign tumor of brown fat is a:**

- A) Hibernoma
- B) Liposarcoma
- C) Fibroma
- D) Lipoma

**Answer: A**

Explanation: Hibernoma = benign brown-fat tumor.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: hibernoma = brown + benign.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to hibernoma = brown + benign.

Difficulty: Application

**\*\*Q27. The lipid droplet of a white adipocyte is bound by:**

- A) A phospholipid bilayer
- B) A phospholipid monolayer
- C) No membrane at all
- D) A double membrane

**Answer: B**

Explanation: A monolayer (tails toward the hydrophobic lipid core).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: droplet = monolayer.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to droplet = monolayer.

Difficulty: Application

**\*\*Q28. With aging, adipose distribution shifts from subcutaneous toward:**

- A) No change
- B) Brown fat everywhere
- C) Visceral (intra-abdominal) fat
- D) Bone marrow fat only

**Answer: C**

**Explanation:** Aging favors visceral fat accumulation (metabolic risk).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: aging    visceral fat.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to aging    visceral fat.

Difficulty: Application

**\*\*Q29. Adipocytes develop through an intermediate stage called the:**

- A) Myoblast
- B) Preadipocyte
- C) Fibroblast exclusively
- D) Chondroblast

**Answer: B**

**Explanation:** Mesenchymal    preadipocyte    adipocyte.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: preadipocyte = intermediate.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to preadipocyte = intermediate.

Difficulty: Application

**\*\*Q30. In routine processing, adipocyte lipid is:**

- A) Converted to protein
- B) Unaffected
- C) Preserved and stained intensely
- D) Dissolved out, leaving an empty vacuole

**Answer: D**

**Explanation:** Alcohol/xylene extract lipid    empty vacuole.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: lipid extracted on H&E.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to lipid extracted on H&E.

Difficulty: Application

**\*\*Q31. Cancer cachexia is driven in part by tumor-secreted:**

- A) Insulin
- B) Adiponectin exclusively
- C) Leptin

D) Cytokines (TNF- ) promoting lipolysis

**Answer: D**

Explanation: TNF- /IL-6 drive lipolysis and wasting.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: cachexia = TNF- .

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to cachexia = TNF- .

Difficulty: Application

**\*\*Q32. Thiazolidinediones improve insulin sensitivity by activating:**

A) Lipoprotein lipase

B) Leptin receptors

C) UCP1

D) PPAR-

**Answer: D**

Explanation: TZDs are PPAR- agonists (adipogenesis/fat redistribution).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: TZD = PPAR- .

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to TZD = PPAR- .

Difficulty: Application

**\*\*Q33. Which change accompanies obesity and contributes to insulin resistance?**

A) Falling adiponectin + rising TNF-

B) Falling TNF-

C) Rising UCP1 in white fat

D) Rising adiponectin

**Answer: A**

Explanation: Obesity low adiponectin + adipose inflammation (TNF- ) insulin resistance.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: obesity = low adiponectin.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to obesity = low adiponectin.

Difficulty: Application

**\*\*Q34. The primary fuel for brown-fat thermogenesis is:**

- A) Fatty acids from lipolysis
- B) Ketone bodies only
- C) Amino acids
- D) Glucose only

**Answer: A**

Explanation: Lipolysis releases fatty acids that both fuel oxidation and activate UCP1.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: BAT fuel = fatty acids.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to BAT fuel = fatty acids.

Difficulty: Application

**\*\*Q35. A tumor shows multivacuolated cells with scalloped nuclei. This indicates:**

- A) Lipoma
- B) Liposarcoma
- C) Hibernoma
- D) Fibroma

**Answer: B**

Explanation: Liposarcoma lipoblasts multivacuolated scalloped nuclei atypical; hibernoma brown fat multivacuolated central nucleus benign.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Liposarcoma lipoblasts multivacuolated scalloped nuclei atypical; hibernoma brown fat multivacuolate

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to Liposarcoma lipoblasts multivacuolated scalloped nuclei atypical; hibernoma brown

Difficulty: Application

## **Part D    Integration**

**\*\*Q36. UCP1 uncoupling results in:**

- A) Heat instead of ATP
- B) Glucose storage
- C) Less oxygen use
- D) More ATP

**Answer: A**

Explanation: Protons bypass ATP synthase    gradient energy released as heat.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: UCP1 = heat.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to UCP1 = heat.

Difficulty: Integration

**\*\*Q37. Adiponectin generally:**

- A) Causes inflammation
- B) Promotes insulin resistance
- C) Increases insulin sensitivity
- D) Stimulates appetite

**Answer: C**

Explanation: Adiponectin is an insulin sensitizer.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: adiponectin = insulin sensitizer.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to adiponectin = insulin sensitizer.

Difficulty: Integration

**\*\*Q38. The master transcription factor for adipocyte differentiation is:**

- A) GATA1
- B) MyoD
- C) PPAR-
- D) Sox9

**Answer: C**

Explanation: PPAR- drives adipogenesis (TZD target). MyoD = muscle; Sox9 = cartilage.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: PPAR- = fat lineage.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to PPAR- = fat lineage.

Difficulty: Integration

**\*\*Q39. UCP1 generates heat by:**

- A) Breaking down glucose anaerobically
- B) Inhibiting the electron transport chain
- C) Increasing ATP synthase
- D) Dissipating the proton gradient, uncoupling oxidation from ATP synthesis

**Answer: D**

**Explanation:** Proton leak through UCP1 uncouples oxidation from ATP → heat.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: UCP1 = proton leak.

Learning Objective mapping: Chapter 6 → Adipose Tissue → LO related to UCP1 = proton leak.

Difficulty: Integration

**\*\*Q40. Brown adipose tissue in adult humans is most abundant in the:**

- A) Supraclavicular and paravertebral regions
- B) Buttocks
- C) It is entirely absent in adults
- D) Subcutaneous abdominal wall

**Answer: A**

**Explanation:** Adult BAT persists at supraclavicular/paravertebral sites.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: adult BAT = supraclavicular.

Learning Objective mapping: Chapter 6 → Adipose Tissue → LO related to adult BAT = supraclavicular.

Difficulty: Integration

**\*\*Q41. Perirenal and perivascular adipose tissue primarily serves:**

- A) Hormone production only
- B) Mechanical cushioning and structural support
- C) Immune surveillance exclusively
- D) Heat generation exclusively

**Answer: B**

**Explanation:** Structural fat cushions and supports organs/vessels.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: structural fat = cushion.

Learning Objective mapping: Chapter 6 → Adipose Tissue → LO related to structural fat = cushion.

Difficulty: Integration

**\*\*Q42. The best histologic clue distinguishing brown from white fat is:**

- A) They are histologically identical
- B) Brown fat lacks a nucleus
- C) White fat has multiple small droplets
- D) Brown fat has multiple small droplets (foamy), white has one large droplet

**Answer: D**

Explanation: Multilocular (brown) vs unilocular (white).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: droplets discriminate.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to droplets discriminate.

Difficulty: Integration

**\*\*Q43. Which is correctly paired regarding adipose lipolysis regulation?**

- A) Insulin hormone-sensitive lipase
- B) Catecholamines hormone-sensitive lipase
- C) Insulin UCP1
- D) Leptin lipoprotein lipase

**Answer: B**

Explanation: Catecholamines activate hormone-sensitive lipase (lipolysis).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: catechol HSL.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to catechol HSL.

Difficulty: Integration

**\*\*Q44. Beige (brite) adipocytes are induced in white depots by:**

- A) Fasting only
- B) Heat
- C) Cold exposure and  $\alpha$ -adrenergic stimulation
- D) Insulin

**Answer: C**

Explanation: Cold/ white beige (UCP1+) conversion.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: beige cold.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to beige cold.

Difficulty: Integration

**\*\*Q45. Which cell type is the adipocyte's precursor lineage marker?**

- A) PPAR-  $\gamma$ -driven mesenchymal lineage
- B) Desmin-positive

- C) GFAP-positive
- D) Keratin-positive

**Answer: A**

Explanation: Adipocytes derive from PPAR- $\gamma$ -driven mesenchymal precursors.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: fat = mesenchymal + PPAR- $\gamma$ .

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to fat = mesenchymal + PPAR- $\gamma$ .

Difficulty: Integration

**\*\*Q46. Which hormone promotes triglyceride storage via lipoprotein lipase?**

- A) Insulin
- B) Glucagon
- C) Cortisol
- D) Epinephrine

**Answer: A**

Explanation: Insulin promotes storage via lipoprotein lipase; catecholamines activate hormone-sensitive lipase lipolysis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Insulin promotes storage via lipoprotein lipase; catecholamines activate hormone-sensitive lipase li

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to Insulin promotes storage via lipoprotein lipase; catecholamines activate hormone

Difficulty: Integration

**\*\*Q47. Cold-induced thermogenesis in brown fat is mediated by:**

- A) Somatic motor neurons
- B) Insulin
- C) Parasympathetic fibers
- D) Sympathetic stimulation

**Answer: C**

Explanation: Cold thermogenesis sympathetic stimulation UCP1 activation in brown fat.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Cold thermogenesis sympathetic stimulation UCP1 activation in brown fat.



Learning Objective mapping: Chapter 6 Adipose Tissue LO related to Cold thermogenesis sympathetic 3 stimulation UCP1 activation in brown fat.

Difficulty: Integration

## Part E High-Difficulty

### **\*\*Q48. Leptin is secreted by:**

- A) Pancreatic cells
- B) Brown adipocytes
- C) White adipocytes
- D) Hepatocytes

#### **Answer: C**

Explanation: White adipocytes secrete leptin (satiety hormone).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: leptin white fat.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to leptin white fat.

Difficulty: High-Difficulty

### **\*\*Q49. Neonates rely on brown fat for:**

- A) Energy storage
- B) Non-shivering thermogenesis
- C) Insulation only
- D) Cushioning

#### **Answer: B**

Explanation: Brown fat generates heat without shivering.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: neonate = non-shivering thermogenesis.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to neonate = non-shivering thermogenesis.

Difficulty: High-Difficulty

### **\*\*Q50. Lipolysis in white adipocytes is stimulated primarily by:**

- A) Catecholamines (epinephrine)
- B) Leptin
- C) Adiponectin
- D) Insulin

#### **Answer: A**

Explanation: Catecholamines hormone-sensitive lipase lipolysis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: catecholamines = lipolysis.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to catecholamines = lipolysis.

Difficulty: High-Difficulty

**\*\*Q51. Adiponectin's metabolic effect is to:**

- A) Promote fat storage only
- B) Decrease insulin sensitivity
- C) Increase insulin sensitivity
- D) Stimulate appetite

**Answer: C**

Explanation: Adiponectin enhances insulin sensitivity.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: adiponectin = sensitizer.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to adiponectin = sensitizer.

Difficulty: High-Difficulty

**\*\*Q52. Excess glucocorticoid (Cushing) redistributes fat toward the:**

- A) Brown fat depots only
- B) No redistribution occurs
- C) Lower limbs
- D) Trunk, face, and dorsocervical region

**Answer: D**

Explanation: Central lipogenesis + peripheral lipolysis    centripetal obesity.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Cushing = central fat.

Learning Objective mapping: Chapter 6    Adipose Tissue    LO related to Cushing = central fat.

Difficulty: High-Difficulty

**\*\*Q53. Compared with white adipocytes, brown adipocytes contain much higher density of:**

- A) Golgi apparatus
- B) Mitochondria
- C) Peroxisomes
- D) Lysosomes

**Answer: B**

Explanation: Mitochondria (for thermogenesis).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: brown = mitochondria.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to brown = mitochondria.

Difficulty: High-Difficulty

**\*\*Q54. Which statement about UCP1 is TRUE?**

A) It is found in white adipocytes only

B) It increases ATP production

C) It is a proton channel in the inner mitochondrial membrane

D) It is activated by insulin

**Answer: C**

Explanation: UCP1 = inner-membrane proton channel (activated by fatty acids, driven by stimulation).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: UCP1 = proton channel.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to UCP1 = proton channel.

Difficulty: High-Difficulty

**\*\*Q55. The neonate's brown fat is concentrated especially in the:**

A) Palms

B) Subcutaneous abdominal wall

C) Interscapular and perirenal regions

D) Buttocks

**Answer: C**

Explanation: Neonatal BAT: interscapular, perirenal, cervical, axillary.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: neonate BAT = interscapular/perirenal.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to neonate BAT = interscapular/perirenal.

Difficulty: High-Difficulty

**\*\*Q56. The lipid in a routine paraffin section of adipose tissue is absent because:**

- A) It is basophilic
- B) It evaporated in the oven
- C) It was never present
- D) Alcohol and xylene dissolved it

**Answer: D**

Explanation: Solvents extract triglyceride during dehydration/clearing.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: solvents extract lipid.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to solvents extract lipid.

Difficulty: High-Difficulty

**\*\*Q57. A cold-stressed neonate maintains temperature without shivering via:**

- A) White-fat storage
- B) Brown-fat UCP1 uncoupling
- C) Muscle glycolysis
- D) Insulin

**Answer: B**

Explanation: BAT UCP1 uncoupling = non-shivering thermogenesis. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: non-shivering = BAT.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to non-shivering = BAT.

Difficulty: High-Difficulty

**\*\*Q58. A patient has moon facies, a buffalo hump, and thin limbs. The likely diagnosis is:**

- A) Cushing syndrome
- B) Lipodystrophy
- C) Hibernoma
- D) Leptin deficiency

**Answer: A**

Explanation: Cushing cortisol excess central obesity moon facies buffalo hump thin limbs striae; cortisol from fasciculata.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Cushing cortisol excess central obesity moon facies buffalo hump thin limbs striae; cortisol from fa

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to Cushing cortisol excess central obesity moon facies buffalo hump thin limbs stri

Difficulty: High-Difficulty

**\*\*Q59. Leptin's primary function is:**

- A) Lowering insulin sensitivity
- B) Increasing appetite
- C) Signaling satiety to the hypothalamus
- D) Stimulating lipolysis

**Answer: C**

Explanation: Leptin satiety signal to hypothalamus, decreases appetite; adiponectin insulin sensitizer.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Adipose Tissue section for decisive features.

Key point: Leptin satiety signal to hypothalamus, decreases appetite; adiponectin insulin sensitizer.

Learning Objective mapping: Chapter 6 Adipose Tissue LO related to Leptin satiety signal to hypothalamus, decreases appetite; adiponectin insulin s

Difficulty: High-Difficulty

# Chapter 7      Nerve Tissue & the Nervous System

## Part A      Recall

**\*\*Q1. Nissl bodies correspond to:**

- A) Mitochondria
- B) Lysosomes
- C) Golgi apparatus
- D) Rough ER and ribosomes

**Answer: D**

Explanation: Nissl = RER + free ribosomes (basophilic). Golgi/mitochondria/lysosomes are other organelles.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Nissl = RER.

Learning Objective mapping: Chapter 7      Nerve Tissue & the Nervous System      LO related to Nissl = RER.

Difficulty: Recall

**\*\*Q2. Astrocytes contribute to the blood brain barrier via:**

- A) Phagocytosis
- B) CSF production only
- C) Myelin formation
- D) End-feet (perivascular feet) on capillaries

**Answer: D**

Explanation: Astrocyte end-feet induce/maintain endothelial tight junctions.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: astrocyte end-feet = BBB.

Learning Objective mapping: Chapter 7      Nerve Tissue & the Nervous System      LO related to astrocyte end-feet = BBB.

Difficulty: Recall

**\*\*Q3. Bipolar neurons are found in the:**

- A) Sympathetic ganglion
- B) Spinal cord anterior horn
- C) Retina and olfactory epithelium
- D) DRG

**Answer: C**

Explanation: Bipolar neurons = retina, olfactory epithelium, vestibulocochlear ganglia.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: bipolar = retina/olfactory.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to bipolar = retina/olfactory.

Difficulty: Recall

**\*\*Q4. One oligodendrocyte can myelinate segments of:**

- A) Only one axon
- B) Multiple axons
- C) No axons
- D) Exactly two

**Answer: B**

Explanation: One oligodendrocyte myelinates multiple axons (many processes).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: oligo = many segments.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to oligo = many segments.

Difficulty: Recall

**\*\*Q5. Bipolar neurons are characteristically found in the:**

- A) Retina, olfactory epithelium, and vestibulocochlear ganglia
- B) Cerebral cortex only
- C) Autonomic ganglia
- D) Anterior horn

**Answer: A**

Explanation: Retina/olfactory/vestibulocochlear.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: bipolar = special senses.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to bipolar = special senses.

Difficulty: Recall

**\*\*Q6. Astrocyte foot processes contribute to the BBB by:**

- A) Producing myelin around capillaries
- B) Phagocytosing pathogens
- C) Forming the tight junctions themselves

D) Ensheathing capillaries and inducing/maintaining endothelial tight junctions

**Answer: D**

Explanation: Astrocytes induce/maintain the endothelial junctions (they do not form them).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: astrocyte = induces BBB.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to astrocyte = induces BBB.

Difficulty: Recall

**\*\*Q7. An electrical synapse is structurally formed by:**

- A) Gap junctions
- B) Tight junctions
- C) Desmosomes
- D) Synaptic vesicles

**Answer: A**

Explanation: Gap junctions (connexons) = electrical synapse.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: electrical = gap junction.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to electrical = gap junction.

Difficulty: Recall

**\*\*Q8. The neurotransmitter at the neuromuscular junction is:**

- A) Dopamine
- B) GABA
- C) Norepinephrine
- D) Acetylcholine

**Answer: D**

Explanation: ACh at the NMJ (a chemical synapse).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: NMJ = ACh.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to NMJ = ACh.

Difficulty: Recall



**\*\*Q9. The pale region where the axon leaves the soma is the:**

- A) Axon hillock
- B) Node of Ranvier
- C) Synaptic bouton
- D) Dendrite

**Answer: A**

Explanation: Axon hillock (no Nissl) = the pale output zone.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: hillock = pale (no Nissl).

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to hillock = pale (no Nissl).

Difficulty: Recall

**\*\*Q10. Which statement about the oligodendrocyte is TRUE?**

- A) It has a neurilemma
- B) It is mesodermal
- C) It myelinates one segment of one axon
- D) It myelinates multiple axons in the CNS

**Answer: D**

Explanation: Oligodendrocyte = multiple CNS axons (no neurilemma).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: oligo = many, CNS.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to oligo = many, CNS.

Difficulty: Recall

**\*\*Q11. A nerve biopsy shows segmental myelin loss with preserved axons. The disease is:**

- A) Multiple sclerosis
- B) Guillain-Barré syndrome
- C) Amyotrophic lateral sclerosis
- D) Poliomyelitis

**Answer: B**

Explanation: Guillain-Barré    PNS segmental demyelination preserved axons; MS CNS demyelination; ALS motor neuron loss.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Guillain-Barré PNS segmental demyelination preserved axons; MS CNS demyelination; ALS motor neuron I

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Guillain-Barré PNS segmental demyelination preserved axons; MS CNS demyelination

Difficulty: Recall

**\*\*Q12. One oligodendrocyte myelinates:**

- A) Multiple axons
- B) No axons
- C) Only unmyelinated axons
- D) One internode of one axon

**Answer: A**

Explanation: Oligodendrocyte myelinates many internodes of multiple axons CNS no neurilemma no regeneration; Schwann one internode one axon PNS neurilemma basal lamina regeneration.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Oligodendrocyte myelinates many internodes of multiple axons CNS no neurilemma no regeneration; Schw

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Oligodendrocyte myelinates many internodes of multiple axons CNS no neurilemma n

Difficulty: Recall

## **Part B Understanding**

**\*\*Q13. Nissl bodies are absent in the:**

- A) Perikaryon
- B) Cell body
- C) Dendrites
- D) Axon and axon hillock

**Answer: D**

Explanation: The axon and hillock lack RER hence their pale staining.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: no Nissl in axon/hillock.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to no Nissl in axon/hillock.

Difficulty: Understanding

**\*\*Q14. A marker for astrocytes is:**

- A) Vimentin only

- B) GFAP
- C) Cytokeratin
- D) Desmin

**Answer: B**

Explanation: GFAP = glial fibrillary acidic protein (astrocytes).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: GFAP = astrocyte.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to GFAP = astrocyte.

Difficulty: Understanding

**\*\*Q15. After axonal injury, dispersion of Nissl substance is called:**

- A) Apoptosis
- B) Gliosis
- C) Chromatolysis
- D) Wallerian degeneration

**Answer: C**

Explanation: Chromatolysis = Nissl dispersal in the soma (regeneration attempt).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: chromatolysis = Nissl dispersal.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to chromatolysis = Nissl dispersal.

Difficulty: Understanding

**\*\*Q16. Nissl bodies visible by light microscopy correspond ultrastructurally to:**

- A) Golgi apparatus
- B) Rough endoplasmic reticulum and free ribosomes
- C) Mitochondria
- D) Lysosomes

**Answer: B**

Explanation: Nissl = RER + ribosomes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Nissl = RER (ultrastructurally).

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Nissl = RER (ultrastructurally).

Difficulty: Understanding

**\*\*Q17. Microglia are embryologically derived from:**

- A) Neuroectoderm
- B) Mesoderm/monocyte lineage
- C) Neural crest
- D) Endoderm

**Answer: B**

Explanation: Microglia = mesoderm (monocyte lineage).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: microglia = mesoderm.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to microglia = mesoderm.

Difficulty: Understanding

**\*\*Q18. Nodes of Ranvier are the sites of:**

- A) Neurotransmitter synthesis
- B) Concentrated voltage-gated sodium channels enabling saltatory conduction
- C) Synapse formation exclusively
- D) Axon-terminal branching

**Answer: B**

Explanation: Na<sup>+</sup> channels concentrate at nodes saltatory conduction.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: node = Na<sup>+</sup> channels.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to node = Na<sup>+</sup> channels.

Difficulty: Understanding

**\*\*Q19. At a chemical synapse, neurotransmitter vesicles are located in the:**

- A) Presynaptic terminal
- B) Synaptic cleft only
- C) Dendrite exclusively
- D) Postsynaptic membrane

**Answer: A**

Explanation: Vesicles are presynaptic; receptors are postsynaptic.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: vesicles = presynaptic.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to vesicles = presynaptic.

Difficulty: Understanding

**\*\*Q20. A single Schwann cell myelinates:**

- A) Multiple axons simultaneously
- B) An entire axon from origin to termination
- C) No axons
- D) One segment (internode) of a single axon

**Answer: D**

Explanation: One Schwann = one internode of one axon.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Schwann = one internode.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to Schwann = one internode.

Difficulty: Understanding

**\*\*Q21. Which cell is the CNS's resident macrophage?**

- A) Ependyma
- B) Astrocyte
- C) Microglia
- D) Oligodendrocyte

**Answer: C**

Explanation: Microglia = CNS phagocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: microglia = macrophage.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to microglia = macrophage.

Difficulty: Understanding

**\*\*Q22. Wallerian degeneration occurs in the:**

- A) Proximal stump
- B) Distal stump
- C) Soma
- D) Nucleus

**Answer: B**

Explanation: Distal stump degenerates (severed from the soma).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Wallerian = distal.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Wallerian = distal.

Difficulty: Understanding

**\*\*Q23. The pale region of the neuron where the axon emerges is the:**

- A) Node of Ranvier
- B) Nissl body
- C) Dendrite
- D) Axon hillock

**Answer: D**

Explanation: Axon hillock pale no Nissl action potential initiation; Nissl = RER basophilic in perikaryon and dendrites.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Axon hillock pale no Nissl action potential initiation; Nissl = RER basophilic in perikaryon and den

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Axon hillock pale no Nissl action potential initiation; Nissl = RER basophilic i

Difficulty: Understanding

**\*\*Q24. After spinal-cord transection, regeneration fails mainly because of:**

- A) Microglial inactivity
- B) Schwann-cell loss
- C) Glial scarring and absence of neurilemma
- D) Excess myelin

**Answer: C**

Explanation: CNS regeneration fails due to glial scarring oligodendrocyte inhibitory factors absence neurilemma; PNS Schwann basal lamina allows regeneration.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: CNS regeneration fails due to glial scarring oligodendrocyte inhibitory factors absence neurilemma;

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to CNS regeneration fails due to glial scarring oligodendrocyte inhibitory factors

Difficulty: Understanding

## Part C Application

**\*\*Q25. Sensory neurons of the dorsal root ganglion are:**

- A) Pseudounipolar
- B) Apolar
- C) Multipolar
- D) Bipolar

**Answer: A**

Explanation: DRG neurons are pseudounipolar (single bifurcating process).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: DRG = pseudounipolar.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to DRG = pseudounipolar.

Difficulty: Application

**\*\*Q26. One Schwann cell myelinates:**

- A) No axon
- B) Many axons
- C) One internode of one axon
- D) The whole nerve

**Answer: C**

Explanation: One Schwann cell = one internode of one axon (with neurilemma).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Schwann = one segment.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Schwann = one segment.

Difficulty: Application

**\*\*Q27. The blood nerve barrier is formed mainly by the:**

- A) Endoneurium
- B) Perineurium
- C) Epineurium
- D) Schwann cell only

**Answer: B**

Explanation: Tight-junctioned perineurial cells form the barrier.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: perineurium = blood nerve barrier.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to perineurium = blood nerve barrier.

Difficulty: Application

**\*\*Q28. After transection of a peripheral axon, the distal segment degenerates this is:**

- A) Apoptosis
- B) Wallerian degeneration
- C) Coagulative necrosis
- D) Liquefactive necrosis

**Answer: B**

Explanation: Wallerian degeneration = distal-stump degeneration.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Wallerian = distal.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Wallerian = distal.

Difficulty: Application

**\*\*Q29. Ependymal cells line the:**

- A) Neuromuscular junction
- B) Peripheral nerve endoneurium
- C) Ventricles and central canal
- D) Dorsal root ganglia

**Answer: C**

Explanation: Ependyma lines ventricles/central canal.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: ependyma = ventricles.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to ependyma = ventricles.

Difficulty: Application

**\*\*Q30. Saltatory conduction is:**

- A) Identical in speed
- B) Impossible in the PNS
- C) Slower than unmyelinated conduction



D) Significantly faster than unmyelinated conduction

**Answer: D**

Explanation: Jumping node-to-node is much faster than continuous conduction.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: saltatory = fast.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to saltatory = fast.

Difficulty: Application

**\*\*Q31. After peripheral-nerve injury, axon regeneration is guided by:**

A) Schwann-cell basal-lamina tubes (bands of Bü ngner)

B) Astrocyte processes

C) Microglial scar

D) Degenerated axon fragments

**Answer: A**

Explanation: Surviving Schwann cells form guiding tubes.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: regeneration = Schwann tubes.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to regeneration = Schwann tubes.

Difficulty: Application

**\*\*Q32. Substantia nigra neurons (degeneration = Parkinson disease) contain the pigment:**

A) Neuromelanin

B) Hemosiderin

C) Bilirubin

D) Lipofuscin only

**Answer: A**

Explanation: Neuromelanin accumulates in substantia nigra neurons.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: nigra = neuromelanin.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to nigra = neuromelanin.

Difficulty: Application

**\*\*Q33. Myelin appears as a clear ring around an axon on H&E because:**

- A) It is basophilic
- B) It is calcified
- C) It is protein-rich
- D) Its lipid is dissolved during processing

**Answer: D**

Explanation: Myelin lipid is extracted by solvents    clear ring.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: myelin clear = lipid extracted.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to myelin clear = lipid extracted.

Difficulty: Application

**\*\*Q34. Chromatolysis after axon injury indicates:**

- A) The soma's regenerative response (Nissl dispersal)
- B) Myelin loss
- C) Microglial activation
- D) Neuron death

**Answer: A**

Explanation: Chromatolysis = Nissl dispersal = regeneration attempt.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: chromatolysis = regeneration attempt.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to chromatolysis = regeneration attempt.

Difficulty: Application

**\*\*Q35. Which glial cell is mesodermal in origin?**

- A) Oligodendrocyte
- B) Microglia
- C) Ependyma
- D) Astrocyte

**Answer: B**

Explanation: Microglia mesodermal monocyte-derived phagocytosis; astrocyte oligodendrocyte ependyma ectodermal.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Microglia mesodermal monocyte-derived phagocytosis; astrocyte oligodendrocyte ependyma ectodermal.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Microglia mesodermal monocyte-derived phagocytosis; astrocyte oligodendrocyte ep

Difficulty: Application

**\*\*Q36. Nodes of Ranvier are essential for:**

- A) Synaptic delay
- B) Wallerian degeneration
- C) Continuous conduction
- D) Saltatory conduction

**Answer: D**

Explanation: Nodes Ranvier Na<sup>+</sup> channels high density regenerate action potential saltatory conduction jumping.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Nodes Ranvier Na<sup>+</sup> channels high density regenerate action potential saltatory conduction jumping.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Nodes Ranvier Na<sup>+</sup> channels high density regenerate action potential saltatory co

Difficulty: Application

## **Part D Integration**

**\*\*Q37. Myelin in the CNS is produced by:**

- A) Astrocytes
- B) Microglia
- C) Schwann cells
- D) Oligodendrocytes

**Answer: D**

Explanation: Oligodendrocytes myelinate CNS axons (many segments each).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: CNS myelin = oligodendrocyte.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to CNS myelin = oligodendrocyte.

Difficulty: Integration

**\*\*Q38. The connective-tissue layer surrounding a nerve fascicle is the:**

- A) Neurilemma
- B) Endoneurium
- C) Perineurium
- D) Epineurium

**Answer: C**

Explanation: Perineurium wraps fascicles (blood nerve barrier).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: perineurium = fascicle.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to perineurium = fascicle.

Difficulty: Integration

**\*\*Q39. CNS regeneration is poor because CNS myelin lacks:**

- A) Axons
- B) Lipid
- C) Neurilemma (Schwann-cell sheath)
- D) Nodes of Ranvier

**Answer: C**

Explanation: No neurilemma + glial scar + myelin inhibitors no CNS regeneration.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: no neurilemma = no CNS regrowth.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to no neurilemma = no CNS regrowth.

Difficulty: Integration

**\*\*Q40. Multipolar neurons are characterized by:**

- A) No axon
- B) Multiple axons
- C) One axon and one dendrite
- D) One axon and multiple dendrites

**Answer: D**

Explanation: Multipolar = 1 axon + many dendrites (motor, Purkinje).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: multipolar = many dendrites.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to multipolar = many dendrites.

Difficulty: Integration

**\*\*Q41. Satellite cells are found in the:**

- A) Central white matter
- B) Spinal cord gray matter
- C) Ganglia
- D) Neuromuscular junction

**Answer: C**

Explanation: Satellite cells surround ganglion cell bodies.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: satellite = ganglia.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to satellite = ganglia.

Difficulty: Integration

**\*\*Q42. Multiple sclerosis primarily targets:**

- A) Schwann cells of the PNS
- B) Oligodendrocytes of the CNS
- C) Satellite cells
- D) Microglia

**Answer: B**

Explanation: MS = CNS (oligodendrocyte) demyelination.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: MS = CNS.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to MS = CNS.

Difficulty: Integration

**\*\*Q43. CNS regeneration after injury is:**

- A) Identical to PNS
- B) Mediated by Schwann cells
- C) Robust and complete
- D) Poor, due to glial scarring and myelin inhibitors

**Answer: D**

Explanation: Glial scar + inhibitory myelin factors    no CNS regrowth.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: CNS = no regeneration.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to CNS = no regeneration.

Difficulty: Integration

**\*\*Q44. Astrocytes contribute to the glial scar after CNS injury via:**

- A) Astrogliosis (reactive astrogliosis)
- B) Wallerian degeneration
- C) Apoptosis
- D) Demyelination

**Answer: A**

Explanation: Reactive astrocyte proliferation = gliosis = CNS scar.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: gliosis = astrocyte scar.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to gliosis = astrocyte scar.

Difficulty: Integration

**\*\*Q45. Which structure provides the blood nerve barrier?**

- A) Myelin
- B) Epineurium
- C) Perineurium
- D) Endoneurium

**Answer: C**

Explanation: Perineurium (tight-junctioned fascicle sheath).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: peri = barrier.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to peri = barrier.

Difficulty: Integration

**\*\*Q46. Which synapse type is faster and bidirectional?**

- A) Electrical
- B) Neuromuscular
- C) Cholinergic
- D) Chemical

**Answer: A**

Explanation: Electrical (gap junctions) = fast, bidirectional.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: electrical = fast + bidirectional.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to electrical = fast + bidirectional.

Difficulty: Integration

**\*\*Q47. The blood nerve barrier is formed by the:**

- A) Perineurium
- B) Endoneurium
- C) Neurilemma
- D) Epineurium

**Answer: A**

Explanation: Blood-nerve barrier perineurium tight junctions; BBB astrocyte foot processes + endothelial tight junctions.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Blood-nerve barrier perineurium tight junctions; BBB astrocyte foot processes + endothelial tight junctions

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Blood-nerve barrier perineurium tight junctions; BBB astrocyte foot processes +

Difficulty: Integration

**\*\*Q48. A GFAP-positive brain tumor is of:**

- A) Astrocytic origin
- B) Oligodendroglial origin
- C) Microglial origin
- D) Neuronal origin

**Answer: A**

Explanation: GFAP glial fibrillary acidic protein astrocyte marker astrocytoma; oligodendroglioma OLIG2.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: GFAP glial fibrillary acidic protein astrocyte marker astrocytoma; oligodendroglioma OLIG2.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to GFAP glial fibrillary acidic protein astrocyte marker astrocytoma; oligodendro

Difficulty: Integration

**Part E High-Difficulty**

**\*\*Q49. The only glial cell derived from mesoderm (monocytes) is:**

- A) Microglia
- B) Ependymal cell
- C) Astrocyte
- D) Oligodendrocyte

**Answer: A**

Explanation: Microglia = mesodermal (monocyte lineage) the CNS's resident macrophages.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: microglia = mesoderm.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to microglia = mesoderm.

Difficulty: High-Difficulty

**\*\*Q50. Ependymal cells are characterized by:**

- A) Satellite function
- B) Myelin production
- C) Cilia and lining of ventricles
- D) Phagocytosis

**Answer: C**

Explanation: Ependyma = ciliated ventricular/central-canal lining.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: ependyma = ciliated lining.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to ependyma = ciliated lining.

Difficulty: High-Difficulty

**\*\*Q51. Satellite cells are found:**

- A) In myelin
- B) In CNS white matter
- C) Surrounding neuron bodies in ganglia
- D) In the ependyma

**Answer: C**

Explanation: Satellite cells surround ganglion cell bodies (PNS).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.



Key point: satellite = ganglia.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to satellite = ganglia.

Difficulty: High-Difficulty

**\*\*Q52. A pseudounipolar neuron's single process:**

- A) Contains no cell body
- B) Bifurcates into peripheral and central branches
- C) Never branches
- D) Lacks an axon

**Answer: B**

Explanation: The process bifurcates (peripheral + central)    DRG sensory neurons.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: pseudounipolar = bifurcates.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to pseudounipolar = bifurcates.

Difficulty: High-Difficulty

**\*\*Q53. The blood brain barrier is formed primarily by tight junctions between:**

- A) Oligodendrocytes
- B) Astrocyte foot processes
- C) Capillary endothelial cells
- D) Neurons

**Answer: C**

Explanation: Endothelial tight junctions form the BBB (induced by astrocyte end-feet).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: BBB = endothelial tight junctions.

Learning Objective mapping: Chapter 7    Nerve Tissue & the Nervous System    LO related to BBB = endothelial tight junctions.

Difficulty: High-Difficulty

**\*\*Q54. Guillain-Barré syndrome primarily affects:**

- A) Peripheral nerve Schwann cells/myelin
- B) Muscle fibers directly
- C) Astrocytes
- D) CNS oligodendrocytes

**Answer: A**

Explanation: GBS = PNS (Schwann) demyelination.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: GBS = PNS.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to GBS = PNS.

Difficulty: High-Difficulty

**\*\*Q55. Autonomic ganglia are the site of synapse between:**

- A) Two sensory neurons
- B) Preganglionic and postganglionic autonomic neurons
- C) Sensory neurons and the spinal cord
- D) Motor neurons and skeletal muscle

**Answer: B**

Explanation: Autonomic ganglia = preganglionic postganglionic synapse.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: autonomic ganglion = pre post synapse.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to autonomic ganglion = pre post synapse.

Difficulty: High-Difficulty

**\*\*Q56. Which cell myelinates a single internode but is present in the PNS?**

- A) Oligodendrocyte
- B) Schwann cell
- C) Astrocyte
- D) Microglia

**Answer: B**

Explanation: Schwann cell = one internode, PNS.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Schwann = one, PNS.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Schwann = one, PNS.

Difficulty: High-Difficulty

**\*\*Q57. A tumor of Schwann cells is a:**

- A) Astrocytoma
- B) Schwannoma
- C) Ependymoma

D) Medulloblastoma

**Answer: B**

Explanation: Schwannoma = Schwann-cell tumor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: schwannoma = Schwann cell.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to schwannoma = Schwann cell.

Difficulty: High-Difficulty

**\*\*Q58. CSF movement within the ventricles is aided by:**

- A) Microglial processes
- B) Oligodendrocyte processes
- C) Astrocyte cilia
- D) Ependymal cell cilia

**Answer: D**

Explanation: Ependymal cilia aid CSF movement. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: ependyma = cilia = CSF flow.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to ependyma = cilia = CSF flow.

Difficulty: High-Difficulty

**\*\*Q59. A sensory neuron whose single process bifurcates is:**

- A) Bipolar
- B) Pseudounipolar
- C) Apolar
- D) Multipolar

**Answer: B**

Explanation: Pseudounipolar sensory neuron single process bifurcates T-shaped, dorsal root ganglion.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Pseudounipolar sensory neuron single process bifurcates T-shaped, dorsal root ganglion.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Pseudounipolar sensory neuron single process bifurcates T-shaped, dorsal root ga

Difficulty: High-Difficulty

**\*\*Q60. Dispersal of Nissl substance after axonal injury is called:**

- A) Pyknosis
- B) Gliosis
- C) Chromatolysis
- D) Wallerian degeneration

**Answer: C**

Explanation: Chromatolysis dispersal Nissl RER after axonal injury, perikaryon swollen, nucleus eccentric.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Nerve Tissue & the Nervous System section for decisive features.

Key point: Chromatolysis dispersal Nissl RER after axonal injury, perikaryon swollen, nucleus eccentric.

Learning Objective mapping: Chapter 7 Nerve Tissue & the Nervous System LO related to Chromatolysis dispersal Nissl RER after axonal injury, perikaryon swollen, nucle

Difficulty: High-Difficulty

# Chapter 8      The Circulatory System

## Part A      Recall

**\*\*Q1. The internal elastic lamina is a feature of the:**

- A) Tunica adventitia
- B) Endothelium only
- C) Tunica intima
- D) Tunica media

**Answer: C**

Explanation: The IEL marks the outer boundary of the tunica intima.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: IEL = intima.

Learning Objective mapping: Chapter 8      The Circulatory System      LO related to IEL = intima.

Difficulty: Recall

**\*\*Q2. Fenestrated capillaries are found in the:**

- A) Renal glomerulus
- B) Skin
- C) Brain
- D) Skeletal muscle

**Answer: A**

Explanation: Glomerular capillaries are fenestrated (filtration).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: glomerulus = fenestrated.

Learning Objective mapping: Chapter 8      The Circulatory System      LO related to glomerulus = fenestrated.

Difficulty: Recall

**\*\*Q3. Atherosclerotic plaques develop primarily in the:**

- A) Tunica adventitia
- B) Vasa vasorum
- C) Tunica media
- D) Tunica intima

**Answer: D**

Explanation: Atherosclerosis begins in the intima (endothelial injury      foam cells).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: athero = intima.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to athero = intima.

Difficulty: Recall

**\*\*Q4. The IEL is most prominent in which vessel type?**

- A) Lymphatics
- B) Capillaries
- C) Muscular (medium) arteries
- D) Veins

**Answer: C**

Explanation: Muscular arteries show the prominent IEL (and EEL).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: IEL prominent = muscular.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to IEL prominent = muscular.

Difficulty: Recall

**\*\*Q5. Arterioles are characterized by:**

- A) A thick adventitia and thin media
- B) A relatively thick smooth-muscle media compared with lumen diameter
- C) Absence of smooth muscle
- D) A fenestrated endothelium

**Answer: B**

Explanation: Arterioles have 1-2 smooth-muscle layers thick relative to their tiny lumen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: arteriole = muscular wall, small lumen.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to arteriole = muscular wall, small lumen.

Difficulty: Recall

**\*\*Q6. The vasa vasorum are located primarily within the:**

- A) Adventitia (and outer media)
- B) Lumen
- C) Endothelium
- D) Intima

**Answer: A**

**Explanation:** Adventitia + outer media.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: vasa = adventitia.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to vasa = adventitia.

Difficulty: Recall

**\*\*Q7. Intercalated-disc gap junctions allow:**

A) Electrical coupling and synchronized contraction

B) Nutrient exchange only

C) Hormone secretion

D) Mechanical strength only

**Answer: A**

**Explanation:** Gap junctions = electrical coupling    synchronized contraction.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: gap = electrical sync.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to gap = electrical sync.

Difficulty: Recall

**\*\*Q8. Portal systems (two capillary beds in series) are found in the hepatic circulation and the:**

A) Cerebral circulation

B) Coronary circulation

C) Hypothalamo-hypophyseal circulation

D) Pulmonary circulation only

**Answer: C**

**Explanation:** Hepatic portal + hypothalamo-hypophyseal portal systems.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: portal = hepatic + hypophyseal.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to portal = hepatic + hypophyseal.

Difficulty: Recall

**\*\*Q9. The thoracic duct differs from lymphatic capillaries by having:**

A) A complete endothelial absence

- B) No valves
- C) Smooth muscle in its wall
- D) RBCs in its lumen

**Answer: C**

Explanation: The thoracic duct has smooth muscle (larger lymphatic duct).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: thoracic duct = muscular.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to thoracic duct = muscular.

Difficulty: Recall

**\*\*Q10. Which is correctly paired regarding capillary types and their locations?**

- A) Continuous capillary glomerulus
- B) Fenestrated capillary kidney glomerulus
- C) Sinusoidal brain
- D) Continuous liver

**Answer: B**

Explanation: Glomerulus = fenestrated. Brain = continuous; liver = sinusoidal.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: glomerulus = fenestrated.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to glomerulus = fenestrated.

Difficulty: Recall

**\*\*Q11. A vessel shows many wavy elastic lamellae in its media. It is a(n):**

- A) Vein
- B) Muscular artery
- C) Elastic artery
- D) Arteriole

**Answer: C**

Explanation: Elastic artery aorta many wavy elastic lamellae media Windkessel recoil; muscular artery many smooth muscle layers.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Elastic artery aorta many wavy elastic lamellae media Windkessel recoil; muscular artery many smooth



Learning Objective mapping: Chapter 8 The Circulatory System LO related to Elastic artery aorta many wavy elastic lamellae media Windkessel recoil; muscula

Difficulty: Recall

**\*\*Q12. Atherosclerosis begins in the:**

- A) Intima
- B) Adventitia
- C) Vasa vasorum
- D) Media

**Answer: A**

Explanation: Atherosclerosis begins in intima subendothelial LDL accumulation foam cells.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Atherosclerosis begins in intima subendothelial LDL accumulation foam cells.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Atherosclerosis begins in intima subendothelial LDL accumulation foam cells.

Difficulty: Recall

## **Part B Understanding**

**\*\*Q13. Vasa vasorum are located in the:**

- A) Media only
- B) Adventitia
- C) Lumen
- D) Intima

**Answer: B**

Explanation: Vasa vasorum supply the adventitia (and outer media).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: vasa = adventitia.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to vasa = adventitia.

Difficulty: Understanding

**\*\*Q14. Sinusoidal capillaries are found in the:**

- A) Brain
- B) Muscle
- C) Lung
- D) Liver and spleen

**Answer: D**

Explanation: Liver/spleen/marrow = sinusoidal (gaps, incomplete BM).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: liver/spleen = sinusoidal.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to liver/spleen = sinusoidal.

Difficulty: Understanding

**\*\*Q15. Pericytes are associated with:**

- A) Capillaries and postcapillary venules
- B) Large veins only
- C) The heart
- D) Elastic arteries

**Answer: A**

Explanation: Pericytes wrap capillaries/postcapillary venules.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: pericyte = capillaries.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to pericyte = capillaries.

Difficulty: Understanding

**\*\*Q16. Fenestrated capillaries with diaphragms are found in endocrine glands and intestinal mucosa to permit:**

- A) Barrier function
- B) Rapid fluid/solute exchange
- C) Phagocytosis
- D) Blood-brain barrier

**Answer: B**

Explanation: Fenestrae permit rapid exchange in endocrine/mucosal tissues.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: fenestrated = exchange.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to fenestrated = exchange.

Difficulty: Understanding

**\*\*Q17. Continuous capillaries forming the blood brain barrier are found in:**

- A) Liver sinusoids

- B) Endocrine glands
- C) Brain, muscle, and skin
- D) Renal glomerulus

**Answer: C**

Explanation: Brain (BBB), muscle, skin = continuous.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: brain = continuous.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to brain = continuous.

Difficulty: Understanding

**\*\*Q18. Lymphatic capillaries differ from blood capillaries in that they:**

- A) Contain RBCs normally
- B) Have thicker walls
- C) Have a continuous basal lamina and tight junctions
- D) Are blind-ended with overlapping endothelial cells lacking a complete basal lamina

**Answer: D**

Explanation: Lymphatics = blind-ended, overlapping endothelium, incomplete BM.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: lymphatic = blind-ended, no complete BM.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to lymphatic = blind-ended, no complete BM.

Difficulty: Understanding

**\*\*Q19. Heart valves are covered by:**

- A) Cartilage
- B) Cardiac muscle
- C) Endothelium
- D) Adipose tissue

**Answer: C**

Explanation: Heart valves = dense CT core covered by endothelium.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: valve = endothelium-covered.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to valve = endothelium-covered.

Difficulty: Understanding

**\*\*Q20. The tunica intima is bounded externally by the:**

- A) External elastic lamina
- B) Internal elastic lamina
- C) Adventitia directly
- D) Vasa vasorum

**Answer: B**

Explanation: The IEL is the intima's outer boundary.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: IEL bounds intima.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to IEL bounds intima.

Difficulty: Understanding

**\*\*Q21. Which layer is thickened in the left ventricle?**

- A) Pericardium
- B) Endocardium
- C) Myocardium
- D) Epicardium

**Answer: C**

Explanation: The LV myocardium is thickest (pumps systemic pressure).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: LV = thick myocardium.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to LV = thick myocardium.

Difficulty: Understanding

**\*\*Q22. The blood supply to a large artery's own wall is provided by:**

- A) Pericytes
- B) Lymphatics
- C) Luminal diffusion only
- D) Vasa vasorum

**Answer: D**

Explanation: Vasa vasorum supply thick vessel walls.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: vasa vasorum = wall supply.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to vasa vasorum = wall supply.

Difficulty: Understanding

**\*\*Q23. The layer destroyed in an aortic aneurysm is the:**

- A) Tunica intima
- B) Tunica media
- C) Tunica adventitia
- D) Endocardium

**Answer: B**

Explanation: Aortic aneurysm media destruction cystic medial necrosis Marfan fibrillin defect.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Aortic aneurysm media destruction cystic medial necrosis Marfan fibrillin defect.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Aortic aneurysm media destruction cystic medial necrosis Marfan fibrillin defect

Difficulty: Understanding

**\*\*Q24. Venous valves are folds of the:**

- A) Media
- B) External elastic lamina
- C) Adventitia
- D) Intima

**Answer: D**

Explanation: Venous valves folds of intima endothelium + connective tissue core prevent backflow.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Venous valves folds of intima endothelium + connective tissue core prevent backflow.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Venous valves folds of intima endothelium + connective tissue core prevent backf

Difficulty: Understanding

## **Part C Application**

**\*\*Q25. The aorta is an example of which artery type?**

- A) Muscular
- B) Elastic
- C) Arteriole
- D) Metarteriole

**Answer: B**

Explanation: The aorta = elastic (conducting) artery with many elastic lamellae.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: aorta = elastic.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to aorta = elastic.

Difficulty: Application

**\*\*Q26. Compared with arteries, veins have:**

- A) No adventitia
- B) Thicker media
- C) Thinner wall, larger irregular lumen, valves
- D) More elastic lamellae

**Answer: C**

Explanation: Veins = thin wall + large irregular lumen + valves.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: vein = thin + valves.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to vein = thin + valves.

Difficulty: Application

**\*\*Q27. Valves in veins are formed by folds of the:**

- A) Adventitia
- B) External elastic lamina
- C) Media
- D) Intima

**Answer: D**

Explanation: Venous valves = intimal folds.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: valve = intima.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to valve = intima.

Difficulty: Application

**\*\*Q28. Pericytes are most similar in function to:**

- A) Fibroblasts
- B) Macrophages

- C) Endothelial cells
- D) Smooth muscle cells

**Answer: D**

Explanation: Pericytes are contractile smooth-muscle-like.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: pericyte = SMC-like.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to pericyte = SMC-like.

Difficulty: Application

**\*\*Q29. Sinusoidal capillaries are characteristic of the:**

- A) Lung alveoli
- B) Brain
- C) Liver, spleen, and bone marrow
- D) Skeletal muscle

**Answer: C**

Explanation: Sinusoids = liver/spleen/marrow.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: sinusoid = liver/spleen.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to sinusoid = liver/spleen.

Difficulty: Application

**\*\*Q30. The endocardium is histologically continuous with the:**

- A) Media of arteries
- B) Endothelium (intima) of blood vessels
- C) Epicardium
- D) Pericardium

**Answer: B**

Explanation: Endocardium = endothelium + subendothelial CT, continuous with the intima.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: endocardium = intima.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to endocardium = intima.

Difficulty: Application

**\*\*Q31. A fatty streak in atherosclerosis is composed of:**

- A) Fibroblasts
- B) Lipid-laden macrophages (foam cells)
- C) Lymphocytes only
- D) Mast cells

**Answer: B**

Explanation: Foam cells (lipid-laden macrophages) form the fatty streak.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: fatty streak = foam cells.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to fatty streak = foam cells.

Difficulty: Application

**\*\*Q32. Varicose veins result from:**

- A) Chronic valvular incompetence with wall dilation/fibrosis
- B) Complete loss of media
- C) Endothelial calcification only
- D) Smooth-muscle atrophy only

**Answer: A**

Explanation: Valve failure pooling dilation wall fibrosis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: varicose = valve failure.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to varicose = valve failure.

Difficulty: Application

**\*\*Q33. A tear in the intima that lets blood split the media describes:**

- A) Aortic dissection
- B) Varicose veins
- C) Arteriosclerosis
- D) Atherosclerosis

**Answer: A**

Explanation: Dissection = intimal tear blood enters/splits the media.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: dissection = intimal tear    media.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to dissection = intimal tear    media.

Difficulty: Application

**\*\*Q34. Which feature distinguishes a lymphatic from a vein microscopically?**

- A) Absence of red blood cells in the lumen
- B) Thicker media
- C) Presence of elastic lamellae
- D) Presence of valves

**Answer: A**

Explanation: Lymphatics lack RBCs; veins contain them.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: no RBC = lymphatic.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to no RBC = lymphatic.

Difficulty: Application

**\*\*Q35. Which capillary type lines the renal glomerulus?**

- A) Lymphatic
- B) Continuous
- C) Fenestrated
- D) Sinusoidal

**Answer: C**

Explanation: Glomerular capillary fenestrated without diaphragm + thick basal lamina + podocyte slit diaphragm filtration.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Glomerular capillary fenestrated without diaphragm + thick basal lamina + podocyte slit diaphragm fi

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to Glomerular capillary fenestrated without diaphragm + thick basal lamina + podocy

Difficulty: Application

**\*\*Q36. The windkessel effect is due to:**

- A) Venous valves
- B) Pericyte contraction
- C) Smooth-muscle contraction
- D) Elastic recoil of the aortic wall

**Answer: D**

Explanation: Windkessel effect elastic recoil aortic wall maintains diastolic flow.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Windkessel effect elastic recoil aortic wall maintains diastolic flow.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Windkessel effect elastic recoil aortic wall maintains diastolic flow.

Difficulty: Application

## **Part D Integration**

**\*\*Q37. A prominent internal AND external elastic lamina is characteristic of:**

- A) Arterioles
- B) Capillaries
- C) Elastic arteries
- D) Muscular arteries

**Answer: D**

Explanation: Muscular arteries have distinct IEL + EEL.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: IEL + EEL = muscular.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to IEL + EEL = muscular.

Difficulty: Integration

**\*\*Q38. Purkinje fibers are characterized by:**

- A) No nucleus
- B) Heavy myelination
- C) Dense myofibrils
- D) Abundant glycogen, few myofibrils, pale cytoplasm

**Answer: D**

Explanation: Purkinje = large, pale, glycogen-rich, few myofibrils.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Purkinje = glycogen.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Purkinje = glycogen.

Difficulty: Integration

**\*\*Q39. Intercalated discs contain:**

- A) Hemidesmosomes only
- B) Synapses
- C) Tight junctions only
- D) Gap junctions and desmosomes

**Answer: D**

Explanation: Intercalated discs = gap junctions (electrical) + desmosomes (mechanical).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: disc = gap + desmosome.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to disc = gap + desmosome.

Difficulty: Integration

**\*\*Q40. Elastic arteries contain numerous elastic lamellae within the:**

- A) Endothelium only
- B) Tunica intima
- C) Tunica media
- D) Tunica adventitia

**Answer: C**

Explanation: The media holds the concentric elastic lamellae.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: lamellae = media.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to lamellae = media.

Difficulty: Integration

**\*\*Q41. Veins, compared with arteries, have:**

- A) Absence of adventitia
- B) An IEL exclusively
- C) A thicker media
- D) A thinner wall, larger lumen, and valves

**Answer: D**

Explanation: Veins = thinner wall + larger lumen + valves.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: vein = thin + valve.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to vein = thin + valve.  
Difficulty: Integration

**\*\*Q42. Purkinje fibers differ from ordinary cardiomyocytes by:**

- A) Absence of gap junctions
- B) Location exclusively in the atria
- C) Smaller size and more myofibrils
- D) Larger size, fewer myofibrils, and abundant glycogen

**Answer: D**

Explanation: Purkinje = larger, pale, glycogen-rich, few myofibrils.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Purkinje = large + pale.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to Purkinje = large + pale.

Difficulty: Integration

**\*\*Q43. The media of muscular arteries is composed predominantly of:**

- A) Circularly arranged smooth muscle
- B) Dense collagen only
- C) Cardiac muscle
- D) Elastic lamellae

**Answer: A**

Explanation: Muscular-artery media = circular smooth muscle.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: media = smooth muscle.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to media = smooth muscle.

Difficulty: Integration

**\*\*Q44. Capillary basal lamina is synthesized jointly by endothelial cells and:**

- A) Platelets
- B) Pericytes
- C) Neurons
- D) Erythrocytes

**Answer: B**

Explanation: Endothelial cells + pericytes make the capillary basal lamina.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: basal lamina = endothelium + pericyte.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to basal lamina = endothelium + pericyte.

Difficulty: Integration

**\*\*Q45. Which statement about the windkessel effect is TRUE?**

- A) It requires valves
- B) It depends on smooth-muscle contraction
- C) It smooths pulsatile flow via elastic recoil
- D) It occurs in veins

**Answer: C**

Explanation: Elastic recoil converts pulsatile to continuous flow.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: windkessel = elastic recoil.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to windkessel = elastic recoil.

Difficulty: Integration

**\*\*Q46. Hyaline arteriosclerosis is characteristically seen in:**

- A) Normal arteries
- B) Chronic hypertension and diabetes
- C) Acute inflammation
- D) Young athletes

**Answer: B**

Explanation: Hyaline arteriosclerosis = hypertension/diabetes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: hyaline arteriosclerosis = HTN/DM.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to hyaline arteriosclerosis = HTN/DM.

Difficulty: Integration

**\*\*Q47. Purkinje fibers are best identified by:**

- A) Peripheral nuclei
- B) Dense myofibrils
- C) Large pale glycogen-rich cytoplasm, subendocardial
- D) Multinucleation

**Answer: C**

Explanation: Purkinje fibers large pale glycogen-rich cytoplasm subendocardial, fewer myofibrils, fast conduction.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Purkinje fibers large pale glycogen-rich cytoplasm subendocardial, fewer myofibrils, fast conduction

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Purkinje fibers large pale glycogen-rich cytoplasm subendocardial, fewer myofibr

Difficulty: Integration

**\*\*Q48. Which vessel lacks smooth muscle entirely?**

- A) Muscular artery
- B) Thoracic duct
- C) Arteriole
- D) Capillary

**Answer: D**

Explanation: Capillary lacks smooth muscle, only endothelium + basal lamina + pericyte; arteriole has smooth muscle.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Capillary lacks smooth muscle, only endothelium + basal lamina + pericyte; arteriole has smooth musc

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Capillary lacks smooth muscle, only endothelium + basal lamina + pericyte; arter

Difficulty: Integration

**Part E High-Difficulty**

**\*\*Q49. The main resistance vessels are:**

- A) Venules
- B) Capillaries
- C) Elastic arteries
- D) Arterioles

**Answer: D**

Explanation: Arterioles (1-2 smooth-muscle layers) control peripheral resistance.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: resistance = arteriole.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to resistance = arteriole.

Difficulty: High-Difficulty

**\*\*Q50. The epicardium contains:**

- A) Elastic lamellae
- B) Endothelium only
- C) Mesothelium + connective tissue + fat
- D) Cardiac muscle only

**Answer: C**

Explanation: Epicardium = mesothelium + CT + fat (visceral pericardium).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: epicardium = mesothelium.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to epicardium = mesothelium.

Difficulty: High-Difficulty

**\*\*Q51. Cystic medial degeneration weakening the aortic media is associated with:**

- A) Varicose veins
- B) Atherosclerosis
- C) Marfan syndrome / aneurysm
- D) Vasculitis

**Answer: C**

Explanation: Medial elastic/SMC degeneration (Marfan) aneurysm.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: media degeneration = aneurysm.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to media degeneration = aneurysm.

Difficulty: High-Difficulty

**\*\*Q52. The windkessel effect depends on the recoil of:**

- A) Smooth muscle
- B) Elastic lamellae
- C) Collagen fibers alone
- D) Endothelium

**Answer: B**

Explanation: Elastic lamellae store/release recoil.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: windkessel = elastic recoil.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to windkessel = elastic recoil.

Difficulty: High-Difficulty

**\*\*Q53. Venous valves are formed by folds of the:**

- A) Intima (endothelium + thin CT core)
- B) Media
- C) Vasa vasorum
- D) Adventitia

**Answer: A**

Explanation: Valves = intimal folds.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: valve = intima.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to valve = intima.

Difficulty: High-Difficulty

**\*\*Q54. Cardiac muscle cells are connected end-to-end by:**

- A) Intercalated discs
- B) Motor end plates
- C) Costameres
- D) Neuromuscular junctions

**Answer: A**

Explanation: Intercalated discs join cardiac myocytes.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: cardiac = intercalated discs.

Learning Objective mapping: Chapter 8    The Circulatory System    LO related to cardiac = intercalated discs.

Difficulty: High-Difficulty

**\*\*Q55. Capillary exchange occurs by diffusion across the:**

- A) Thin endothelial cell layer and basal lamina
- B) Media
- C) Valves
- D) Adventitia

**Answer: A**



**Explanation:** Exchange = across endothelium + basal lamina.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: exchange across endothelium.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to exchange across endothelium.

Difficulty: High-Difficulty

**\*\*Q56. Which vessel type has NO smooth muscle in its wall?**

- A) Capillary
- B) Muscular artery
- C) Venule (small)
- D) Arteriole

**Answer: A**

**Explanation:** Capillaries = endothelium + basal lamina only (no smooth muscle).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: capillary = no smooth muscle.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to capillary = no smooth muscle.

Difficulty: High-Difficulty

**\*\*Q57. The subendocardial location of Purkinje fibers makes them vulnerable to:**

- A) Endocarditis
- B) Ischemia (and arrhythmia after infarction)
- C) Hypertension
- D) Atherosclerosis

**Answer: B**

**Explanation:** Subendocardial Purkinje fibers are ischemic-sensitive arrhythmias.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Purkinje = ischemia-sensitive.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Purkinje = ischemia-sensitive.

Difficulty: High-Difficulty

**\*\*Q58. The tunica media of the aorta weakens in aneurysm due to:**

- A) Loss of elastic lamellae and smooth muscle

- B) Valve incompetence
- C) Endothelial hyperplasia
- D) Intimal foam cells

**Answer: A**

Explanation: Aneurysm = media elastic/SMC loss. Practice Test

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: aneurysm = media loss.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to aneurysm = media loss.

Difficulty: High-Difficulty

**\*\*Q59. The blood nerve-barrier analogue in the circulatory system is best represented by:**

- A) Continuous capillaries of the brain
- B) Sinusoids of the liver
- C) Venous valves
- D) Fenestrated capillaries

**Answer: A**

Explanation: BBB continuous capillaries brain tight junctions astrocyte foot processes; liver sinusoid discontinuous; kidney glomerulus fenestrated.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: BBB continuous capillaries brain tight junctions astrocyte foot processes; liver sinusoid discontinuous

Learning Objective mapping: Chapter 8 The Circulatory System LO related to BBB continuous capillaries brain tight junctions astrocyte foot processes; liver

Difficulty: High-Difficulty

**\*\*Q60. A thin-walled vessel with valves and no red blood cells is a:**

- A) Vein
- B) Lymphatic
- C) Muscular artery
- D) Sinusoid

**Answer: B**

Explanation: Lymphatic thin-walled valves no RBC, no basal lamina, drainage; vein has RBC.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Circulatory System section for decisive features.

Key point: Lymphatic thin-walled valves no RBC, no basal lamina, drainage; vein has RBC.

Learning Objective mapping: Chapter 8 The Circulatory System LO related to Lymphatic thin-walled valves no RBC, no basal lamina, drainage; vein has RBC.

Difficulty: High-Difficulty

# Chapter 9      Blood

## Part A      Recall

**\*\*Q1. A reticulocyte is identified by residual:**

- A) Golgi
- B) DNA
- C) RNA / ribosomes
- D) Mitochondria

**Answer: C**

Explanation: Reticulocytes retain ribosomal RNA (supravital stain). DNA is extruded with the nucleus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: reticulocyte = RNA.

Learning Objective mapping: Chapter 9      Blood      LO related to reticulocyte = RNA.

Difficulty: Recall

**\*\*Q2. The nucleus of a monocyte is typically:**

- A) Bilobed
- B) Kidney- or horseshoe-shaped
- C) Multilobed (5 lobes)
- D) Round and dense

**Answer: B**

Explanation: Monocyte = kidney/horseshoe, lacy nucleus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: monocyte = kidney nucleus.

Learning Objective mapping: Chapter 9      Blood      LO related to monocyte = kidney nucleus.

Difficulty: Recall

**\*\*Q3. Hypersegmented neutrophils ( 6 lobes) suggest:**

- A) Acute bacterial infection
- B) Vitamin B12 / folate deficiency
- C) CML
- D) Allergy

**Answer: B**

Explanation: Hypersegmentation = impaired DNA synthesis (B12/folate).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: hypersegmented = B12/folate.

Learning Objective mapping: Chapter 9    Blood    LO related to hypersegmented = B12/folate.

Difficulty: Recall

**\*\*Q4. Reticulocytes are identified by supravital staining that reveals:**

- A) Residual ribosomal RNA (reticular network)
- B) Hemoglobin H inclusions
- C) Basophilic stippling only on H&E
- D) Nuclear remnants

**Answer: A**

Explanation: The reticular network = residual RNA.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: reticulocyte = RNA network.

Learning Objective mapping: Chapter 9    Blood    LO related to reticulocyte = RNA network.

Difficulty: Recall

**\*\*Q5. Eosinophil granules stain intensely with eosin due to:**

- A) Histamine primarily
- B) Major basic protein and other cationic proteins
- C) Heparin
- D) Hemoglobin

**Answer: B**

Explanation: Cationic major basic protein binds the acidic dye eosin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: eosinophil = major basic protein.

Learning Objective mapping: Chapter 9    Blood    LO related to eosinophil = major basic protein.

Difficulty: Recall

**\*\*Q6. The normal adult platelet count is approximately:**

- A) 150,000 400,000/ L
- B) 4 6 million/ L
- C) 1,000 2,000/ L
- D) 4,000 11,000/ L

**Answer: A**

Explanation: 150,000 400,000/ L.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: platelets 150 400k.

Learning Objective mapping: Chapter 9 Blood LO related to platelets 150 400k.

Difficulty: Recall

**\*\*Q7. In iron-deficiency anemia, RBCs appear:**

- A) Normocytic and normochromic
- B) Microcytic and hypochromic with increased central pallor
- C) Macrocytic
- D) Target cells exclusively

**Answer: B**

Explanation: Microcytic + hypochromic + central pallor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: iron = microcytic hypochromic.

Learning Objective mapping: Chapter 9 Blood LO related to iron = microcytic hypochromic.

Difficulty: Recall

**\*\*Q8. The buffy coat after centrifugation contains:**

- A) Plasma
- B) Leukocytes and platelets
- C) Erythrocytes only
- D) Serum only

**Answer: B**

Explanation: Buffy coat = WBCs + platelets (between plasma and RBCs).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: buffy coat = WBCs + platelets.

Learning Objective mapping: Chapter 9 Blood LO related to buffy coat = WBCs + platelets.

Difficulty: Recall

**\*\*Q9. Teardrop-shaped RBCs (dacrocytes) suggest:**

- A) Sickle cell
- B) Spherocytosis
- C) Iron deficiency
- D) Myelofibrosis

**Answer: D**

**Explanation:** Dacrocytes = marrow fibrosis (myelofibrosis).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: teardrop = myelofibrosis.

Learning Objective mapping: Chapter 9    Blood    LO related to teardrop = myelofibrosis.

Difficulty: Recall

**\*\*Q10. Which cell is the blood counterpart of the connective-tissue mast cell?**

- A) Monocyte
- B) Neutrophil
- C) Basophil
- D) Eosinophil

**Answer: C**

**Explanation:** Basophil (histamine/heparin) = blood mast-cell analogue.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: basophil    mast cell.

Learning Objective mapping: Chapter 9    Blood    LO related to basophil    mast cell.

Difficulty: Recall

**\*\*Q11. A smear shows cells with bilobed nuclei and red-orange granules. These are:**

- A) Eosinophils
- B) Basophils
- C) Monocytes
- D) Neutrophils

**Answer: A**

**Explanation:** Eosinophil bilobed nucleus red-orange granules major basic protein parasitic allergy.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Eosinophil bilobed nucleus red-orange granules major basic protein parasitic allergy.

Learning Objective mapping: Chapter 9    Blood    LO related to Eosinophil bilobed nucleus red-orange granules major basic protein parasitic all

Difficulty: Recall

**\*\*Q12. Platelet dense granules contain:**

- A) Fibrinogen
- B) ADP and calcium
- C) Von Willebrand factor
- D) Histamine

**Answer: B**

Explanation: Platelet dense granules ADP, calcium, serotonin; alpha granules fibrinogen, VWF, growth factors.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Platelet dense granules ADP, calcium, serotonin; alpha granules fibrinogen, VWF, growth factors.

Learning Objective mapping: Chapter 9 Blood LO related to Platelet dense granules ADP, calcium, serotonin; alpha granules fibrinogen, VWF,

Difficulty: Recall

## Part B Understanding

**\*\*Q13. The stain used to demonstrate reticulocytes is:**

- A) Wright-Giemsa only
- B) PAS
- C) H&E
- D) Supravital stain (new methylene blue)

**Answer: D**

Explanation: Supravital dyes precipitate the ribosomal network.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: reticulocyte = supravital.

Learning Objective mapping: Chapter 9 Blood LO related to reticulocyte = supravital.

Difficulty: Understanding

**\*\*Q14. Specific (secondary) granules of neutrophils first appear at the:**

- A) Myeloblast
- B) Promyelocyte
- C) Myelocyte
- D) Metamyelocyte

**Answer: C**

Explanation: Specific granules appear at the myelocyte stage.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: specific = myelocyte.

Learning Objective mapping: Chapter 9    Blood    LO related to specific = myelocyte.

Difficulty: Understanding

**\*\*Q15. A "left shift" in the WBC count refers to:**

- A) Increased eosinophils
- B) Decreased neutrophils
- C) Increased lymphocytes
- D) Increased band/immature neutrophils

**Answer: D**

Explanation: Left shift = increased band (immature) neutrophils.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: left shift = bands.

Learning Objective mapping: Chapter 9    Blood    LO related to left shift = bands.

Difficulty: Understanding

**\*\*Q16. The normal reticulocyte count is approximately:**

- A) 0.5 2%
- B) 10 20%
- C) 30 40%
- D) 50%

**Answer: A**

Explanation: Normal reticulocytes = 0.5 2% of RBCs.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: reticulocyte    1%.

Learning Objective mapping: Chapter 9    Blood    LO related to reticulocyte    1%.

Difficulty: Understanding

**\*\*Q17. Basophils contain granules rich in histamine and:**

- A) Major basic protein
- B) Lysozyme exclusively
- C) Myeloperoxidase
- D) Heparin

**Answer: D**

Explanation: Basophils = histamine + heparin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: basophil = histamine/heparin.

Learning Objective mapping: Chapter 9    Blood    LO related to basophil = histamine/heparin.

Difficulty: Understanding

**\*\*Q18. Neutrophils arrive at sites of acute bacterial infection within:**

- A) They never leave the bloodstream
- B) Minutes to hours
- C) Several weeks
- D) Months

**Answer: B**

Explanation: Neutrophils are the first responders (minutes hours).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: neutrophil = first responder.

Learning Objective mapping: Chapter 9    Blood    LO related to neutrophil = first responder.

Difficulty: Understanding

**\*\*Q19. Sickle cell disease results from a point mutation in the gene encoding:**

- A) Fetal hemoglobin
- B) Heme synthase
- C) Alpha-globin
- D) Beta-globin

**Answer: D**

Explanation:  $\beta$ -globin (Glu → Val)    HbS.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: sickle =  $\beta$ -globin.

Learning Objective mapping: Chapter 9    Blood    LO related to sickle =  $\beta$ -globin.

Difficulty: Understanding

**\*\*Q20. Serum lacks:**

- A) Clotting factors, particularly fibrinogen
- B) Electrolytes
- C) Water
- D) Red blood cells only

**Answer: A**

**Explanation:** Serum = plasma   clotting factors.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: serum = no clotting factors.

Learning Objective mapping: Chapter 9   Blood   LO related to serum = no clotting factors.

Difficulty: Understanding

**\*\*Q21. Atypical lymphocytes on a smear are classically seen in:**

- A) Iron deficiency
- B) Lead poisoning
- C) Bacterial pneumonia
- D) Infectious mononucleosis (EBV)

**Answer: D**

**Explanation:** EBV   atypical (reactive) lymphocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: atypical lymphs = EBV.

Learning Objective mapping: Chapter 9   Blood   LO related to atypical lymphs = EBV.

Difficulty: Understanding

**\*\*Q22. Reticulocytosis is most characteristic of:**

- A) Aplastic anemia
- B) Hemolytic anemia
- C) Iron deficiency (untreated)
- D) B12 deficiency

**Answer: B**

**Explanation:** Hemolysis   marrow compensation   reticulocytosis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: hemolysis = reticulocytosis.

Learning Objective mapping: Chapter 9   Blood   LO related to hemolysis = reticulocytosis.

Difficulty: Understanding

**\*\*Q23. A patient with splenectomy would most likely show:**

- A) Left shift
- B) Basophilic stippling
- C) Howell-Jolly bodies
- D) Hypersegmented neutrophils

### Answer: C

Explanation: Splenectomy    Howell-Jolly bodies nuclear remnants, Heinz bodies, target cells, encapsulated bacteria risk.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Splenectomy    Howell-Jolly bodies nuclear remnants, Heinz bodies, target cells, encapsulated bacteria

Learning Objective mapping: Chapter 9    Blood    LO related to Splenectomy    Howell-Jolly bodies nuclear remnants, Heinz bodies, target cells,

Difficulty: Understanding

### **\*\*Q24. A left shift refers to increased:**

- A) Lymphocytes
- B) Monocytes
- C) Segmented neutrophils
- D) Band neutrophils

### Answer: D

Explanation: Left shift increased band neutrophils immature neutrophils indicates acute bacterial infection.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Left shift increased band neutrophils immature neutrophils indicates acute bacterial infection.

Learning Objective mapping: Chapter 9    Blood    LO related to Left shift increased band neutrophils immature neutrophils indicates acute bacterial infection

Difficulty: Understanding

## Part C    Application

### **\*\*Q25. Eosinophils characteristically have:**

- A) Round nucleus + thin rim
- B) Multilobed nucleus + neutral granules
- C) Bilobed nucleus + red-orange granules
- D) Obscured nucleus + blue granules

### Answer: C

Explanation: Eosinophil = bilobed + red-orange granules (major basic protein).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: eosinophil = bilobed + red.

Learning Objective mapping: Chapter 9    Blood    LO related to eosinophil = bilobed + red.

Difficulty: Application

**\*\*Q26. Azurophilic (primary) granules of neutrophils are essentially:**

- A) Glycogen
- B) Secretory vesicles
- C) Lysosomes (myeloperoxidase)
- D) Mitochondria

**Answer: C**

Explanation: Primary (azurophilic) granules = lysosomes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: azurophilic = lysosomes.

Learning Objective mapping: Chapter 9    Blood    LO related to azurophilic = lysosomes.

Difficulty: Application

**\*\*Q27. Howell-Jolly bodies in RBCs are seen after:**

- A) Splenectomy
- B) G6PD deficiency
- C) Thalassemia only
- D) Iron deficiency

**Answer: A**

Explanation: Nuclear remnants persist without the spleen's pitting function.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Howell-Jolly = splenectomy.

Learning Objective mapping: Chapter 9    Blood    LO related to Howell-Jolly = splenectomy.

Difficulty: Application

**\*\*Q28. Howell-Jolly bodies are classically seen following:**

- A) Thymectomy
- B) Splenectomy
- C) Bone marrow transplant
- D) Liver transplant

**Answer: B**

Explanation: Splenectomy    no pitting    nuclear remnants persist.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Howell-Jolly = splenectomy.

Learning Objective mapping: Chapter 9    Blood    LO related to Howell-Jolly = splenectomy.

Difficulty: Application

**\*\*Q29. Lymphocytes are characterized microscopically by:**

- A) A large round nucleus with scant cytoplasm
- B) Prominent cytoplasmic granules always
- C) Absence of a nucleus
- D) Abundant cytoplasm and a multilobed nucleus

**Answer: A**

Explanation: Lymphocyte = large round nucleus + thin cytoplasmic rim.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: lymphocyte = round + rim.

Learning Objective mapping: Chapter 9    Blood    LO related to lymphocyte = round + rim.

Difficulty: Application

**\*\*Q30. Eosinophilia is classically associated with:**

- A) Acute bacterial infection
- B) Parasitic infections and allergic conditions
- C) Viral infection exclusively
- D) Chronic renal failure

**Answer: B**

Explanation: Eosinophils    parasites + allergy.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: eosinophilia = parasite/allergy.

Learning Objective mapping: Chapter 9    Blood    LO related to eosinophilia = parasite/allergy.

Difficulty: Application

**\*\*Q31. ABO blood-group antigens are:**

- A) Carbohydrate (glycoprotein/glycolipid) antigens
- B) Lipids only
- C) Nucleic acids
- D) Proteins only

**Answer: A**

Explanation: ABO antigens = carbohydrate determinants on glycoproteins/glycolipids.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: ABO = carbohydrate.

Learning Objective mapping: Chapter 9    Blood    LO related to ABO = carbohydrate.

Difficulty: Application

**\*\*Q32. Acute leukemia is defined by a bone-marrow blast percentage of at least:**

- A) 5%
- B) 20%
- C) 50%
- D) 90%

**Answer: B**

Explanation: 20% blasts = acute leukemia (WHO).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: acute leukemia = 20% blasts.

Learning Objective mapping: Chapter 9    Blood    LO related to acute leukemia = 20% blasts.

Difficulty: Application

**\*\*Q33. Target cells are associated with:**

- A) Lead poisoning only
- B) Iron deficiency only
- C) Liver disease, thalassemia, and post-splenectomy
- D) B12 deficiency only

**Answer: C**

Explanation: Target cells = liver disease/thalassemia/post-splenectomy.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: target cells = liver/thalassemia/splenectomy.

Learning Objective mapping: Chapter 9    Blood    LO related to target cells = liver/thalassemia/splenectomy.

Difficulty: Application

**\*\*Q34. The central pallor of a normal RBC occupies approximately:**

- A) The entire cell
- B) Half the diameter
- C) No pallor
- D) One-third of the diameter

**Answer: D**

**Explanation:** Normal central pallor      diameter (biconcave disc).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: pallor      = normal.

Learning Objective mapping: Chapter 9      Blood      LO related to pallor      = normal.

Difficulty: Application

**\*\*Q35. Reticulocytes contain residual:**

- A) RNA
- B) Hemoglobin only
- C) Platelet granules
- D) DNA

**Answer: A**

**Explanation:** Reticulocyte residual RNA polyribosomes supravital stain, 1-2 days maturation to RBC.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Reticulocyte residual RNA polyribosomes supravital stain, 1-2 days maturation to RBC.

Learning Objective mapping: Chapter 9      Blood      LO related to Reticulocyte residual RNA polyribosomes supravital stain, 1-2 days maturation to

Difficulty: Application

**\*\*Q36. Basophilic stippling is classically due to:**

- A) Iron deficiency
- B) Splenectomy
- C) Lead poisoning
- D) B12 deficiency

**Answer: C**

**Explanation:** Basophilic stippling ribosomal aggregates lead poisoning impaired pyrimidine 5'-nucleotidase.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Basophilic stippling ribosomal aggregates lead poisoning impaired pyrimidine 5'-nucleotidase.

Learning Objective mapping: Chapter 9      Blood      LO related to Basophilic stippling ribosomal aggregates lead poisoning impaired pyrimidine 5'-

Difficulty: Application



## Part D Integration

**\*\*Q37. The granules that obscure the nucleus are seen in:**

- A) Basophils
- B) Monocytes
- C) Neutrophils
- D) Eosinophils

**Answer: A**

Explanation: Basophil's dense blue-purple granules obscure the nucleus.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: obscured nucleus = basophil.

Learning Objective mapping: Chapter 9 Blood LO related to obscured nucleus = basophil.

Difficulty: Integration

**\*\*Q38. Platelet -granules contain:**

- A) ADP and serotonin
- B) Fibrinogen and von Willebrand factor
- C) Calcium only
- D) Histamine

**Answer: B**

Explanation: -granules = fibrinogen, vWF, PF4.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: -granule = fibrinogen/vWF.

Learning Objective mapping: Chapter 9 Blood LO related to -granule = fibrinogen/vWF.

Difficulty: Integration

**\*\*Q39. The membrane protein responsible for RBC deformability is:**

- A) Anion exchanger only
- B) Glycophorin only
- C) Hemoglobin
- D) Spectrin

**Answer: D**

Explanation: Spectrin (with ankyrin/actin) = the deformable skeleton.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: spectrin = deformability.

Learning Objective mapping: Chapter 9    Blood    LO related to spectrin = deformability.

Difficulty: Integration

**\*\*Q40. The normal lifespan of a circulating erythrocyte is approximately:**

- A) 1 year
- B) 7 days
- C) 10 days
- D) 120 days

**Answer: D**

Explanation: ~120 days.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: RBC = 120 days.

Learning Objective mapping: Chapter 9    Blood    LO related to RBC = 120 days.

Difficulty: Integration

**\*\*Q41. Monocytes characteristically have a nucleus that is:**

- A) Absent
- B) Multilobed
- C) Kidney/horseshoe-shaped
- D) Round and small

**Answer: C**

Explanation: Monocyte = kidney/horseshoe nucleus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: monocyte = kidney.

Learning Objective mapping: Chapter 9    Blood    LO related to monocyte = kidney.

Difficulty: Integration

**\*\*Q42. Microcytic hypochromic anemia is most classically caused by:**

- A) Chronic hemolysis
- B) Vitamin B12 deficiency
- C) Iron deficiency
- D) Folate deficiency

**Answer: C**

Explanation: Iron deficiency    microcytic hypochromic.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: iron deficiency = microcytic.

Learning Objective mapping: Chapter 9    Blood    LO related to iron deficiency = microcytic.

Difficulty: Integration

**\*\*Q43. Blood type O is the universal donor because the RBCs lack:**

- A) Plasma proteins
- B) Rh antigen only
- C) Both A and B surface antigens
- D) All surface antigens including Rh

**Answer: C**

Explanation: Type O = no A or B antigens.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: O = no A/B.

Learning Objective mapping: Chapter 9    Blood    LO related to O = no A/B.

Difficulty: Integration

**\*\*Q44. Toxic granulation and D hle bodies in neutrophils suggest:**

- A) Iron deficiency
- B) B12 deficiency
- C) Normal physiology
- D) Severe bacterial infection/systemic inflammation

**Answer: D**

Explanation: Toxic granulation/D hle bodies = severe infection/inflammation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: toxic granulation = severe infection.

Learning Objective mapping: Chapter 9    Blood    LO related to toxic granulation = severe infection.

Difficulty: Integration

**\*\*Q45. Which granule contains myeloperoxidase?**

- A) -granule
- B) Specific (secondary)
- C) Azurophilic (primary)
- D) Dense ( )

**Answer: C**

Explanation: Azurophilic (primary) granules = lysosomes with myeloperoxidase.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: MPO = azurophilic.

Learning Objective mapping: Chapter 9    Blood    LO related to MPO = azurophilic.

Difficulty: Integration

**\*\*Q46. Which statement about platelets is TRUE?**

- A) They are whole cells
- B) They contain hemoglobin
- C) They have a nucleus
- D) They are anucleate fragments with - and dense granules

**Answer: D**

Explanation: Platelets = anucleate fragments with + dense granules.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: platelet = anucleate fragment.

Learning Objective mapping: Chapter 9    Blood    LO related to platelet = anucleate fragment.

Difficulty: Integration

**\*\*Q47. Hypersegmented neutrophils indicate:**

- A) Lead poisoning
- B) Iron deficiency
- C) Bacterial infection
- D) B12/folate deficiency

**Answer: D**

Explanation: Hypersegmented neutrophils >5 lobes indicate megaloblastic anemia B12/folate deficiency impaired DNA synthesis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Hypersegmented neutrophils >5 lobes indicate megaloblastic anemia B12/folate deficiency impaired DNA

Learning Objective mapping: Chapter 9    Blood    LO related to Hypersegmented neutrophils >5 lobes indicate megaloblastic anemia B12/folate def

Difficulty: Integration

**\*\*Q48. Serum differs from plasma by lacking:**

- A) Albumin
- B) Fibrinogen
- C) Electrolytes
- D) Globulins

**Answer: B**

Explanation: Serum = plasma minus fibrinogen and clotting factors after clotting.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Serum = plasma minus fibrinogen and clotting factors after clotting.

Learning Objective mapping: Chapter 9 Blood LO related to Serum = plasma minus fibrinogen and clotting factors after clotting.

Difficulty: Integration

## **Part E High-Difficulty**

**\*\*Q49. The largest leukocyte in peripheral blood is the:**

- A) Neutrophil
- B) Lymphocyte
- C) Monocyte
- D) Eosinophil

**Answer: C**

Explanation: Monocyte (15–20 μm) is the largest.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: largest = monocyte.

Learning Objective mapping: Chapter 9 Blood LO related to largest = monocyte.

Difficulty: High-Difficulty

**\*\*Q50. Platelet dense ( ) granules contain:**

- A) ADP, ATP, Ca<sup>2+</sup>, serotonin
- B) Fibrinogen
- C) Lactoferrin
- D) vWF

**Answer: A**

Explanation: Dense granules = ADP/ATP/Ca<sup>2+</sup>/serotonin.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: dense = ADP/Ca<sup>2+</sup>.

Learning Objective mapping: Chapter 9 Blood LO related to dense = ADP/Ca<sup>2+</sup>.

Difficulty: High-Difficulty

**\*\*Q51. Serum differs from plasma by the absence of:**

- A) Electrolytes
- B) Albumin
- C) Fibrinogen and clotting factors
- D) Globulins

**Answer: C**

Explanation: Serum = plasma minus fibrinogen/clotting factors.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: serum = no fibrinogen.

Learning Objective mapping: Chapter 9    Blood    LO related to serum = no fibrinogen.

Difficulty: High-Difficulty

**\*\*Q52. Neutrophils are characterized by a nucleus with:**

- A) 2-5 lobes connected by thin strands
- B) A bilobed spectacle shape
- C) No nucleus
- D) A single round shape

**Answer: A**

Explanation: Neutrophil = 2-5 (classically 3-5) lobes.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: neutrophil = multi-lobed.

Learning Objective mapping: Chapter 9    Blood    LO related to neutrophil = multi-lobed.

Difficulty: High-Difficulty

**\*\*Q53. Platelets are derived from the cytoplasm of:**

- A) Megakaryocytes
- B) Lymphoblasts
- C) Myeloblasts
- D) Erythroblasts

**Answer: A**

Explanation: Platelets = megakaryocyte fragments.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: platelet    megakaryocyte.

Learning Objective mapping: Chapter 9    Blood    LO related to platelet    megakaryocyte.

Difficulty: High-Difficulty

**\*\*Q54. Macrocytic (megaloblastic) anemia is classically caused by:**

- A) Iron
- B) Vitamin B12 or folate deficiency
- C) Vitamin C
- D) Copper

**Answer: B**

Explanation: B12/folate    macrocytic + hypersegmented neutrophils.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: macrocytic = B12/folate.

Learning Objective mapping: Chapter 9    Blood    LO related to macrocytic = B12/folate.

Difficulty: High-Difficulty

**\*\*Q55. Basophilic stippling of RBCs is classically associated with:**

- A) Iron deficiency
- B) Lead poisoning
- C) Vitamin C deficiency
- D) Normal aging

**Answer: B**

Explanation: Lead inhibits pyrimidine 5'-nucleotidase    RNA aggregates    stippling.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: stippling = lead.

Learning Objective mapping: Chapter 9    Blood    LO related to stippling = lead.

Difficulty: High-Difficulty

**\*\*Q56. Which is correctly paired regarding leukocyte granules and mediators?**

- A) Neutrophil    major basic protein
- B) Monocyte    hemoglobin
- C) Eosinophil    histamine
- D) Basophil    histamine and heparin

**Answer: D**

Explanation: Basophil = histamine + heparin. Eosinophil = major basic protein.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: basophil = histamine/heparin.

Learning Objective mapping: Chapter 9    Blood    LO related to basophil = histamine/heparin.

Difficulty: High-Difficulty

**\*\*Q57. The normal adult RBC count is approximately:**

- A) 4 6 million/ L
- B) 150,000 400,000/ L
- C) 1 2 million/ L
- D) 4,000 11,000/ L

**Answer: A**

Explanation: RBCs    4 6 million/ L (4,000 11,000 = WBCs; 150 400k = platelets).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: RBC = millions/ L.

Learning Objective mapping: Chapter 9    Blood    LO related to RBC = millions/ L.

Difficulty: High-Difficulty

**\*\*Q58. A patient with severe B12 deficiency would most likely show:**

- A) Spherocytes and reticulocytosis
- B) Teardrop cells and left shift
- C) Microcytic RBCs and target cells
- D) Macrocytic RBCs and hypersegmented neutrophils

**Answer: D**

Explanation: B12 deficiency = macrocytic + hypersegmented neutrophils. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: B12 = macrocytic + hypersegmented.

Learning Objective mapping: Chapter 9    Blood    LO related to B12 = macrocytic + hypersegmented.

Difficulty: High-Difficulty

**\*\*Q59. The largest circulating leukocyte is the:**

- A) Monocyte
- B) Neutrophil
- C) Basophil
- D) Lymphocyte

**Answer: A**

Explanation: Monocyte largest circulating leukocyte kidney-shaped nucleus, precursor to macrophage.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Monocyte largest circulating leukocyte kidney-shaped nucleus, precursor to macrophage.

Learning Objective mapping: Chapter 9 Blood LO related to Monocyte largest circulating leukocyte kidney-shaped nucleus, precursor to macro

Difficulty: High-Difficulty

**\*\*Q60. Microcytic hypochromic anemia is most characteristic of:**

- A) Acute blood loss
- B) Iron deficiency
- C) B12 deficiency
- D) Folate deficiency

**Answer: B**

Explanation: Microcytic hypochromic iron deficiency impaired hemoglobin synthesis; B12/folate macrocytic.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Blood section for decisive features.

Key point: Microcytic hypochromic iron deficiency impaired hemoglobin synthesis; B12/folate macrocytic.

Learning Objective mapping: Chapter 9 Blood LO related to Microcytic hypochromic iron deficiency impaired hemoglobin synthesis; B12/folate

Difficulty: High-Difficulty

# Chapter 10 Hemopoiesis

## Part A Recall

**\*\*Q1. The hematopoietic stem-cell marker is:**

- A) CD34
- B) CD19
- C) CD8
- D) CD3

**Answer: A**

Explanation: CD34 marks the pluripotent HSC. CD3 = T cells; CD19 = B cells.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: HSC = CD34.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to HSC = CD34.

Difficulty: Recall

**\*\*Q2. The erythroblast stage that extrudes its nucleus is the:**

- A) Orthochromatic (acidophilic) erythroblast
- B) Proerythroblast
- C) Basophilic erythroblast
- D) Polychromatophilic erythroblast

**Answer: A**

Explanation: The orthochromatic normoblast extrudes its pyknotic nucleus → reticulocyte.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: extrusion = orthochromatic.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to extrusion = orthochromatic.

Difficulty: Recall

**\*\*Q3. A hypocellular marrow replaced by fat indicates:**

- A) Aplastic anemia
- B) Hemolytic anemia
- C) Leukemoid reaction
- D) CML

**Answer: A**

Explanation: Fat-replaced hypocellular marrow = aplastic anemia.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: hypocellular = aplasia.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to hypocellular = aplasia.

Difficulty: Recall

**\*\*Q4. Erythropoietin is produced mainly by:**

- A) Hepatocytes (in adults)
- B) Splenic macrophages
- C) Bone-marrow stromal cells
- D) Renal peritubular interstitial cells

**Answer: D**

Explanation: EPO = kidney (peritubular interstitial cells).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: EPO = kidney.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to EPO = kidney.

Difficulty: Recall

**\*\*Q5. The most primitive hematopoietic cell is the:**

- A) Pluripotent hematopoietic stem cell
- B) Reticulocyte
- C) Promyelocyte
- D) Myeloblast

**Answer: A**

Explanation: The pluripotent HSC (CD34+).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: HSC = pluripotent.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to HSC = pluripotent.

Difficulty: Recall

**\*\*Q6. Extramedullary hematopoiesis in adults most commonly occurs in the:**

- A) Kidney and lung
- B) Liver and spleen
- C) Thymus and tonsils
- D) Skin

**Answer: B**

Explanation: Liver + spleen (fetal sites) resume production.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: extramedullary = liver/spleen.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to extramedullary = liver/spleen.

Difficulty: Recall

**\*\*Q7. Aplastic anemia results from failure to produce:**

- A) Lymphocytes only
- B) RBCs only
- C) All three lineages (pancytopenia)
- D) Platelets only

**Answer: C**

Explanation: Aplastic anemia = pancytopenia (all three lineages).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: aplasia = pancytopenia.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to aplasia = pancytopenia.

Difficulty: Recall

**\*\*Q8. Specific (secondary) granules first appear at which stage?**

- A) Band cell
- B) Segmented neutrophil only
- C) Myeloblast
- D) Myelocyte

**Answer: D**

Explanation: Specific granules = myelocyte.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: specific = myelocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to specific = myelocyte.

Difficulty: Recall

**\*\*Q9. The cytoplasmic color change (blue pink) during erythropoiesis reflects:**

- A) Loss of mitochondria
- B) Decreasing RNA and increasing hemoglobin
- C) Nuclear extrusion

D) Iron accumulation

**Answer: B**

Explanation: RNA (blue) decreases as hemoglobin (pink) accumulates.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: blue pink = RNA Hb.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to blue pink = RNA Hb.

Difficulty: Recall

**\*\*Q10. Which is TRUE of the orthochromatic erythroblast?**

A) It is the last nucleated stage before extrusion

B) It is basophilic

C) It contains no hemoglobin

D) It divides actively

**Answer: A**

Explanation: Orthochromatic = last nucleated stage (pink, pyknotic nucleus).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: orthochromatic = last nucleated.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to orthochromatic = last nucleated.

Difficulty: Recall

**\*\*Q11. A marrow aspirate shows a very large cell with a single multilobed polyploid nucleus. This is a:**

A) Megakaryocyte

B) Myeloblast

C) Plasma cell

D) Osteoclast

**Answer: A**

Explanation: Megakaryocyte very large multilobed polyploid nucleus produces platelets from demarcation membranes.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Megakaryocyte very large multilobed polyploid nucleus produces platelets from demarcation membranes.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Megakaryocyte very large multilobed polyploid nucleus produces platelets from de

Difficulty: Recall

**\*\*Q12. The orthochromatic erythroblast:**

- A) Divides actively
- B) Extrudes its nucleus to become a reticulocyte
- C) Contains no hemoglobin
- D) Is the earliest erythroid stage

**Answer: B**

Explanation: Orthochromatic erythroblast eosinophilic hemoglobin-rich extrudes nucleus to become reticulocyte.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Orthochromatic erythroblast eosinophilic hemoglobin-rich extrudes nucleus to become reticulocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Orthochromatic erythroblast eosinophilic hemoglobin-rich extrudes nucleus to become reticulocyte.

Difficulty: Recall

**Part B Understanding**

**\*\*Q13. The normal myeloid:erythroid (M:E) ratio is approximately:**

- A) 10:1
- B) 1:3
- C) 3:1
- D) 1:1

**Answer: C**

Explanation: M:E = 3:1 (myeloid predominates).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: M:E = 3:1.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to M:E = 3:1.

Difficulty: Understanding

**\*\*Q14. Erythroblastic islands are organized around a central:**

- A) Megakaryocyte
- B) Plasma cell
- C) Lymphocyte
- D) Macrophage

**Answer: D**

Explanation: A central macrophage supplies iron and phagocytoses extruded nuclei.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: island = central macrophage.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to island = central macrophage.

Difficulty: Understanding

**\*\*Q15. Teardrop RBCs (dacrocytes) and extramedullary hematopoiesis suggest:**

- A) Sick cell
- B) Spherocytosis
- C) Iron deficiency
- D) Myelofibrosis

**Answer: D**

Explanation: Myelofibrosis teardrops + extramedullary hematopoiesis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: teardrops = myelofibrosis.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to teardrops = myelofibrosis.

Difficulty: Understanding

**\*\*Q16. Bone-marrow hematopoiesis occurs primarily adjacent to:**

- A) Arterioles
- B) Sinusoids
- C) Lymphatics
- D) Venules only

**Answer: B**

Explanation: Hematopoiesis occurs near sinusoids (release of mature cells).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: marrow = sinusoids.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to marrow = sinusoids.

Difficulty: Understanding

**\*\*Q17. G-CSF stimulates production of:**

- A) Lymphocytes
- B) Erythrocytes
- C) Neutrophils
- D) Platelets

**Answer: C**

**Explanation:** G-CSF = neutrophil production.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: G-CSF = neutrophils.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to G-CSF = neutrophils.

Difficulty: Understanding

**\*\*Q18. Bone-marrow stroma includes reticular cells, adipocytes, and:**

A) Osteoclasts only

B) Sinusoidal endothelial cells and macrophages

C) Cardiac myocytes

D) Hepatocytes

**Answer: B**

**Explanation:** Stroma = reticular cells + adipocytes + endothelial cells + macrophages.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: stroma = reticular + sinusoids.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to stroma = reticular + sinusoids.

Difficulty: Understanding

**\*\*Q19. The earliest identifiable granulocyte precursor is the:**

A) Band cell

B) Promyelocyte

C) Myeloblast

D) Metamyelocyte

**Answer: C**

**Explanation:** Myeloblast = earliest granulocyte precursor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: first = myeloblast.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to first = myeloblast.

Difficulty: Understanding

**\*\*Q20. The yolk sac is active during:**

A) Throughout gestation equally

B) Only after birth

C) The first trimester (from week 3)



D) The third trimester exclusively

**Answer: C**

Explanation: Yolk sac = first trimester (blood islands, week 3).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: yolk sac = first trimester.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to yolk sac = first trimester.

Difficulty: Understanding

**\*\*Q21. Which cell is the largest in the bone marrow?**

- A) Myeloblast
- B) Megakaryocyte
- C) Proerythroblast
- D) Plasma cell

**Answer: B**

Explanation: The megakaryocyte is the largest marrow cell.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: largest = megakaryocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to largest = megakaryocyte.

Difficulty: Understanding

**\*\*Q22. Renal anemia is best explained by:**

- A) Erythropoietin deficiency
- B) Vitamin B12 deficiency
- C) Thrombopoietin excess
- D) Iron deficiency

**Answer: A**

Explanation: Kidney failure EPO deficiency anemia.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: renal anemia = EPO.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to renal anemia = EPO.

Difficulty: Understanding

**\*\*Q23. The last dividing granulocyte precursor is the:**

- A) Band cell

- B) Promyelocyte
- C) Myelocyte
- D) Metamyelocyte

**Answer: B**

Explanation: Myelocyte last dividing stage of granulopoiesis, after that metamyelocyte band segmented non-dividing.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Myelocyte last dividing stage of granulopoiesis, after that metamyelocyte band segmented non-dividing

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Myelocyte last dividing stage of granulopoiesis, after that metamyelocyte band s

Difficulty: Understanding

**\*\*Q24. A leukemoid reaction is distinguished from CML by a:**

- A) Low reticulocyte count
- B) High blast count
- C) Low LAP score
- D) High LAP score

**Answer: D**

Explanation: Leukemoid reaction high LAP score vs CML low LAP score; BCR-ABL negative vs positive.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Leukemoid reaction high LAP score vs CML low LAP score; BCR-ABL negative vs positive.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Leukemoid reaction high LAP score vs CML low LAP score; BCR-ABL negative vs posi

Difficulty: Understanding

## Part C Application

**\*\*Q25. Primary (azurophilic) granules first appear in the:**

- A) Myelocyte
- B) Metamyelocyte
- C) Myeloblast
- D) Promyelocyte

**Answer: D**

Explanation: Azurophilic granules appear at the promyelocyte.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: primary granules = promyelocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to primary granules = promyelocyte.

Difficulty: Application

**\*\*Q26. Megakaryocytes are characterized by:**

- A) A polyploid nucleus via endomitosis
- B) No nucleus
- C) A bilobed nucleus
- D) A diploid nucleus

**Answer: A**

Explanation: Endomitosis polyploid multilobed nucleus.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: megakaryocyte = polyploid.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to megakaryocyte = polyploid.

Difficulty: Application

**\*\*Q27. Acute leukemia is defined by marrow blasts of at least:**

- A) 5%
- B) 20%
- C) 50%
- D) 90%

**Answer: B**

Explanation: 20% blasts (WHO criterion).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: acute leukemia = 20% blasts.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to acute leukemia = 20% blasts.

Difficulty: Application

**\*\*Q28. Thrombopoietin acts primarily on:**

- A) Myeloblasts
- B) Lymphoblasts
- C) Erythroid progenitors
- D) Megakaryocytes

**Answer: D**

Explanation: Thrombopoietin megakaryocytes platelets.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: TPO = megakaryocytes.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to TPO = megakaryocytes.

Difficulty: Application

**\*\*Q29. The first morphologically recognizable erythroid precursor is the:**

- A) Proerythroblast
- B) Orthochromatic normoblast
- C) Erythrocyte
- D) Reticulocyte

**Answer: A**

Explanation: Proerythroblast = first recognizable erythroid cell.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: first = proerythroblast.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to first = proerythroblast.

Difficulty: Application

**\*\*Q30. IL-3 and GM-CSF act on:**

- A) No role in hematopoiesis
- B) Late, lineage-committed stages only
- C) Early multipotent progenitors (multiple lineages)
- D) Post-mitotic cells only

**Answer: C**

Explanation: IL-3/GM-CSF = early, multi-lineage stimulation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: IL-3/GM-CSF = early.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to IL-3/GM-CSF = early.

Difficulty: Application

**\*\*Q31. Thrombopoietin is produced mainly by the:**

- A) Bone-marrow stroma
- B) Liver
- C) Spleen
- D) Kidney

**Answer: B**

**Explanation:** Thrombopoietin = liver (EPO = kidney).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: TPO = liver.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to TPO = liver.

Difficulty: Application

**\*\*Q32. A leukemoid reaction is distinguished from CML by a \_\_\_ LAP score:**

- A) The LAP score is not useful
- B) Absent
- C) Low
- D) High

**Answer: D**

**Explanation:** Leukemoid = high LAP; CML = low LAP.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: LAP: leukemoid high, CML low.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to LAP: leukemoid high, CML low.

Difficulty: Application

**\*\*Q33. In hemolytic anemia, the marrow M:E ratio:**

- A) Stays constant
- B) Becomes 10:1
- C) Increases
- D) Decreases (erythroid hyperplasia)

**Answer: D**

**Explanation:** Erythroid hyperplasia M:E ratio falls below 3:1.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: hemolysis = M:E.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to hemolysis = M:E.

Difficulty: Application

**\*\*Q34. The polyploidy of the megakaryocyte is achieved by:**

- A) Phagocytosis
- B) Meiosis
- C) Endomitosis

D) Cell fusion

**Answer: C**

Explanation: Endomitosis (DNA replication without cytokinesis).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: endomitosis = megakaryocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to endomitosis = megakaryocyte.

Difficulty: Application

**\*\*Q35. A hypocellular, fat-replaced marrow with pancytopenia indicates:**

- A) Myelofibrosis
- B) CML
- C) Acute leukemia
- D) Aplastic anemia

**Answer: D**

Explanation: Aplastic anemia hypocellular fat-replaced marrow pancytopenia due to stem cell failure.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Aplastic anemia hypocellular fat-replaced marrow pancytopenia due to stem cell failure.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Aplastic anemia hypocellular fat-replaced marrow pancytopenia due to stem cell f

Difficulty: Application

**\*\*Q36. Primary (azurophilic) granules first appear at the:**

- A) Promyelocyte
- B) Myelocyte
- C) Metamyelocyte
- D) Myeloblast

**Answer: A**

Explanation: Primary azurophilic granules first appear at promyelocyte, secondary specific granules at myelocyte.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Primary azurophilic granules first appear at promyelocyte, secondary specific granules at myelocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Primary azurophilic granules first appear at promyelocyte, secondary specific gr

Difficulty: Application

## Part D Integration

**\*\*Q37. The last granulocyte precursor capable of mitosis is the:**

- A) Metamyelocyte
- B) Band cell
- C) Promyelocyte
- D) Myelocyte

**Answer: D**

Explanation: The myelocyte is the last dividing stage.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: last mitosis = myelocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to last mitosis = myelocyte.

Difficulty: Integration

**\*\*Q38. During erythropoiesis, cell size:**

- A) Doubles
- B) Increases
- C) Decreases
- D) Stays constant

**Answer: C**

Explanation: Erythroid precursors shrink as they mature.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: size .

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to size .

Difficulty: Integration

**\*\*Q39. Yellow bone marrow is composed mainly of:**

- A) Megakaryocytes
- B) Hematopoietic cells
- C) Adipocytes
- D) Fibroblasts

**Answer: C**

Explanation: Yellow marrow = adipocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: yellow = fat.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to yellow = fat.

Difficulty: Integration

**\*\*Q40. In the second trimester, the primary site of hematopoiesis is the:**

- A) Yolk sac
- B) Liver and spleen
- C) Bone marrow exclusively
- D) Thymus

**Answer: B**

Explanation: Second trimester = liver + spleen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: fetal = liver/spleen.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to fetal = liver/spleen.

Difficulty: Integration

**\*\*Q41. The nucleus is extruded at which erythroid stage?**

- A) It is never extruded
- B) Proerythroblast
- C) Orthochromatic (normoblast) stage
- D) Basophilic erythroblast

**Answer: C**

Explanation: Orthochromatic normoblast extrusion reticulocyte.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: extrusion = orthochromatic.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to extrusion = orthochromatic.

Difficulty: Integration

**\*\*Q42. A left shift in peripheral blood is characterized by increased:**

- A) Segmented neutrophils
- B) Band forms (immature neutrophils)
- C) Lymphocytes
- D) Eosinophils

**Answer: B**

Explanation: Left shift = increased bands (immature neutrophils).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.



C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: left shift = bands.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to left shift = bands.

Difficulty: Integration

**\*\*Q43. Colony-forming-unit assays are named according to the:**

- A) Site of biopsy
- B) Lineage(s) of mature cells the colony produces
- C) Patient's blood type
- D) Size of the colony

**Answer: B**

Explanation: CFU names reflect the lineage (CFU-E, CFU-GM ).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: CFU = lineage.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to CFU = lineage.

Difficulty: Integration

**\*\*Q44. The marrow biopsy that preserves architecture is the:**

- A) Peripheral smear
- B) Bone scan
- C) Fine-needle aspirate
- D) Core (trephine) biopsy

**Answer: D**

Explanation: Core (trephine) biopsy preserves architecture.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: core biopsy = architecture.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to core biopsy = architecture.

Difficulty: Integration

**\*\*Q45. Which correctly orders granulopoiesis?**

- A) Myeloblast metamyelocyte promyelocyte myelocyte
- B) Myeloblast promyelocyte myelocyte metamyelocyte
- C) Promyelocyte myeloblast myelocyte metamyelocyte
- D) Myelocyte promyelocyte metamyelocyte band

**Answer: B**

Explanation: Myeloblast promyelocyte myelocyte metamyelocyte band segmented.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: know the order.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to know the order.

Difficulty: Integration

**\*\*Q46. Basophilia in the peripheral blood is most characteristic of:**

- A) Iron deficiency
- B) Aplastic anemia
- C) Acute bacterial infection
- D) Chronic myeloid leukemia

**Answer: D**

Explanation: CML basophilia (with all-stage granulocytosis).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: basophilia = CML.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to basophilia = CML.

Difficulty: Integration

**\*\*Q47. Teardrop-shaped red cells and splenomegaly suggest:**

- A) Lead poisoning
- B) Iron deficiency
- C) Myelofibrosis
- D) B12 deficiency

**Answer: C**

Explanation: Myelofibrosis marrow fibrosis extramedullary hematopoiesis teardrop RBC splenomegaly.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Myelofibrosis marrow fibrosis extramedullary hematopoiesis teardrop RBC splenomegaly.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Myelofibrosis marrow fibrosis extramedullary hematopoiesis teardrop RBC splenome

Difficulty: Integration

**\*\*Q48. Specific (secondary) granules first appear in the:**

- A) Myeloblast

- B) Promyelocyte
- C) Myelocyte
- D) Band

**Answer: C**

Explanation: Specific granules appear at the myelocyte.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: specific granules = myelocyte.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to specific granules = myelocyte.

Difficulty: High-Difficulty

**\*\*Q49. In the adult, the main site of hematopoiesis is the:**

- A) Bone marrow
- B) Yolk sac
- C) Liver
- D) Spleen

**Answer: A**

Explanation: Adult hematopoiesis = bone marrow.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: adult = marrow.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to adult = marrow.

Difficulty: High-Difficulty

**\*\*Q50. The metamyelocyte nucleus is:**

- A) Indented / kidney-shaped
- B) Fully segmented
- C) Absent
- D) Round with nucleoli

**Answer: A**

Explanation: Metamyelocyte = kidney/indented nucleus (no division).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: metamyelocyte = kidney nucleus.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to metamyelocyte = kidney nucleus.

Difficulty: High-Difficulty

**\*\*Q51. By birth, the primary site of hematopoiesis is the:**

- A) Spleen
- B) Lymph nodes
- C) Liver
- D) Bone marrow

**Answer: D**

Explanation: Birth onward = bone marrow.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: birth = marrow.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to birth = marrow.

Difficulty: High-Difficulty

**\*\*Q52. Megakaryocytes undergo nuclear replication without cell division, called:**

- A) Endomitosis
- B) Meiosis
- C) Amitosis
- D) Mitosis

**Answer: A**

Explanation: Endomitosis = DNA replication without cytokinesis.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: endomitosis = polyploid.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to endomitosis = polyploid.

Difficulty: High-Difficulty

**\*\*Q53. The normal M:E ratio is:**

- A) Approximately 2:1 to 4:1 (myeloid predominant)
- B) 1:10
- C) 10:1 in favor of erythroid
- D) 1:1

**Answer: A**

Explanation: M:E 2:1 4:1 (myeloid dominant).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: M:E = myeloid-dominant.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to M:E = myeloid-dominant.

Difficulty: High-Difficulty

**\*\*Q54. In myelodysplastic syndromes, the marrow typically shows:**

- A) Absence of blasts entirely
- B) Complete aplasia
- C) A hypercellular marrow with ineffective, dysplastic hematopoiesis
- D) Normal maturation

**Answer: C**

Explanation: MDS = hypercellular + ineffective/dysplastic hematopoiesis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: MDS = dysplastic.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to MDS = dysplastic.

Difficulty: High-Difficulty

**\*\*Q55. Which growth factor is correctly paired with its source?**

- A) EPO kidney
- B) Thrombopoietin kidney
- C) G-CSF kidney
- D) EPO liver

**Answer: A**

Explanation: EPO = kidney. TPO = liver; G-CSF = stroma/endothelium.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: EPO = kidney.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to EPO = kidney.

Difficulty: High-Difficulty

**\*\*Q56. The extruded erythroblast nucleus is phagocytosed by:**

- A) Neutrophils
- B) The central macrophage of the erythroblastic island
- C) Platelets
- D) Megakaryocytes

**Answer: B**

Explanation: The island macrophage phagocytoses extruded nuclei.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: macrophage = nucleus disposal.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to macrophage = nucleus disposal.

Difficulty: High-Difficulty

**\*\*Q57. Which statement about hematopoiesis is TRUE?**

- A) It occurs in lymph nodes as the primary site
- B) It is confined to the yolk sac in adults
- C) It shifts from yolk sac liver/spleen bone marrow
- D) It never leaves the bone marrow after birth

**Answer: C**

Explanation: Yolk sac liver/spleen marrow (and extramedullary return in disease).  
Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: hematopoiesis migrates.

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to hematopoiesis migrates.

Difficulty: High-Difficulty

**\*\*Q58. Erythropoietin is produced by the:**

- A) Bone marrow
- B) Liver
- C) Kidney
- D) Spleen

**Answer: C**

Explanation: Erythropoietin produced by peritubular interstitial cells kidney in response to hypoxia, stimulates erythropoiesis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Hemopoiesis section for decisive features.

Key point: Erythropoietin produced by peritubular interstitial cells kidney in response to hypoxia, stimulates

Learning Objective mapping: Chapter 10 Hemopoiesis LO related to Erythropoietin produced by peritubular interstitial cells kidney in response to

Difficulty: High-Difficulty

# Chapter 11 The Immune System & Lymphoid Organs

## Part A Recall

**\*\*Q1. Hassall's (thymic) corpuscles are found in the:**

- A) Tonsil crypts
- B) Lymph node medulla
- C) Thymic medulla
- D) Spleen white pulp

**Answer: C**

Explanation: Hassall's corpuscles = thymic medulla (concentric epithelial whorls).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Hassall's = thymus.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Hassall's = thymus.

Difficulty: Recall

**\*\*Q2. Old and damaged RBCs are removed in the spleen by the:**

- A) Red-pulp cords and sinusoids
- B) Marginal zone only
- C) Central artery
- D) White-pulp PALS

**Answer: A**

Explanation: Red pulp (cords + sinusoids) filters old/damaged RBCs.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: red pulp = RBC filter.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to red pulp = RBC filter.

Difficulty: Recall

**\*\*Q3. DiGeorge syndrome is characterized by absence of the:**

- A) Thymus
- B) Lymph nodes
- C) Tonsils

D) Spleen

**Answer: A**

Explanation: DiGeorge (22q11) = absent thymus (+ parathyroids).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: DiGeorge = no thymus.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to DiGeorge = no thymus.

Difficulty: Recall

**\*\*Q4. Germinal centers are the primary site of:**

- A) Erythropoiesis
- B) Platelet storage
- C) T-cell maturation
- D) B-cell proliferation and affinity maturation

**Answer: D**

Explanation: Germinal centers = B-cell proliferation + affinity maturation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: germinal center = B.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to germinal center = B.

Difficulty: Recall

**\*\*Q5. Hassall's corpuscles are composed of:**

- A) Concentric epithelial reticular cells
- B) Lymphocytes
- C) Macrophages
- D) Plasma cells

**Answer: A**

Explanation: Hassall's = concentric epithelial reticular cells.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Hassall's = epithelial whorls.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Hassall's = epithelial whorls.

Difficulty: Recall

**\*\*Q6. The PALS is composed predominantly of:**

- A) Plasma cells
- B) Macrophages exclusively
- C) B lymphocytes
- D) T lymphocytes

**Answer: D**

Explanation: PALS = T lymphocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: PALS = T.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to PALS = T.

Difficulty: Recall

**\*\*Q7. Natural killer (NK) cells lack:**

- A) Cytoplasmic granules
- B) A nucleus
- C) Any surface receptors
- D) Antigen-specific T- or B-cell receptors

**Answer: D**

Explanation: NK cells lack antigen-specific (T/B) receptors.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: NK = no specific receptor.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to NK = no specific receptor.

Difficulty: Recall

**\*\*Q8. The thymus undergoes involution (adipose replacement) most pronounced:**

- A) It never involutes
- B) Only in disease
- C) In infancy
- D) After puberty, continuing through adult life

**Answer: D**

Explanation: Thymic involution begins after puberty.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: thymus involutes post-puberty.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to thymus involutes post-puberty.

Difficulty: Recall

**\*\*Q9. Which best describes the marginal zone?**

- A) The thymic cortex
- B) The tonsillar crypt
- C) The medulla of the node
- D) The border between white and red pulp (antigen capture)

**Answer: D**

Explanation: Marginal zone = white/red pulp border (antigen capture).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: marginal zone = border.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to marginal zone = border.

Difficulty: Recall

**\*\*Q10. Which is TRUE of the tonsils?**

- A) They contain red and white pulp
- B) They contain Hassall's corpuscles
- C) They have a complete capsule and afferent lymphatics
- D) They have crypts and no afferent lymphatics

**Answer: D**

Explanation: Tonsils = crypts, no afferent lymphatics, partial capsule.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: tonsil = crypts.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to tonsil = crypts.

Difficulty: Recall

**\*\*Q11. A lymphoid organ shows Hassall's corpuscles and no germinal centers. It is the:**

- A) Lymph node
- B) Thymus
- C) Spleen
- D) Tonsil

**Answer: B**

Explanation: Thymus Hassall's corpuscles concentric keratinized epithelial reticular cells, no germinal centers, blood-thymus barrier.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Thymus Hassall's corpuscles concentric keratinized epithelial reticular cells, no germinal centers,

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Thymus Hassall's corpuscles concentric keratinized epithelial reticular cells, n

Difficulty: Recall

**\*\*Q12. Post-splenectomy smears classically show:**

- A) Howell-Jolly bodies
- B) Left shift
- C) Basophilic stippling
- D) Hypersegmented neutrophils

**Answer: A**

Explanation: Post-splenectomy Howell-Jolly bodies nuclear remnants, Heinz bodies, target cells, encapsulated bacteria risk.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Post-splenectomy Howell-Jolly bodies nuclear remnants, Heinz bodies, target cells, encapsulated bact

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Post-splenectomy Howell-Jolly bodies nuclear remnants, Heinz bodies, target cell  
Difficulty: Recall

## Part B Understanding

### **\*\*Q13. The thymus characteristically lacks:**

- A) A medulla
- B) A cortex
- C) Lymphoid follicles / germinal centers
- D) Epithelial cells

#### **Answer: C**

Explanation: The thymus has no germinal centers (no B-cell response).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: thymus = no germinal centers.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to thymus = no germinal centers.

Difficulty: Understanding

### **\*\*Q14. The spleen differs from lymph nodes in that it:**

- A) Has NO afferent lymphatics (filters blood)
- B) Lacks lymphocytes
- C) Has Hassall's corpuscles
- D) Has afferent lymphatics

#### **Answer: A**

Explanation: The spleen has no afferent lymphatics it filters blood.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: spleen = blood filter.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to spleen = blood filter.

Difficulty: Understanding

### **\*\*Q15. Medullary cords of the lymph node contain mainly:**

- A) Plasma cells and macrophages
- B) HEV
- C) Germinal centers

D) Immature T cells

**Answer: A**

Explanation: Medullary cords = plasma cells + macrophages.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: cords = plasma cells.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to cords = plasma cells.

Difficulty: Understanding

**\*\*Q16. The thymus provides the microenvironment for maturation of:**

- A) T lymphocytes
- B) NK cells
- C) Plasma cells
- D) B lymphocytes

**Answer: A**

Explanation: Thymus = T-cell maturation.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: thymus = T.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to thymus = T.

Difficulty: Understanding

**\*\*Q17. The blood thymus barrier separates the cortex from:**

- A) The capsule
- B) Hassall's corpuscles
- C) The medulla
- D) The blood (antigens)

**Answer: D**

Explanation: The barrier shields the cortex from blood-borne antigen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: barrier = antigen shield.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to barrier = antigen shield.

Difficulty: Understanding

**\*\*Q18. In a lymph node, primary follicles of the outer cortex are:**

- A) Composed of macrophages only
- B) T-cell zones
- C) B-cell zones
- D) Devoid of lymphocytes

**Answer: C**

Explanation: Outer cortex follicles = B-cell zones.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: cortex = B.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to cortex = B.

Difficulty: Understanding

**\*\*Q19. Peyer's patches are an example of:**

- A) A primary lymphoid organ
- B) Gut-associated lymphoid tissue (GALT)
- C) The thymus
- D) Bone-marrow-derived tissue

**Answer: B**

Explanation: Peyer's patches = GALT.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Peyer's = GALT.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Peyer's = GALT.

Difficulty: Understanding

**\*\*Q20. Germinal centers develop in response to:**

- A) Antigenic stimulation (active B proliferation/selection)

- B) Absence of antigen
- C) Bacterial toxins exclusively
- D) T-independent antigens only

**Answer: A**

Explanation: Germinal centers = antigen-driven B proliferation.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: germinal center = antigen-driven.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to germinal center = antigen-driven.

Difficulty: Understanding

**\*\*Q21. B cells mature in the:**

- A) Bone marrow
- B) Spleen
- C) Peyer's patches
- D) Thymus

**Answer: A**

Explanation: B-cell maturation = bone marrow.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: B cells = marrow.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to B cells = marrow.

Difficulty: Understanding

**\*\*Q22. Thymic T-cell negative selection occurs mainly in the:**

- A) Blood vessels
- B) Cortex
- C) Medulla
- D) Capsule

**Answer: C**

Explanation: Negative selection (self-tolerance) = medulla.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: negative selection = medulla.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to negative selection = medulla.

Difficulty: Understanding

**\*\*Q23. The T-cell zone of the spleen is the:**

- A) Marginal zone
- B) Red pulp
- C) PALS
- D) Follicle

**Answer: C**

Explanation: Spleen T-cell zone periarteriolar lymphoid sheath PALS around central artery; B-cell zone follicles.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Spleen T-cell zone periarteriolar lymphoid sheath PALS around central artery; B-cell zone follicles.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Spleen T-cell zone periarteriolar lymphoid sheath PALS around central artery; B-

Difficulty: Understanding

**\*\*Q24. Viral infection classically causes:**

- A) Follicular hyperplasia
- B) Paracortical hyperplasia
- C) Red-pulp expansion
- D) Thymic involution

**Answer: B**

Explanation: Viral infection paracortical T-cell hyperplasia; bacterial follicular B-cell hyperplasia.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Viral infection paracortical T-cell hyperplasia; bacterial follicular B-cell hyperplasia.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Viral infection paracortical T-cell hyperplasia; bacterial follicular B-cell hyp

Difficulty: Understanding



## Part C Application

### **\*\*Q25. Germinal centers are a feature of:**

- A) The PALS
- B) Primary follicles
- C) Secondary follicles
- D) The thymus

#### **Answer: C**

Explanation: Secondary follicles (with germinal centers) = active B-cell response.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: germinal center = secondary follicle.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to germinal center = secondary follicle.

Difficulty: Application

### **\*\*Q26. M cells (microfold cells) function to:**

- A) Kill bacteria directly
- B) Secrete IgA
- C) Sample and transport luminal antigens
- D) Produce mucus

#### **Answer: C**

Explanation: M cells transcytose (sample) luminal antigen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: M cell = antigen sampling.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to M cell = antigen sampling.

Difficulty: Application

### **\*\*Q27. Palatine tonsils are histologically identified by:**

- A) PALS
- B) Red pulp
- C) Hassall's corpuscles
- D) Lymphoid follicles with surface crypts

#### **Answer: D**

**Explanation:** Tonsils = follicles + surface crypts.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: tonsil = crypts.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to tonsil = crypts.

Difficulty: Application

**\*\*Q28. HEVs are found in the:**

- A) Thymic medulla
- B) Bone-marrow sinusoids
- C) Splenic red pulp
- D) Lymph-node paracortex

**Answer: D**

**Explanation:** HEVs = node paracortex.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: HEV = node paracortex.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to HEV = node paracortex.

Difficulty: Application

**\*\*Q29. Positive and negative T-cell selection occurs in the thymic:**

- A) Capsule
- B) Septa only
- C) Medulla only
- D) Cortex and corticomedullary junction

**Answer: D**

**Explanation:** Positive (cortex) + negative (medulla/junction) selection.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: selection = cortex + junction.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to selection = cortex + junction.

Difficulty: Application

**\*\*Q30. Afferent lymphatics deliver lymph into a node at the:**

- A) Convex (subcapsular) surface
- B) Medulla directly
- C) Efferent vessel
- D) Hilum

**Answer: A**

Explanation: Afferent = convex surface.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: afferent = convex.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to afferent = convex.

Difficulty: Application

**\*\*Q31. The tonsils are distinguished by:**

- A) Hassall's corpuscles
- B) A complete capsule and afferent lymphatics like nodes
- C) Crypts + epithelial infiltration + no afferent lymphatics
- D) Red and white pulp

**Answer: C**

Explanation: Tonsils = crypts, no afferent lymphatics, partial capsule.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: tonsil = crypts, no afferent lymphatics.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to tonsil = crypts, no afferent lymphatics.

Difficulty: Application

**\*\*Q32. SCID results in profound deficiency of:**

- A) Neutrophils and eosinophils
- B) T and B lymphocyte function
- C) RBCs and platelets
- D) Complement only

**Answer: B**

Explanation: SCID = T + B lymphocyte failure.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: SCID = T + B.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to SCID = T + B.

Difficulty: Application

**\*\*Q33. Which structure is the diagnostic marker of the thymus?**

- A) PALS
- B) Germinal center
- C) Hassall's corpuscle
- D) HEV

**Answer: C**

Explanation: Hassall's corpuscle = thymus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Hassall's = thymus marker.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Hassall's = thymus marker.

Difficulty: Application

**\*\*Q34. Which is a B-cell zone?**

- A) Thymic cortex
- B) Node paracortex
- C) Node outer-cortex follicles
- D) Splenic PALS

**Answer: C**

Explanation: Node outer-cortex follicles = B-cell zone.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: follicles = B.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to follicles = B.

Difficulty: Application

**\*\*Q35. Lymphocytes enter a lymph node from the blood via:**

- A) Efferent lymphatics
- B) The subcapsular sinus
- C) Afferent lymphatics
- D) High endothelial venules

**Answer: D**

Explanation: Lymphocytes enter lymph node from blood via high endothelial venules HEV in paracortex.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Lymphocytes enter lymph node from blood via high endothelial venules HEV in paracortex.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Lymphocytes enter lymph node from blood via high endothelial venules HEV in para

Difficulty: Application

**\*\*Q36. M cells function to:**

- A) Secrete IgA
- B) Sample luminal antigen
- C) Produce mucus
- D) Phagocytose bacteria

**Answer: B**

Explanation: M cells sample luminal antigen transport to underlying lymphoid tissue in Peyer's patches.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: M cells sample luminal antigen transport to underlying lymphoid tissue in Peyer's patches.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to M cells sample luminal antigen transport to underlying lymphoid tissue in Peyer'

Difficulty: Application

## **Part D Integration**

**\*\*Q37. High endothelial venules (HEV) are located in the lymph-node:**

- A) Cortex follicles
- B) Paracortex (T-cell zone)
- C) Medullary cords
- D) Subcapsular sinus

**Answer: B**

Explanation: HEVs are in the paracortex (T zone).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: HEV = paracortex.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to HEV = paracortex.

Difficulty: Integration

**\*\*Q38. Peyer's patches are located mainly in the:**

- A) Stomach
- B) Ileum
- C) Esophagus
- D) Colon only

**Answer: B**

Explanation: Peyer's patches = ileum (GALT).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Peyer's = ileum.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Peyer's = ileum.

Difficulty: Integration

**\*\*Q39. Paracortical hyperplasia of a lymph node is typical of:**

- A) Bacterial infection
- B) Viral infection (T-cell response)
- C) Allergy
- D) Folate deficiency

**Answer: B**

Explanation: Viral infection T-cell (paracortical) hyperplasia.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: paracortical = viral.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to paracortical = viral.

Difficulty: Integration

**\*\*Q40. Primary lymphoid organs include the thymus and:**

- A) Lymph nodes
- B) Tonsils
- C) Spleen
- D) Bone marrow

**Answer: D**

Explanation: Primary = thymus + bone marrow.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: primary = marrow + thymus.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to primary = marrow + thymus.

Difficulty: Integration

**\*\*Q41. The splenic white pulp is responsible for:**

- A) RBC destruction
- B) Immune surveillance of blood-borne antigens
- C) Bile production
- D) Erythropoiesis in adults

**Answer: B**

Explanation: White pulp = immune surveillance (blood-borne antigens).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: white pulp = immune.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to white pulp = immune.

Difficulty: Integration

**\*\*Q42. Efferent lymphatics exit a node at the:**

- A) Convex surface
- B) Hilum
- C) Subcapsular sinus
- D) Cortex

**Answer: B**

Explanation: Efferent = hilum.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: efferent = hilum.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to efferent = hilum.

Difficulty: Integration

**\*\*Q43. Asplenic patients are at risk of encapsulated-bacteria infection because the spleen provides:**

- A) T-cell maturation only
- B) Antibody-mediated clearance and phagocytosis of encapsulated organisms
- C) RBC production
- D) No immune function

**Answer: B**

Explanation: Spleen = clearance + IgM to encapsulated bacteria.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: asplenia = encapsulated risk.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to asplenia = encapsulated risk.

Difficulty: Integration

**\*\*Q44. The mucosal immune system relies heavily on which immunoglobulin in secretions?**

- A) IgE
- B) IgM exclusively
- C) IgG
- D) Secretory IgA

**Answer: D**

Explanation: Secretory IgA = mucosal defense.



A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: mucosal = IgA.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to mucosal = IgA.

Difficulty: Integration

**\*\*Q45. Lymphocytes exit the blood into the node via:**

- A) High endothelial venules
- B) Efferent lymphatics
- C) The subcapsular sinus
- D) Afferent lymphatics

**Answer: A**

Explanation: Lymphocytes enter via HEVs.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: entry = HEV.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to entry = HEV.

Difficulty: Integration

**\*\*Q46. The appendix is a component of:**

- A) The thymus
- B) The spleen
- C) Primary lymphoid organs
- D) GALT (MALT)

**Answer: D**

Explanation: Appendix = GALT.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: appendix = GALT.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to appendix = GALT.

Difficulty: Integration

**\*\*Q47. A newborn with hypocalcemia and absent T cells most likely has:**

- A) DiGeorge syndrome
- B) Splenectomy
- C) Thymoma
- D) SCID

**Answer: A**

Explanation: DiGeorge 22q11 deletion thymic aplasia absent T cells hypocalcemia parathyroid aplasia cardiac anomalies.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: DiGeorge 22q11 deletion thymic aplasia absent T cells hypocalcemia parathyroid aplasia cardiac anomaly

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to DiGeorge 22q11 deletion thymic aplasia absent T cells hypocalcemia parathyroid a

Difficulty: Integration

**\*\*Q48. Peyer's patches are located in the:**

- A) Colon
- B) Stomach
- C) Duodenum
- D) Ileum

**Answer: D**

Explanation: Peyer's patches ileum, part of MALT, B-cell follicles with M cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Peyer's patches ileum, part of MALT, B-cell follicles with M cells.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Peyer's patches ileum, part of MALT, B-cell follicles with M cells.

Difficulty: Integration

## **Part E High-Difficulty**

**\*\*Q49. The periarteriolar lymphoid sheath (PALS) of the spleen is rich in:**

- A) B cells

- B) T cells
- C) Macrophages only
- D) Plasma cells only

**Answer: B**

Explanation: PALS = T cells (around the central artery).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: PALS = T.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to PALS = T.

Difficulty: High-Difficulty

**\*\*Q50. The blood thymus barrier is located in the thymic:**

- A) Cortex
- B) Capsule
- C) Hilum
- D) Medulla

**Answer: A**

Explanation: The blood thymus barrier is in the cortex.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: barrier = cortex.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to barrier = cortex.

Difficulty: High-Difficulty

**\*\*Q51. After splenectomy, RBCs may show:**

- A) Sickle shape
- B) Howell-Jolly bodies and target cells
- C) Teardrop shape
- D) Spherocytes only

**Answer: B**

Explanation: Post-splenectomy Howell-Jolly bodies + target cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: splenectomy = Howell-Jolly.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to splenectomy = Howell-Jolly.

Difficulty: High-Difficulty

**\*\*Q52. Secondary lymphoid organs include the spleen, lymph nodes, and:**

- A) Bone marrow
- B) MALT
- C) Thymic cortex
- D) Yolk sac

**Answer: B**

Explanation: Secondary = node + spleen + MALT.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: secondary = node/spleen/MALT.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to secondary = node/spleen/MALT.

Difficulty: High-Difficulty

**\*\*Q53. The splenic red pulp is responsible for:**

- A) Platelet production
- B) T-cell maturation
- C) Filtration/destruction of aged RBCs
- D) Antibody class-switching only

**Answer: C**

Explanation: Red pulp = RBC filtration/destruction.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: red pulp = RBC filter.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to red pulp = RBC filter.

Difficulty: High-Difficulty

**\*\*Q54. Follicular dendritic cells in germinal centers function to:**

- A) Produce antibodies
- B) Present antigen to T cells in the thymus
- C) Trap and display antigen to B cells (affinity maturation)
- D) Phagocytose RBCs

**Answer: C**

Explanation: FDCs display antigen to B cells    affinity maturation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: FDC = antigen display to B.

Learning Objective mapping: Chapter 11    The Immune System & Lymphoid Organs    LO related to FDC = antigen display to B.

Difficulty: High-Difficulty

**\*\*Q55. Interdigitating dendritic cells (antigen-presenting to T cells) are concentrated in the:**

- A) B-cell follicles
- B) Paracortex of lymph nodes
- C) Splenic red pulp
- D) Thymic cortex only

**Answer: B**

Explanation: Interdigitating dendritic cells = paracortex (T zone).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: IDC = paracortex.

Learning Objective mapping: Chapter 11    The Immune System & Lymphoid Organs    LO related to IDC = paracortex.

Difficulty: High-Difficulty

**\*\*Q56. The spleen's central artery is surrounded by:**

- A) Medullary cords
- B) Red pulp
- C) PALS (white pulp)
- D) Hassall's corpuscles

**Answer: C**

Explanation: The central artery is surrounded by the PALS.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: central artery + PALS.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to central artery + PALS.

Difficulty: High-Difficulty

**\*\*Q57. A node's subcapsular sinus receives:**

- A) Afferent lymph
- B) Arterial blood
- C) Venous blood
- D) Efferent lymph

**Answer: A**

Explanation: Subcapsular sinus receives afferent lymph.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: subcapsular = afferent.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to subcapsular = afferent.

Difficulty: High-Difficulty

**\*\*Q58. Which infection pattern enlarges the T-cell zone?**

- A) Encapsulated bacteria
- B) Viral infection
- C) Toxoid response
- D) Allergic reaction

**Answer: B**

Explanation: Viral infection T-cell (paracortical) enlargement. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: viral = T-zone.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to viral = T-zone.

Difficulty: High-Difficulty

**\*\*Q59. The spleen filters:**

- A) Bile
- B) Lymph
- C) Blood
- D) CSF

**Answer: C**

Explanation: Spleen filters blood no afferent lymphatics; lymph node filters lymph.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Spleen filters blood no afferent lymphatics; lymph node filters lymph.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Spleen filters blood no afferent lymphatics; lymph node filters lymph.

Difficulty: High-Difficulty

**\*\*Q60. Germinal centers are the site of:**

- A) B-cell affinity maturation
- B) RBC destruction
- C) Platelet production
- D) T-cell maturation

**Answer: A**

Explanation: Germinal centers B-cell affinity maturation somatic hypermutation class switching.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Immune System & Lymphoid Organs section for decisive features.

Key point: Germinal centers B-cell affinity maturation somatic hypermutation class switching.

Learning Objective mapping: Chapter 11 The Immune System & Lymphoid Organs LO related to Germinal centers B-cell affinity maturation somatic hypermutation class switchin

Difficulty: High-Difficulty

# Chapter 12 Digestive Tract

## Part A Recall

**\*\*Q1. The submucosal (Meissner's) plexus is located in the:**

- A) Submucosa
- B) Muscularis externa
- C) Serosa
- D) Mucosa

**Answer: A**

Explanation: Meissner's plexus = submucosa (secretion).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Meissner = submucosa.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Meissner = submucosa.

Difficulty: Recall

**\*\*Q2. Peyer's patches are most prominent in the:**

- A) Duodenum
- B) Jejunum
- C) Ileum
- D) Colon

**Answer: C**

Explanation: Peyer's patches = ileum.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Peyer's = ileum.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Peyer's = ileum.

Difficulty: Recall

**\*\*Q3. Barrett esophagus shows replacement of squamous epithelium by:**

- A) Transitional epithelium
- B) Stratified squamous keratinized
- C) Intestinal-type columnar epithelium with goblet cells
- D) Pseudostratified epithelium

**Answer: C**

Explanation: Barrett = intestinal columnar metaplasia (goblet cells).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Barrett = goblet cells.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Barrett = goblet cells.

Difficulty: Recall

**\*\*Q4. Parietal cells are recognized by their:**

- A) Mucous-filled apical cytoplasm
- B) Argentaffin granules
- C) Basophilic cytoplasm
- D) Eosinophilic "fried-egg" appearance with abundant mitochondria

**Answer: D**

Explanation: Parietal = eosinophilic, mitochondria-rich.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: parietal = mitochondria.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to parietal = mitochondria.

Difficulty: Recall

**\*\*Q5. Chief cells secrete:**

- A) Gastrin
- B) HCl and intrinsic factor
- C) Pepsinogen
- D) Mucus exclusively

**Answer: C**

Explanation: Chief = pepsinogen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: chief = pepsinogen.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to chief = pepsinogen.

Difficulty: Recall

**\*\*Q6. Crypts of Lieberkühn are found in the:**

- A) Esophagus
- B) Duodenal submucosa
- C) Stomach only
- D) Small and large intestine

**Answer: D**

Explanation: Crypts of Lieberkühn = small + large intestine.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: crypts = intestines.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to crypts = intestines.

Difficulty: Recall

**\*\*Q7. The stomach's third, innermost oblique muscle layer aids:**

- A) Mechanical churning and mixing
- B) Sphincter closure only
- C) Absorption
- D) Peristalsis of liquids only

**Answer: A**

Explanation: Oblique layer = churning/mixing.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: oblique = mixing.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to oblique = mixing.

Difficulty: Recall

**\*\*Q8. The muscularis mucosae functions to:**

- A) Produce local mucosal movements (secretion/absorption)
- B) Form the primary peristaltic wave
- C) Anchor the tract
- D) Propel food through the tract

**Answer: A**

Explanation: Muscularis mucosae = local mucosal movements.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: muscularis mucosae = local.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to muscularis mucosae = local.

Difficulty: Recall

**\*\*Q9. The duodenum's alkaline mucus neutralizes acid via:**

- A) Parietal cells
- B) Paneth cells
- C) Brunner's glands
- D) Goblet cells

**Answer: C**

**Explanation:** Brunner's glands    alkaline mucus/bicarbonate.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Brunner's = alkaline.

Learning Objective mapping: Chapter 12    Digestive Tract    LO related to Brunner's = alkaline.

Difficulty: Recall

**\*\*Q10. H. pylori gastritis is characterized by:**

- A) Granulomas
- B) Goblet-cell metaplasia
- C) Neutrophils in the lamina propria/epithelium
- D) Villous atrophy

**Answer: C**

**Explanation:** H. pylori gastritis = neutrophilic infiltrate.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: H. pylori = neutrophils.

Learning Objective mapping: Chapter 12    Digestive Tract    LO related to H. pylori = neutrophils.

Difficulty: Recall

**\*\*Q11. A biopsy shows villi and mucous glands in the submucosa. This is the:**

- A) Ileum
- B) Duodenum
- C) Jejunum
- D) Colon

**Answer: B**

**Explanation:** Duodenum villi + Brunner's glands mucous alkaline submucosa protects from acid; jejunum no Brunner, ileum Peyer's patches.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Duodenum villi + Brunner's glands mucous alkaline submucosa protects from acid; jejunum no Brunner,

Learning Objective mapping: Chapter 12    Digestive Tract    LO related to Duodenum villi + Brunner's glands mucous alkaline submucosa protects from acid;

Difficulty: Recall

**\*\*Q12. Meissner's plexus is located in the:**

- A) Mucosa
- B) Submucosa
- C) Muscularis externa
- D) Serosa

**Answer: B**

Explanation: Meissner's submucosal plexus parasympathetic secretion; Auerbach's myenteric plexus between circular and longitudinal muscle motility.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Meissner's submucosal plexus parasympathetic secretion; Auerbach's myenteric plexus between circular

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Meissner's submucosal plexus parasympathetic secretion; Auerbach's myenteric plexus

Difficulty: Recall

## Part B Understanding

**\*\*Q13. The epithelium of the esophagus is:**

- A) Simple columnar
- B) Stratified squamous non-keratinized
- C) Pseudostratified ciliated
- D) Transitional

**Answer: B**

Explanation: Esophagus = stratified squamous non-keratinized.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: esophagus = squamous.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to esophagus = squamous.

Difficulty: Understanding

**\*\*Q14. Paneth cells are located at the:**

- A) Base of the crypts of Lieberk hn
- B) Submucosa
- C) Muscularis mucosae
- D) Villus tip

**Answer: A**

Explanation: Paneth cells = crypt base.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Paneth = crypt base.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Paneth = crypt base.

Difficulty: Understanding

**\*\*Q15. The muscularis mucosae is a component of the:**

- A) Submucosa
- B) Mucosa
- C) Muscularis externa
- D) Serosa

**Answer: B**

Explanation: Muscularis mucosae = part of the mucosa.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: muscularis mucosae = mucosa.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to muscularis mucosae = mucosa.

Difficulty: Understanding

**\*\*Q16. The muscularis mucosae lies:**

- A) In the serosa
- B) Between the mucosa and submucosa
- C) Surrounding the GI tract externally
- D) Within the lamina propria only

**Answer: B**

Explanation: Muscularis mucosae = between mucosa and submucosa.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: muscularis mucosae = mucosa-submucosa border.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to muscularis mucosae = mucosa-submucosa border.

Difficulty: Understanding

**\*\*Q17. Gastrin is produced by G cells in the:**

- A) Esophagus
- B) Ileum
- C) Fundic glands
- D) Pyloric antrum

**Answer: D**

Explanation: G cells = pyloric antrum (gastrin).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: G cells = antrum.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to G cells = antrum.

Difficulty: Understanding

**\*\*Q18. Enteroendocrine cells release their granules toward the:**

- A) Apical surface
- B) Basal surface (lamina propria)
- C) Crypt base
- D) Lumen only

**Answer: B**

Explanation: Enteroendocrine cells = basal release (toward lamina propria/blood).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: enteroendocrine = basal.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to enteroendocrine = basal.

Difficulty: Understanding

**\*\*Q19. Teniae coli are thickened bands of:**

- A) Muscularis mucosae
- B) Submucosa
- C) Circular muscle
- D) Longitudinal muscle

**Answer: D**

Explanation: Teniae coli = 3 longitudinal muscle bands.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: taeniae = longitudinal.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to taeniae = longitudinal.

Difficulty: Understanding

**\*\*Q20. Chief-cell zymogen granules and parietal-cell mitochondria are features of glands in the gastric:**

- A) Cardia
- B) Fundus and body
- C) Pylorus
- D) Antrum exclusively

**Answer: B**

Explanation: Fundus/body = parietal + chief cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: fundus/body = both.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to fundus/body = both.

Difficulty: Understanding

**\*\*Q21. Which is TRUE of the stomach's surface epithelium?**

- A) Simple columnar (mucous cells)
- B) Pseudostratified
- C) Transitional
- D) Stratified squamous

**Answer: A**

Explanation: Stomach surface = simple columnar mucous cells.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: stomach = simple columnar.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to stomach = simple columnar.

Difficulty: Understanding

**\*\*Q22. Which cells secrete gastrin?**

- A) Parietal cells
- B) G cells
- C) Chief cells
- D) Paneth cells

**Answer: B**

Explanation: G cells = gastrin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: G cell = gastrin.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to G cell = gastrin.

Difficulty: Understanding

**\*\*Q23. Pink "fried-egg" cells in gastric glands are:**

- A) G cells
- B) Paneth cells

- C) Chief cells
- D) Parietal cells

**Answer: D**

Explanation: Parietal cells pink fried-egg central nucleus eosinophilic HCl + intrinsic factor; chief cells basophilic pepsinogen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Parietal cells pink fried-egg central nucleus eosinophilic HCl + intrinsic factor; chief cells basophilic pepsinogen

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Parietal cells pink fried-egg central nucleus eosinophilic HCl + intrinsic factor

Difficulty: Understanding

**\*\*Q24. Transmural inflammation with non-caseating granulomas is characteristic of:**

- A) Celiac disease
- B) Barrett esophagus
- C) Ulcerative colitis
- D) Crohn's disease

**Answer: D**

Explanation: Crohn's transmural inflammation non-caseating granulomas skip lesions, cobblestoning, anywhere GI mouth to anus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Crohn's transmural inflammation non-caseating granulomas skip lesions, cobblestoning, anywhere GI mouth to anus

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Crohn's transmural inflammation non-caseating granulomas skip lesions, cobblestoning

Difficulty: Understanding

## Part C Application

**\*\*Q25. Parietal cells are characterized by:**

- A) Basophilic cytoplasm
- B) Eosinophilic "fried-egg" cytoplasm
- C) Mucin-filled cytoplasm
- D) Dark-staining granules

**Answer: B**

Explanation: Parietal = eosinophilic "fried-egg" (mitochondria).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: parietal = pink.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to parietal = pink.

Difficulty: Application

**\*\*Q26. The large intestine differs from the small intestine by:**

- A) Having Brunner's glands
- B) Lacking goblet cells
- C) Having taller villi
- D) Lacking villi

**Answer: D**

Explanation: Colon = no villi.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: no villi = colon.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to no villi = colon.

Difficulty: Application

**\*\*Q27. Intrinsic factor is secreted by:**

- A) G cells
- B) Mucous neck cells
- C) Chief cells
- D) Parietal cells

**Answer: D**

Explanation: Parietal cells = HCl + intrinsic factor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: IF = parietal.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to IF = parietal.

Difficulty: Application

**\*\*Q28. Paneth cells function in:**

- A) Hormone secretion
- B) Mucus production
- C) Nutrient absorption
- D) Innate immune defense (lysozyme/defensins)

**Answer: D**

Explanation: Paneth cells = innate defense.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Paneth = defense.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Paneth = defense.

Difficulty: Application

**\*\*Q29. Plicae circulares are most prominent in the:**

- A) Esophagus
- B) Stomach
- C) Jejunum
- D) Colon

**Answer: C**

Explanation: Plicae circulares = jejunum (prominent).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: plicae = jejunum.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to plicae = jejunum.

Difficulty: Application

**\*\*Q30. The large intestine is distinguished by the absence of:**

- A) Villi
- B) Crypts of Lieberkühn
- C) A muscularis externa
- D) Goblet cells

**Answer: A**

Explanation: Colon = no villi.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: colon = no villi.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to colon = no villi.

Difficulty: Application

**\*\*Q31. The appendix wall is characterized by:**

- A) Heavy lymphoid (GALT) infiltration
- B) Squamous epithelium
- C) Absent muscularis mucosae
- D) Extensive muscularis thickening

**Answer: A**

Explanation: Appendix = lymphoid-rich wall.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: appendix = GALT.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to appendix = GALT.

Difficulty: Application

**\*\*Q32. Hirschsprung disease = absence of ganglion cells in the:**

- A) Stomach only
- B) Mucosa only
- C) Myenteric and submucosal plexuses of the distal colon
- D) Small intestine exclusively

**Answer: C**

Explanation: Hirschsprung = aganglionic distal colon (both plexuses).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Hirschsprung = no ganglia.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Hirschsprung = no ganglia.

Difficulty: Application

**\*\*Q33. Ulcerative colitis characteristically shows:**

- A) Villous atrophy
- B) Aganglionosis
- C) Transmural inflammation with granulomas
- D) Mucosal inflammation with crypt abscesses

**Answer: D**

Explanation: UC = mucosal + crypt abscesses.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: UC = mucosal + crypt abscesses.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to UC = mucosal + crypt abscesses.

Difficulty: Application

**\*\*Q34. The ileum is the primary site of:**

- A) Brunner's glands
- B) Peyer's patches
- C) Taenia coli

D) Esophageal glands

**Answer: B**

Explanation: Ileum = Peyer's patches.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: ileum = Peyer's.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to ileum = Peyer's.

Difficulty: Application

**\*\*Q35. The colon is distinguished by:**

A) Brunner's glands

B) Peyer's patches

C) Tall villi

D) Absence of villi

**Answer: D**

Explanation: Colon distinguished by absence of villi, presence of crypts, goblet cells, taenia coli; small intestine has villi.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Colon distinguished by absence of villi, presence of crypts, goblet cells, taenia coli; small intestine

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Colon distinguished by absence of villi, presence of crypts, goblet cells, taeni

Difficulty: Application

**\*\*Q36. Chief (zymogenic) cells secrete:**

A) Pepsinogen

B) Intrinsic factor

C) Gastrin

D) HCl

**Answer: A**

Explanation: Chief cells = pepsinogen.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: chief = pepsinogen.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to chief = pepsinogen.

Difficulty: Integration

**\*\*Q37. Goblet cells are most abundant in the:**

- A) Duodenum
- B) Stomach
- C) Colon
- D) Esophagus

**Answer: C**

Explanation: Colon = goblet-rich.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: colon = goblets.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to colon = goblets.

Difficulty: Integration

**\*\*Q38. Taenia coli are a feature of the:**

- A) Esophagus
- B) Small intestine
- C) Large intestine
- D) Stomach

**Answer: C**

Explanation: Taenia coli = colon.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: taenia coli = colon.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to taenia coli = colon.

Difficulty: Integration

**\*\*Q39. The esophagus is lined along most of its length by:**

- A) Pseudostratified columnar
- B) Simple columnar
- C) Stratified squamous non-keratinized
- D) Transitional

**Answer: C**

Explanation: Esophagus = stratified squamous non-keratinized.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: esophagus = squamous.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to esophagus = squamous.

Difficulty: Integration

**\*\*Q40. Villi are a distinguishing feature of the:**

- A) Stomach
- B) Small intestine
- C) Colon
- D) Esophagus

**Answer: B**

Explanation: Villi = small intestine.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: villi = small intestine.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to villi = small intestine.

Difficulty: Integration

**\*\*Q41. Goblet cells increase progressively from:**

- A) Duodenum to rectum (colon has the most)
- B) Rectum to duodenum
- C) They remain constant
- D) They are absent in the colon

**Answer: A**

Explanation: Goblet cells from duodenum colon (most).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: goblets distally.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to goblets distally.

Difficulty: Integration

**\*\*Q42. Antigen sampling in the gut is mediated by:**

- A) Chief cells
- B) Paneth cells
- C) M (microfold) cells
- D) Goblet cells

**Answer: C**

Explanation: M cells = antigen sampling.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: M cell = sampler.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to M cell = sampler.

Difficulty: Integration

**\*\*Q43. Celiac disease shows:**

- A) Villous atrophy + crypt hyperplasia + intraepithelial lymphocytes
- B) Normal villi
- C) Loss of goblet cells only
- D) Villous hypertrophy

**Answer: A**

Explanation: Celiac = villous atrophy + crypt hyperplasia + IELs.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: celiac = villous atrophy.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to celiac = villous atrophy.

Difficulty: Integration

**\*\*Q44. Crohn's disease characteristically shows:**

- A) Crypt abscesses only
- B) Transmural inflammation + non-caseating granulomas
- C) Mucosal inflammation only
- D) Villous atrophy

**Answer: B**

Explanation: Crohn's = transmural + granulomas.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Crohn's = transmural + granulomas.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Crohn's = transmural + granulomas.

Difficulty: Integration

**\*\*Q45. Paneth-cell granules contain:**

- A) HCl
- B) Lysozyme and defensins
- C) Pepsinogen
- D) Intrinsic factor

**Answer: B**

Explanation: Paneth granules = lysozyme + defensins.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Paneth = lysozyme/defensins.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Paneth = lysozyme/defensins.

Difficulty: Integration

**\*\*Q46. Villous atrophy with crypt hyperplasia and intraepithelial lymphocytes indicates:**

- A) Hirschsprung disease
- B) Crohn's disease
- C) Celiac disease
- D) Ulcerative colitis

**Answer: C**

Explanation: Celiac disease villous atrophy crypt hyperplasia intraepithelial lymphocytes gluten-sensitive, anti-tTG.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Celiac disease villous atrophy crypt hyperplasia intraepithelial lymphocytes gluten-sensitive, anti-

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Celiac disease villous atrophy crypt hyperplasia intraepithelial lymphocytes glu

Difficulty: Integration

**\*\*Q47. The upper third of the esophageal muscularis is:**

- A) Mixed
- B) Smooth muscle
- C) Skeletal muscle
- D) Cardiac muscle

**Answer: C**

Explanation: Esophagus upper third skeletal muscle, middle mixed, lower third smooth muscle; Auerbach and Meissner throughout.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Esophagus upper third skeletal muscle, middle mixed, lower third smooth muscle; Auerbach and Meissner

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Esophagus upper third skeletal muscle, middle mixed, lower third smooth muscle;

Difficulty: Integration

**Part E High-Difficulty**



**\*\*Q48. Brunner's glands are a diagnostic feature of the:**

- A) Jejunum
- B) Ileum
- C) Stomach
- D) Duodenum

**Answer: D**

Explanation: Brunner's = duodenal submucosa.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Brunner's = duodenum.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Brunner's = duodenum.

Difficulty: High-Difficulty

**\*\*Q49. The jejunum is distinguished by:**

- A) Peyer's patches
- B) Tall villi without Brunner's or Peyer's
- C) Taeniae coli
- D) Brunner's glands

**Answer: B**

Explanation: Jejunum = tall villi, no Brunner's/Peyer's.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: jejunum = neither.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to jejunum = neither.

Difficulty: High-Difficulty

**\*\*Q50. The appendix is histologically characterized by:**

- A) Brunner's glands
- B) Taeniae coli
- C) Tall villi
- D) Abundant lymphoid tissue

**Answer: D**

Explanation: Appendix = lymphoid-rich (GALT).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: appendix = lymphoid.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to appendix = lymphoid.

Difficulty: High-Difficulty

**\*\*Q51. The stomach mucosa secretes a protective:**

- A) Alkaline mucous layer
- B) Intrinsic factor uniformly
- C) Pepsinogen uniformly
- D) Hydrochloric acid layer

**Answer: A**

Explanation: Surface mucous cells    alkaline mucus (protective layer).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: stomach = mucous barrier.

Learning Objective mapping: Chapter 12    Digestive Tract    LO related to stomach = mucous barrier.

Difficulty: High-Difficulty

**\*\*Q52. Brunner's glands are located in the:**

- A) Colon
- B) Jejunum
- C) Duodenum (submucosa)
- D) Ileum

**Answer: C**

Explanation: Brunner's = duodenal submucosa.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Brunner's = duodenum.

Learning Objective mapping: Chapter 12    Digestive Tract    LO related to Brunner's = duodenum.

Difficulty: High-Difficulty

**\*\*Q53. The muscularis externa has an inner circular and outer:**

- A) Oblique layer only
- B) Radial layer
- C) Circular layer
- D) Longitudinal layer

**Answer: D**

Explanation: Muscularis externa = inner circular + outer longitudinal.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: inner circular + outer longitudinal.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to inner circular + outer longitudinal.

Difficulty: High-Difficulty

**\*\*Q54. Achalasia results from degeneration of neurons in the:**

- A) Myenteric (Auerbach's) plexus
- B) Vagus nerve exclusively
- C) Dorsal root ganglia
- D) Meissner's plexus only

**Answer: A**

Explanation: Achalasia = myenteric (Auerbach's) plexus degeneration.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: achalasia = Auerbach's.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to achalasia = Auerbach's.

Difficulty: High-Difficulty

**\*\*Q55. Which cell has a brush border?**

- A) Paneth cell
- B) Chief cell
- C) Goblet cell
- D) Enterocyte

**Answer: D**

Explanation: Enterocyte = brush border (microvilli).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: enterocyte = brush border.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to enterocyte = brush border.

Difficulty: High-Difficulty

**\*\*Q56. The muscularis externa of the upper esophagus is:**

- A) Skeletal muscle
- B) Cardiac muscle
- C) Mixed only at the pylorus
- D) Smooth muscle

**Answer: A**

Explanation: Upper esophagus = skeletal muscle.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: upper esophagus = skeletal.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to upper esophagus = skeletal.

Difficulty: High-Difficulty

**\*\*Q57. The appendix most closely resembles which organ histologically?**

- A) Stomach
- B) Duodenum
- C) Colon (with abundant lymphoid tissue)
- D) Ileum only

**Answer: C**

Explanation: Appendix = colon-like + lymphoid-rich. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: appendix colon + GALT.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to appendix colon + GALT.

Difficulty: High-Difficulty

**\*\*Q58. Which cells are located at the crypt base with eosinophilic granules?**

- A) Chief cells
- B) Goblet cells
- C) Paneth cells
- D) Enterocytes

**Answer: C**

Explanation: Paneth cells crypt base eosinophilic granules lysozyme defensins, regulate microbiome.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Digestive Tract section for decisive features.

Key point: Paneth cells crypt base eosinophilic granules lysozyme defensins, regulate microbiome.

Learning Objective mapping: Chapter 12 Digestive Tract LO related to Paneth cells crypt base eosinophilic granules lysozyme defensins, regulate micro

Difficulty: High-Difficulty

# Chapter 13      Organs Associated with the Digestive Tract

## Part A      Recall

**\*\*Q1. The parotid gland is predominantly:**

- A) Sebaceous
- B) Mucous
- C) Serous
- D) Mixed

**Answer: C**

Explanation: Parotid = purely serous.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: parotid = serous.

Learning Objective mapping: Chapter 13      Organs Associated with the Digestive Tract      LO related to parotid = serous.

Difficulty: Recall

**\*\*Q2. Blood in the liver flows from the:**

- A) Central vein to portal triad
- B) Portal triad to central vein
- C) Bile duct to sinusoid
- D) Hepatic vein to portal vein

**Answer: B**

Explanation: Blood = portal triad      central vein.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: blood      central vein.

Learning Objective mapping: Chapter 13      Organs Associated with the Digestive Tract      LO related to blood      central vein.

Difficulty: Recall

**\*\*Q3. Serous demilunes are characteristic of:**

- A) Parotid
- B) Submandibular (mixed) glands
- C) Pancreas

D) Liver

**Answer: B**

Explanation: Demilunes = mixed (submandibular) glands.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: demilunes = submandibular.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to demilunes = submandibular.

Difficulty: Recall

**\*\*Q4. Zone 1 (periportal) hepatocytes are most susceptible to:**

- A) Directly-acting toxins delivered via portal blood
- B) Acetaminophen toxicity
- C) Alcohol-related fatty change
- D) Ischemia/hypoxia

**Answer: A**

Explanation: Zone 1 first receives portal blood    direct toxins.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Zone 1 = direct toxins.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to Zone 1 = direct toxins.

Difficulty: Recall

**\*\*Q5. The liver acinus is organized around the:**

- A) Gallbladder
- B) Central vein
- C) Line connecting two portal triads (blood flow from that axis)
- D) Bile canaliculus

**Answer: C**

Explanation: Acinus = the portal-triad axis (blood supply/metabolism).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: acinus = triad axis.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to acinus = triad axis.

Difficulty: Recall

**\*\*Q6. Hepatic sinusoids differ from typical capillaries by being:**

- A) Continuous with a complete basal lamina
- B) Fenestrated/discontinuous, lacking a complete basal lamina
- C) Devoid of Kupffer cells
- D) Lined entirely by hepatocytes

**Answer: B**

Explanation: Sinusoids = discontinuous, incomplete basal lamina.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: sinusoid = discontinuous.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to sinusoid = discontinuous.

Difficulty: Recall

**\*\*Q7. Islets of Langerhans are most numerous in the:**

- A) Tail
- B) Uncinate process
- C) Distributed evenly
- D) Head

**Answer: A**

Explanation: Islets = tail.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: islets = tail.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to islets = tail.

Difficulty: Recall

**\*\*Q8. Kupffer cells are located adherent to the:**

- A) Portal-vein wall exclusively

- B) Hepatocyte plates
- C) Bile canaliculi
- D) Sinusoidal endothelium (luminal)

**Answer: D**

Explanation: Kupffer = adherent to sinusoidal endothelium.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Kupffer = sinusoid lining.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to Kupffer = sinusoid lining.

Difficulty: Recall

**\*\*Q9. The hepatic artery and portal vein both empty into the:**

- A) Sinusoids
- B) Bile ducts
- C) Space of Disse
- D) Central vein

**Answer: A**

Explanation: Both vessels    sinusoids.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: artery + vein    sinusoid.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to artery + vein    sinusoid.

Difficulty: Recall

**\*\*Q10. Which is TRUE of the space of Disse?**

- A) It is inside the bile canaliculus
- B) It contains Kupffer cells only
- C) It contains bile
- D) It is between hepatocytes and sinusoidal endothelium

**Answer: D**

Explanation: Space of Disse = between hepatocytes and sinusoid.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.



features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Disse = hepatocyte-sinusoid gap.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Disse = hepatocyte-sinusoid gap.

Difficulty: Recall

**\*\*Q11. A gland shows centroacinar cells and no myoepithelial cells. It is the:**

- A) Pancreas
- B) Submandibular
- C) Sublingual
- D) Parotid

**Answer: A**

Explanation: Pancreas exocrine centroacinar cells (ductal) no myoepithelial; salivary glands have myoepithelial.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Pancreas exocrine centroacinar cells (ductal) no myoepithelial; salivary glands have myoepithelial.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Pancreas exocrine centroacinar cells (ductal) no myoepithelial; salivary glands

Difficulty: Recall

**\*\*Q12. Glucagon is secreted by:**

- A) cells
- B) cells
- C) PP cells
- D) cells

**Answer: A**

Explanation: Alpha cells peripheral islet glucagon; beta central insulin; delta somatostatin; PP pancreatic polypeptide.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Alpha cells peripheral islet glucagon; beta central insulin; delta somatostatin; PP pancreatic polyp

Learning Objective mapping: Chapter 13   Organs Associated with the Digestive Tract   LO related to Alpha cells peripheral islet glucagon; beta central insulin; delta somatostatin;  
Difficulty: Recall

## Part B   Understanding

### **\*\*Q13. Which structure is diagnostic of the exocrine pancreas?**

- A) Striated ducts only
- B) Myoepithelial cells
- C) Centroacinar cells
- D) Serous demilunes

#### **Answer: C**

Explanation: Centroacinar cells = pancreas.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: centroacinar = pancreas.

Learning Objective mapping: Chapter 13   Organs Associated with the Digestive Tract   LO related to centroacinar = pancreas.

Difficulty: Understanding

### **\*\*Q14. The hepatic zone most vulnerable to ischemia and toxins is:**

- A) Zone 1 (periportal)
- B) Zone 3 (centrilobular)
- C) Zone 2
- D) The portal triad

#### **Answer: B**

Explanation: Zone 3 (centrilobular, low O<sub>2</sub>) is most vulnerable.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Zone 3 = vulnerable.

Learning Objective mapping: Chapter 13   Organs Associated with the Digestive Tract   LO related to Zone 3 = vulnerable.

Difficulty: Understanding

### **\*\*Q15. Vitamin A is stored in the liver by:**

- A) Hepatic stellate (Ito) cells
- B) Hepatocytes only
- C) Endothelial cells

D) Kupffer cells

**Answer: A**

Explanation: Stellate (Ito) cells store vitamin A.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Ito = vitamin A.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Ito = vitamin A.

Difficulty: Understanding

**\*\*Q16. Kupffer cells are located within the:**

- A) Portal triad
- B) Hepatic sinusoids
- C) Central vein
- D) Bile canaliculi

**Answer: B**

Explanation: Kupffer cells = sinusoid lumen.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Kupffer = sinusoid.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Kupffer = sinusoid.

Difficulty: Understanding

**\*\*Q17. Hepatic stellate cells normally store:**

- A) Bile
- B) Vitamin A (retinoids)
- C) Iron exclusively
- D) Glycogen only

**Answer: B**

Explanation: Stellate cells store vitamin A.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Ito = vitamin A.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Ito = vitamin A.

Difficulty: Understanding

**\*\*Q18. The gallbladder concentrates bile by transporting:**

- A) Bile acids out of the lumen
- B) Sodium (with water following osmotically)
- C) Bilirubin into the lumen
- D) Cholesterol into the lumen

**Answer: B**

Explanation: Na transport water follows concentration.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: gallbladder = Na pump.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to gallbladder = Na pump.

Difficulty: Understanding

**\*\*Q19. Beta cells secrete:**

- A) Somatostatin
- B) Pancreatic polypeptide
- C) Glucagon
- D) Insulin

**Answer: D**

Explanation: cells = insulin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: = insulin.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to = insulin.

Difficulty: Understanding

**\*\*Q20. Chronic alcohol use produces fatty change most prominently in:**

- A) Zone 3 (centrilobular)

- B) Zone 2 only
- C) All zones equally
- D) Zone 1

**Answer: A**

Explanation: Alcoholic fatty change = Zone 3.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: alcohol = Zone 3.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to alcohol = Zone 3.

Difficulty: Understanding

**\*\*Q21. Which is TRUE of the hepatocyte's orientation?**

- A) It has no polarity
- B) It faces blood on one side and bile on the other
- C) It faces bile on both sides
- D) It faces blood on both sides

**Answer: B**

Explanation: Hepatocytes are polarized: sinusoidal (blood) pole + canalicular (bile) pole.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: hepatocyte = dual polarity.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to hepatocyte = dual polarity.

Difficulty: Understanding

**\*\*Q22. Cholecystitis classically shows:**

- A) Prominent Rokitansky-Aschoff sinuses and inflammation
- B) Centroacinar cells
- C) Regenerative nodules
- D) Villous atrophy

**Answer: A**

Explanation: Cholecystitis    prominent Rokitansky-Aschoff sinuses.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: cholecystitis = Rokitansky-Aschoff.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to cholecystitis = Rokitansky-Aschoff.

Difficulty: Understanding

**\*\*Q23. The portal triad contains:**

- A) Hepatic vein, artery, lymphatic
- B) Bile duct, central vein, artery
- C) Central vein, sinusoid, hepatocyte
- D) Hepatic artery, portal vein, bile duct

**Answer: D**

Explanation: Portal triad: hepatic artery (small thick), portal vein (large thin), bile duct (cuboidal epithelium).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Portal triad: hepatic artery (small thick), portal vein (large thin), bile duct (cuboidal epithelium)

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Portal triad: hepatic artery (small thick), portal vein (large thin), bile duct

Difficulty: Understanding

**\*\*Q24. Cirrhosis results from fibrosis driven by:**

- A) Endothelial cells
- B) Kupffer cells
- C) Stellate (Ito) cells
- D) Hepatocytes

**Answer: C**

Explanation: Cirrhosis fibrosis driven by stellate Ito cells vitamin A storage activated to myofibroblast collagen production.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Cirrhosis fibrosis driven by stellate Ito cells vitamin A storage activated to myofibroblast collagen

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Cirrhosis fibrosis driven by stellate Ito cells vitamin A storage activated to m

Difficulty: Understanding

## Part C Application

**\*\*Q25. The pancreas differs from salivary glands by lacking:**

- A) Myoepithelial cells
- B) Ducts
- C) Zymogen granules
- D) Acini

**Answer: A**

Explanation: The pancreas has no myoepithelial cells.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: pancreas = no myoepithelial.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to pancreas = no myoepithelial.

Difficulty: Application

**\*\*Q26. Kupffer cells are:**

- A) Stellate cells
- B) Hepatocytes
- C) Macrophages in the sinusoids
- D) Bile-duct cells

**Answer: C**

Explanation: Kupffer = sinusoidal macrophages.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Kupffer = macrophage.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to Kupffer = macrophage.

Difficulty: Application

**\*\*Q27. The exocrine pancreas secretes enzymes from:**

- A) Centroacinar cells
- B) Islet beta cells
- C) Acinar cells
- D) Ductal cells

**Answer: C**

Explanation: Acinar cells secrete the zymogens.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: acinar cells = enzymes.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to acinar cells = enzymes.

Difficulty: Application

**\*\*Q28. Activation of hepatic stellate cells drives:**

- A) Bile secretion
- B) Portal hypertension without fibrosis
- C) Hepatocyte regeneration
- D) Liver fibrosis

**Answer: D**

Explanation: Stellate activation    collagen    fibrosis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: stellate = fibrosis.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to stellate = fibrosis.

Difficulty: Application

**\*\*Q29. Exocrine acinar cells show basal basophilia due to:**

- A) Mitochondria
- B) Rough endoplasmic reticulum
- C) Lysosomes
- D) Peroxisomes

**Answer: B**

Explanation: Acinar basal basophilia = RER.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.



Key point: acinar RER = blue.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to acinar RER = blue.

Difficulty: Application

**\*\*Q30. Alpha cells secrete:**

- A) Glucagon
- B) Somatostatin
- C) Gastrin
- D) Insulin

**Answer: A**

Explanation:    cells = glucagon.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point:    = glucagon.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to = glucagon.

Difficulty: Application

**\*\*Q31. Gallstones most commonly form from:**

- A) Water
- B) Bilirubin exclusively
- C) Bile acids
- D) Cholesterol

**Answer: D**

Explanation: Most gallstones = cholesterol.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: stones = cholesterol.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to stones = cholesterol.

Difficulty: Application

**\*\*Q32. Insulin is released by cells located in the islet:**

- A) Center
- B) Entirely scattered
- C) Capsule

D) Periphery

**Answer: A**

Explanation: (insulin) cells = islet center.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: = center.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to = center.

Difficulty: Application

**\*\*Q33. Which is a serous gland?**

- A) Goblet cell
- B) Sublingual
- C) Parotid
- D) Gallbladder

**Answer: C**

Explanation: Parotid = serous.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: parotid = serous.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to parotid = serous.

Difficulty: Application

**\*\*Q34. Bile flows toward the:**

- A) Central vein
- B) Portal triad (bile duct)
- C) Hepatic vein
- D) Sinusoid

**Answer: B**

Explanation: Bile flows toward portal triad bile duct (opposite blood flow toward central vein).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Bile flows toward portal triad bile duct (opposite blood flow toward central vein).

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Bile flows toward portal triad bile duct (opposite blood flow toward central vein)

Difficulty: Application

**\*\*Q35. The sublingual gland is predominantly:**

- A) Serous
- B) Mucous
- C) Mixed
- D) Endocrine

**Answer: B**

Explanation: Sublingual predominantly mucous, submandibular mixed, parotid serous.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Sublingual predominantly mucous, submandibular mixed, parotid serous.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Sublingual predominantly mucous, submandibular mixed, parotid serous.

Difficulty: Application

## **Part D Integration**

**\*\*Q36. Insulin is secreted by:**

- A) cells
- B) PP cells
- C) cells
- D) cells

**Answer: D**

Explanation: cells (central, most numerous).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: insulin = .

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to insulin = .

Difficulty: Integration

**\*\*Q37. The gallbladder is unique in lacking:**

- A) Mucosa
- B) Submucosa and muscularis mucosae
- C) Muscularis
- D) Serosa

**Answer: B**

Explanation: Gallbladder = no submucosa, no muscularis mucosae.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: gallbladder = no submucosa.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to gallbladder = no submucosa.

Difficulty: Integration

**\*\*Q38. Bile canaliculi carry bile toward the:**

- A) Portal triad (bile duct)
- B) Hepatic vein
- C) Sinusoid
- D) Central vein

**Answer: A**

Explanation: Bile flows toward the portal triad.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: bile    triad.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to bile    triad.

Difficulty: Integration

**\*\*Q39. The classic hepatic lobule is organized around a:**

- A) Hepatic artery
- B) Portal triad
- C) Central vein
- D) Bile duct

**Answer: C**

Explanation: Classic lobule = central vein (center).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: lobule center = central vein.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to lobule center = central vein.

Difficulty: Integration

**\*\*Q40. Bile canaliculi are formed by:**

- A) Kupffer cells
- B) The space of Disse
- C) A separate duct system
- D) Grooves between adjacent hepatocyte membranes

**Answer: D**

Explanation: Canaliculi = grooves between hepatocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: canaliculus = hepatocyte groove.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to canaliculus = hepatocyte groove.

Difficulty: Integration

**\*\*Q41. Centroacinar cells mark the beginning of the:**

- A) Common bile duct
- B) Islet of Langerhans
- C) Intercalated duct
- D) Interlobular duct

**Answer: C**

Explanation: Centroacinar = intercalated-duct origin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: centroacinar = intercalated duct.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to centroacinar = intercalated duct.

Difficulty: Integration

**\*\*Q42. Acute pancreatitis results from premature activation of:**

- A) Glucagon
- B) Bile acids
- C) Insulin
- D) Trypsinogen to trypsin

**Answer: D**

Explanation: Premature trypsinogen → trypsin → autodigestion.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: pancreatitis = trypsin.

Learning Objective mapping: Chapter 13 → Organs Associated with the Digestive Tract → LO related to pancreatitis = trypsin.

Difficulty: Integration

**\*\*Q43. Which cell has striated ducts?**

- A) Pancreas
- B) Salivary glands
- C) Liver
- D) Gallbladder

**Answer: B**

Explanation: Striated ducts = salivary glands (ion transport).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: striated duct = salivary.

Learning Objective mapping: Chapter 13 → Organs Associated with the Digestive Tract → LO related to striated duct = salivary.

Difficulty: Integration

**\*\*Q44. Which cell type is responsible for liver fibrosis?**

- A) Endothelial cell
- B) Kupffer cell
- C) Stellate (Ito) cell
- D) Hepatocyte

**Answer: C**

Explanation: Stellate cells → collagen → fibrosis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: stellate = fibrosis.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to stellate = fibrosis.

Difficulty: Integration

**\*\*Q45. The hepatic acinus Zone 1 is characterized by:**

- A) Absence of hepatocytes
- B) Only bile ducts
- C) Lowest oxygen
- D) Highest oxygen (periportal)

**Answer: D**

Explanation: Zone 1 = periportal, highest O<sub>2</sub>.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Zone 1 = high O<sub>2</sub>.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to Zone 1 = high O<sub>2</sub>.

Difficulty: Integration

**\*\*Q46. The hepatic zone most vulnerable to ischemia and acetaminophen is:**

- A) Zone 1
- B) Zone 3
- C) Zone 2
- D) The portal triad

**Answer: B**

Explanation: Zone 3 perivenular centrilobular oxygen-poor vulnerable to ischemia acetaminophen centrilobular necrosis; Zone 1 periportal vulnerable to bile, oxygen-rich.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Zone 3 perivenular centrilobular oxygen-poor vulnerable to ischemia acetaminophen centrilobular necr

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Zone 3 perivenular centrilobular oxygen-poor vulnerable to ischemia acetaminophe

Difficulty: Integration

**\*\*Q47. Acute pancreatitis is caused by:**

- A) Bile-duct obstruction only
- B) Glucagon excess
- C) Insulin overproduction
- D) Premature trypsinogen activation

**Answer: D**

Explanation: Acute pancreatitis premature trypsinogen activation autodigestion, often gallstone or alcohol.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Acute pancreatitis premature trypsinogen activation autodigestion, often gallstone or alcohol.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Acute pancreatitis premature trypsinogen activation autodigestion, often gallsto

Difficulty: Integration

## **Part E High-Difficulty**

**\*\*Q48. The portal triad consists of:**

- A) Hepatic vein, artery, lymphatic
- B) Bile duct, central vein, artery
- C) Central vein, sinusoid, hepatocyte
- D) Hepatic artery, portal vein, bile duct

**Answer: D**

Explanation: Portal triad = hepatic artery + portal vein + bile duct.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: triad = artery + vein + duct.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to triad = artery + vein + duct.

Difficulty: High-Difficulty



**\*\*Q49. Rokitansky-Aschoff sinuses are found in the:**

- A) Stomach
- B) Liver
- C) Gallbladder
- D) Pancreas

**Answer: C**

Explanation: Rokitansky-Aschoff sinuses = gallbladder.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Rokitansky-Aschoff = gallbladder.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to Rokitansky-Aschoff = gallbladder.

Difficulty: High-Difficulty

**\*\*Q50. The portal lobule is centered on the:**

- A) Portal vein exclusively
- B) Central vein
- C) Bile duct of the portal triad
- D) Hepatic artery only

**Answer: C**

Explanation: Portal lobule = bile duct (bile drainage).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: portal lobule = bile duct.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to portal lobule = bile duct.

Difficulty: High-Difficulty

**\*\*Q51. The space of Disse is the site of:**

- A) Bile-canalculus formation
- B) Bile flow
- C) Plasma filtration and nutrient exchange with hepatocytes
- D) RBC destruction exclusively

**Answer: C**

Explanation: Space of Disse = exchange between sinusoid and hepatocyte.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Disse = exchange.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to Disse = exchange.

Difficulty: High-Difficulty

**\*\*Q52. Pancreatic enzymes are stored as zymogens to prevent:**

- A) Enzyme degradation
- B) Autodigestion of the pancreas
- C) Enzyme denaturation
- D) Excess insulin release

**Answer: B**

Explanation: Zymogens prevent autodigestion.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: zymogen = protection.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to zymogen = protection.

Difficulty: High-Difficulty

**\*\*Q53. Cirrhosis is defined by diffuse fibrosis with:**

- A) A single tumor mass
- B) Absence of hepatocytes
- C) Normal lobules only
- D) Regenerative nodules surrounded by fibrous septa

**Answer: D**

Explanation: Cirrhosis = fibrosis + regenerative nodules.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: cirrhosis = nodules + fibrosis.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to cirrhosis = nodules + fibrosis.

Difficulty: High-Difficulty

**\*\*Q54. Mumps classically affects the:**

- A) Parotid gland
- B) Sublingual gland
- C) Pancreas
- D) Submandibular gland

**Answer: A**

Explanation: Mumps = parotid inflammation.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: mumps = parotid.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to mumps = parotid.

Difficulty: High-Difficulty

**\*\*Q55. The bile duct in a portal triad is lined by:**

- A) Transitional epithelium
- B) Endothelium only
- C) Squamous epithelium
- D) Cuboidal epithelium

**Answer: D**

Explanation: Bile duct = cuboidal epithelium.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: bile duct = cuboidal.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to bile duct = cuboidal.

Difficulty: High-Difficulty

**\*\*Q56. Which pair is correctly matched regarding pancreatic islet cells and hormones?**

- A) cell    insulin
- B) cell    glucagon
- C) PP cell    somatostatin
- D) cell    insulin

**Answer: A**

Explanation: cell = insulin. Practice Test

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: = insulin.

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to = insulin.

Difficulty: High-Difficulty

**\*\*Q57. The gallbladder lacks:**

- A) Submucosa and muscularis mucosae
- B) Muscularis
- C) Serosa
- D) Mucosa

**Answer: A**

Explanation: Gallbladder lacks submucosa and muscularis mucosae, has mucosa, muscularis, serosa/adventitia, Rokitansky-Aschoff sinuses.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Gallbladder lacks submucosa and muscularis mucosae, has mucosa, muscularis, serosa/adventitia, Rokit

Learning Objective mapping: Chapter 13 Organs Associated with the Digestive Tract LO related to Gallbladder lacks submucosa and muscularis mucosae, has mucosa, muscularis, sero

Difficulty: High-Difficulty

**\*\*Q58. Most gallstones are composed of:**

- A) Protein
- B) Bilirubin
- C) Cholesterol
- D) Calcium

**Answer: C**

Explanation: Most gallstones cholesterol, mixed; pigment bilirubin in hemolysis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Organs Associated with the Digestive Tract section for decisive features.

Key point: Most gallstones cholesterol, mixed; pigment bilirubin in hemolysis.

Learning Objective mapping: Chapter 13    Organs Associated with the Digestive Tract    LO related to  
Most gallstones cholesterol, mixed; pigment bilirubin in hemolysis.

Difficulty: High-Difficulty

# Chapter 14 The Respiratory System

## Part A Recall

**\*\*Q1. The trachea is characterized by:**

- A) C-shaped hyaline cartilage rings
- B) No cartilage
- C) Elastic cartilage only
- D) Irregular cartilage plates

**Answer: A**

Explanation: Trachea = C-shaped hyaline rings.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: trachea = C-rings.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to trachea = C-rings.

Difficulty: Recall

**\*\*Q2. The cell type covering ~95% of the alveolar surface is:**

- A) club cell
- B) Type II pneumocyte
- C) Type I pneumocyte
- D) Alveolar macrophage

**Answer: C**

Explanation: Type I = ~95% of the surface.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type I = 95% surface.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type I = 95% surface.

Difficulty: Recall

**\*\*Q3. Alveolar macrophages are also called:**

- A) Goblet cells
- B) club cells
- C) Dust cells
- D) Type II cells

**Answer: C**

Explanation: Alveolar macrophages = dust cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: dust cell = macrophage.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to dust cell = macrophage.

Difficulty: Recall

**\*\*Q4. Type II pneumocytes secrete:**

- A) Histamine
- B) Elastase
- C) Mucus
- D) Surfactant

**Answer: D**

Explanation: Type II = surfactant.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type II = surfactant.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type II = surfactant.

Difficulty: Recall

**\*\*Q5. The true vocal cords are covered by stratified squamous epithelium to:**

- A) Increase mucus
- B) Withstand mechanical stress (vibration)
- C) Gas exchange
- D) Increase ciliary clearance

**Answer: B**

Explanation: Vocal cords = stratified squamous (mechanical stress).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: vocal cords = squamous.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to vocal cords = squamous.

Difficulty: Recall

**\*\*Q6. Gas exchange begins at the:**

- A) Larynx
- B) Segmental bronchi
- C) Respiratory bronchioles

D) Trachea

**Answer: C**

Explanation: Respiratory bronchioles = first gas exchange.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: respiratory bronchiole = exchange begins.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to respiratory bronchiole = exchange begins.

Difficulty: Recall

**\*\*Q7. Emphysema causes:**

A) Increased surface area

B) Permanent airspace enlargement with surface-area loss

C) Alveolar fibrosis

D) Normal architecture

**Answer: B**

Explanation: Emphysema = enlarged airspaces, lost surface area.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: emphysema = surface loss.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to emphysema = surface loss.

Difficulty: Recall

**\*\*Q8. In chronic bronchitis, the Reid index is:**

A) Unchanged

B) Not applicable

C) Decreased

D) Increased (gland hypertrophy)

**Answer: D**

Explanation: Chronic bronchitis Reid index (gland hypertrophy).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: chronic bronchitis = Reid index.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to chronic bronchitis = Reid index.

Difficulty: Recall



**\*\*Q9. The trachealis muscle is located:**

- A) In the adventitia
- B) In the lamina propria
- C) Anterior to the cartilage
- D) Posteriorly, bridging the C-shaped ring

**Answer: D**

Explanation: Trachealis = posterior, bridging the C.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: trachealis = posterior.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to trachealis = posterior.

Difficulty: Recall

**\*\*Q10. Which structure has hyaline cartilage in C-shaped form?**

- A) Bronchus
- B) Trachea
- C) Bronchiole
- D) Alveolar duct

**Answer: B**

Explanation: Trachea = C-shaped hyaline cartilage.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: trachea = C-shaped.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to trachea = C-shaped.

Difficulty: Recall

**\*\*Q11. An airway with no cartilage and club cells is a:**

- A) Bronchiole
- B) Bronchus
- C) Alveolar duct
- D) Trachea

**Answer: A**

Explanation: Bronchiole no cartilage no glands club cells more smooth muscle; bronchus has cartilage plates and glands.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Bronchiole no cartilage no glands club cells more smooth muscle; bronchus has cartilage plates and g

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Bronchiole no cartilage no glands club cells more smooth muscle; bronchus has ca

Difficulty: Recall

**\*\*Q12. Bronchi differ from bronchioles in that bronchi have:**

- A) Goblet cells
- B) Smooth muscle
- C) Cartilage plates and glands
- D) Ciliated epithelium

**Answer: C**

Explanation: Bronchi = cartilage plates + glands; bronchioles = none.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: bronchi = cartilage.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to bronchi = cartilage.

Difficulty: Understanding

**\*\*Q13. Type II pneumocytes are characterized by:**

- A) Cuboidal shape with lamellar bodies
- B) Cilia
- C) Mucin granules
- D) Flat squamous shape

**Answer: A**

Explanation: Type II = cuboidal + lamellar bodies.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type II = lamellar bodies.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type II = lamellar bodies.

Difficulty: Understanding

**\*\*Q14. "Heart-failure cells" are found in the:**

- A) Trachea
- B) Pleura
- C) Bronchi
- D) Alveoli

**Answer: D**

Explanation: Hemosiderin-laden macrophages = alveoli.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: heart-failure cells = alveoli.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to heart-failure cells = alveoli.

Difficulty: Understanding

**\*\*Q15. Cartilage support is lost at the level of the:**

- A) Larynx
- B) Trachea
- C) Bronchioles
- D) Main bronchi

**Answer: C**

Explanation: Cartilage ends at the bronchioles.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: bronchioles = no cartilage.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to bronchioles = no cartilage.

Difficulty: Understanding

**\*\*Q16. The blood air barrier includes type I cytoplasm, fused basal lamina, and:**

- A) Type II cytoplasm
- B) Capillary endothelial cytoplasm
- C) Smooth muscle
- D) Cartilage

**Answer: B**

Explanation: Barrier = Type I + fused BL + endothelium (+ surfactant).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: barrier = endothelium.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to barrier = endothelium.

Difficulty: Understanding

**\*\*Q17. Cartilage support becomes progressively:**

- A) More abundant distally
- B) Reduced to plates, then absent in bronchioles

- C) Unchanged
- D) Absent even in the trachea

**Answer: B**

Explanation: Cartilage    distally    absent in bronchioles.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: cartilage    distally.

Learning Objective mapping: Chapter 14    The Respiratory System    LO related to cartilage distally.

Difficulty: Understanding

**\*\*Q18. The pleura is lined by:**

- A) Endothelium
- B) Mesothelium
- C) Endocardium
- D) Urothelium

**Answer: B**

Explanation: Pleura = mesothelium.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: pleura = mesothelium.

Learning Objective mapping: Chapter 14    The Respiratory System    LO related to pleura = mesothelium.

Difficulty: Understanding

**\*\*Q19. Ciliary beat serves:**

- A) Surfactant production
- B) Sound production
- C) Gas exchange
- D) Mucociliary clearance (mucus toward the pharynx)

**Answer: D**

Explanation: Cilia = mucociliary clearance.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: cilia = clearance.

Learning Objective mapping: Chapter 14    The Respiratory System    LO related to cilia = clearance.

Difficulty: Understanding

**\*\*Q20. Which is TRUE of the terminal bronchiole?**

- A) It has cartilage
- B) It has glands
- C) It has alveoli
- D) It is the last conducting airway (no alveoli)

**Answer: D**

Explanation: Terminal bronchiole = last conducting (no alveoli, no cartilage).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: terminal = last conducting.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to terminal = last conducting.

Difficulty: Understanding

**\*\*Q21. Hyaline membranes lining atelectatic alveoli characterize:**

- A) NRDS
- B) Emphysema
- C) Pneumonia
- D) Asthma

**Answer: A**

Explanation: Hyaline membranes + atelectasis = NRDS.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: hyaline membranes = NRDS.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to hyaline membranes = NRDS.

Difficulty: Understanding

**\*\*Q22. Surfactant is produced by:**

- A) Type II pneumocytes
- B) Dust cells
- C) club cells
- D) Type I pneumocytes

**Answer: A**

Explanation: Type II pneumocytes cuboidal surfactant + progenitor for type I; type I flat gas exchange 95% surface.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type II pneumocytes cuboidal surfactant + progenitor for type I; type I flat gas exchange 95% surfac

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type II pneumocytes cuboidal surfactant + progenitor for type I; type I flat gas

Difficulty: Understanding

**\*\*Q23. Emphysema is characterized by:**

- A) Smooth-muscle hypertrophy
- B) Hemosiderin macrophages
- C) Alveolar-wall destruction
- D) Hyaline membranes

**Answer: C**

Explanation: Emphysema alveolar wall destruction loss elastic recoil, smoking, centrilobular or panacinar.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Emphysema alveolar wall destruction loss elastic recoil, smoking, centrilobular or panacinar.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Emphysema alveolar wall destruction loss elastic recoil, smoking, centrilobular

Difficulty: Understanding

## **Part C Application**

**\*\*Q24. Bronchioles are defined by the absence of:**

- A) Cartilage and glands
- B) Epithelium
- C) club cells
- D) Smooth muscle

**Answer: A**

Explanation: Bronchioles = no cartilage, no glands.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: bronchiole = no cartilage.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to bronchiole = no cartilage.

Difficulty: Application

**\*\*Q25. The blood air barrier includes all EXCEPT:**

- A) Fused basal laminae
- B) Smooth muscle
- C) Surfactant layer
- D) Type I pneumocyte

**Answer: B**

Explanation: The barrier has no smooth muscle.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: barrier = no muscle.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to barrier = no muscle.

Difficulty: Application

**\*\*Q26. The olfactory epithelium contains:**

- A) club cells
- B) Paneth cells
- C) Type I pneumocytes
- D) Bipolar olfactory neurons

**Answer: D**

Explanation: Olfactory epithelium = bipolar sensory neurons.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: olfactory = bipolar neurons.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to olfactory = bipolar neurons.

Difficulty: Application

**\*\*Q27. club cells are notable for:**

- A) Cartilage formation
- B) Gas exchange
- C) Xenobiotic metabolism and surfactant-like secretion
- D) Mucus production only

**Answer: C**

Explanation: club cells = detox + surfactant-like secretion.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: club cell = detox.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to club cell = detox.

Difficulty: Application

**\*\*Q28. Alveolar macrophages derive from:**

- A) Type II pneumocytes

- B) Lymphocytes
- C) Monocytes
- D) Neutrophils

**Answer: C**

Explanation: Dust cells = monocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: macrophage = monocyte.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to macrophage monocyte.

Difficulty: Application

**\*\*Q29. Pores of Kohn allow:**

- A) RBC passage
- B) Blood flow between alveoli
- C) Collateral air circulation
- D) Lymphatic drainage only

**Answer: C**

Explanation: Pores of Kohn = collateral air circulation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: pores of Kohn = collateral air.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to pores of Kohn = collateral air.

Difficulty: Application

**\*\*Q30. BALT is found in the mucosa of the:**

- A) Bronchi
- B) Trachea only
- C) Nasal cavity only
- D) Alveoli

**Answer: A**

Explanation: BALT = bronchi (mucosa).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: BALT = bronchi.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to BALT = bronchi.

Difficulty: Application



**\*\*Q31. Pulmonary fibrosis impairs gas exchange by:**

- A) Cartilage loss
- B) Surfactant excess
- C) Airway obstruction
- D) Thickening the blood air barrier

**Answer: D**

Explanation: Fibrosis thickens the barrier      diffusion.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: fibrosis = thick barrier.

Learning Objective mapping: Chapter 14    The Respiratory System    LO related to fibrosis = thick barrier.

Difficulty: Application

**\*\*Q32. A brown, Prussian-blue-positive macrophage in the alveolus is a:**

- A) Type II cell
- B) Dust cell
- C) Heart-failure cell
- D) club cell

**Answer: C**

Explanation: Hemosiderin (Prussian-blue positive) = heart-failure cell.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: heart-failure cell = hemosiderin.

Learning Objective mapping: Chapter 14    The Respiratory System    LO related to heart-failure cell = hemosiderin.

Difficulty: Application

**\*\*Q33. The mucociliary escalator fails in:**

- A) Kartagener syndrome (immotile cilia)
- B) Emphysema
- C) Fibrosis
- D) NRDS

**Answer: A**

Explanation: Kartagener = immotile cilia    failed clearance.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Kartagener = failed escalator.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Kartagener = failed escalator.

Difficulty: Application

**\*\*Q34. A premature infant with atelectasis and hyaline membranes has:**

- A) Emphysema
- B) Bronchitis
- C) Asthma
- D) NRDS

**Answer: D**

Explanation: NRDS neonatal respiratory distress syndrome surfactant deficiency premature infant atelectasis hyaline membranes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: NRDS neonatal respiratory distress syndrome surfactant deficiency premature infant atelectasis hyali

Learning Objective mapping: Chapter 14 The Respiratory System LO related to NRDS neonatal respiratory distress syndrome surfactant deficiency premature infa

Difficulty: Application

**\*\*Q35. club cells are found in the:**

- A) Bronchioles
- B) Alveoli
- C) Nasal cavity
- D) Trachea

**Answer: A**

Explanation: club cells = bronchioles.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: club cell = bronchiole.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to club cell = bronchiole.

Difficulty: Integration

**\*\*Q36. The respiratory epithelium of the trachea is:**

- A) Simple columnar
- B) Simple squamous
- C) Pseudostratified ciliated columnar with goblet cells
- D) Stratified squamous

**Answer: C**

Explanation: Respiratory epithelium = pseudostratified ciliated columnar + goblet.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: airway = pseudostratified ciliated.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to airway = pseudostratified ciliated.

Difficulty: Integration

**\*\*Q37. Neonatal RDS is due to surfactant deficiency from:**

- A) club cells
- B) Goblet cells
- C) Type I pneumocytes
- D) Type II pneumocytes

**Answer: D**

Explanation: NRDS = immature Type II no surfactant.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: NRDS = Type II.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to NRDS = Type II.

Difficulty: Integration

**\*\*Q38. The respiratory epithelium transitions to simple cuboidal at the:**

- A) Alveoli directly
- B) Terminal bronchioles
- C) Larynx only
- D) Trachea only

**Answer: B**

Explanation: Terminal bronchioles simple cuboidal (with club cells).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: terminal bronchiole = cuboidal.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to terminal bronchiole = cuboidal.

Difficulty: Integration

**\*\*Q39. Pulmonary surfactant is composed mainly of:**

- A) Phospholipids (dipalmitoylphosphatidylcholine)
- B) Collagen
- C) Elastin

D) Mucus glycoproteins

**Answer: A**

Explanation: Surfactant = phospholipid (DPPC).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: surfactant = DPPC.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to surfactant = DPPC.

Difficulty: Integration

**\*\*Q40. Type I pneumocytes are characterized by:**

A) Ciliated surfaces

B) Cuboidal shape with lamellar bodies

C) Extremely thin, flattened squamous morphology for diffusion

D) High mitotic activity

**Answer: C**

Explanation: Type I = thin, flat, diffusion-optimized.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type I = flat.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type I = flat.

Difficulty: Integration

**\*\*Q41. Asthma shows smooth-muscle hypertrophy, goblet hyperplasia, and:**

A) Cartilage loss

B) Pleural fibrosis only

C) Alveolar-wall destruction

D) Thickened basement membrane + eosinophils

**Answer: D**

Explanation: Asthma = muscle + goblet + eosinophils + thickened BM.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: asthma = muscle + eosinophils.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to asthma = muscle + eosinophils.

Difficulty: Integration

**\*\*Q42. The primary stimulus for increased respiratory rate is:**

- A) Blood oxygen
- B) Blood carbon dioxide (via CSF pH)
- C) Blood pH via peripheral receptors only
- D) Hemoglobin concentration

**Answer: B**

Explanation: CO<sub>2</sub> (via central chemoreceptors) is the primary drive.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: CO<sub>2</sub> = respiratory drive.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to CO<sub>2</sub> = respiratory drive.

Difficulty: Integration

**\*\*Q43. Which is a protease/antiprotease imbalance disease?**

- A) Emphysema (1-antitrypsin deficiency)
- B) NRDS
- C) Bronchitis
- D) Asthma

**Answer: A**

Explanation: Emphysema = elastase vs 1-antitrypsin imbalance.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: emphysema = 1-antitrypsin.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to emphysema = 1-antitrypsin.

Difficulty: Integration

**\*\*Q44. Which is correctly paired regarding pulmonary cells and surfactant?**

- A) club cell surfactant only
- B) Type I surfactant
- C) Type II surfactant
- D) Dust cell surfactant

**Answer: C**

Explanation: Type II = surfactant.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type II = surfactant.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type II = surfactant.

Difficulty: Integration

**\*\*Q45. The cell covering ~95% of the alveolar surface is the:**

- A) club cell
- B) Type I pneumocyte
- C) Type II pneumocyte
- D) Dust cell

**Answer: B**

Explanation: Type I pneumocyte flat 95% surface gas exchange; Type II cuboidal surfactant progenitor.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type I pneumocyte flat 95% surface gas exchange; Type II cuboidal surfactant progenitor.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type I pneumocyte flat 95% surface gas exchange; Type II cuboidal surfactant pro

Difficulty: Integration

**\*\*Q46. Hemosiderin-laden alveolar macrophages indicate:**

- A) NRDS
- B) Emphysema
- C) Asthma
- D) Heart failure (chronic congestion)

**Answer: D**

Explanation: Heart failure chronic congestion hemosiderin-laden macrophages heart failure cells in alveoli.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Heart failure chronic congestion hemosiderin-laden macrophages heart failure cells in alveoli.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Heart failure chronic congestion hemosiderin-laden macrophages heart failure c

Difficulty: Integration

## **Part E High-Difficulty**

**\*\*Q47. The first structures of the respiratory portion are the:**

- A) Terminal bronchioles
- B) Respiratory bronchioles
- C) Bronchi

D) Alveolar ducts

**Answer: B**

Explanation: Respiratory bronchioles = first with alveoli.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: respiratory bronchiole = first.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to respiratory bronchiole = first.

Difficulty: High-Difficulty

**\*\*Q48. Emphysema is characterized histologically by:**

- A) Pleural fibrosis
- B) Excess cartilage
- C) Destruction of alveolar walls
- D) Goblet-cell metaplasia only

**Answer: C**

Explanation: Emphysema = alveolar-wall destruction.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: emphysema = wall destruction.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to emphysema = wall destruction.

Difficulty: High-Difficulty

**\*\*Q49. The olfactory epithelium contains bipolar neurons and:**

- A) Sustentacular and basal cells
- B) Chief cells
- C) Parietal cells
- D) Goblet cells

**Answer: A**

Explanation: Olfactory = bipolar neurons + sustentacular + basal cells.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: olfactory = bipolar + sustentacular + basal.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to olfactory = bipolar + sustentacular + basal.

Difficulty: High-Difficulty

**\*\*Q50. NRDS results from insufficient surfactant due to immaturity of:**

- A) Alveolar macrophages
- B) club cells
- C) Type I pneumocytes
- D) Type II pneumocytes

**Answer: D**

Explanation: NRDS = immature Type II.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: NRDS = Type II.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to NRDS = Type II.

Difficulty: High-Difficulty

**\*\*Q51. Type II pneumocytes are more numerous than type I because they are:**

- A) Larger
- B) Smaller cuboidal cells that are also progenitors
- C) Non-dividing
- D) Absent in normal lung

**Answer: B**

Explanation: Type II = smaller cuboidal + progenitors (more numerous).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type II = more numerous + progenitor.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type II = more numerous + progenitor.

Difficulty: High-Difficulty

**\*\*Q52. The nasal conchae function to:**

- A) Warm and humidify inhaled air
- B) Perform gas exchange
- C) Detect odors only
- D) Produce surfactant

**Answer: A**

Explanation: Conchae = warm + humidify air.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: conchae = air conditioning.



Learning Objective mapping: Chapter 14 The Respiratory System LO related to conchae = air conditioning.

Difficulty: High-Difficulty

**\*\*Q53. Which cell regenerates the alveolar epithelium after injury?**

- A) Type I
- B) Type II (progenitor)
- C) Dust cell
- D) club cell only

**Answer: B**

Explanation: Type II cells regenerate both types.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Type II = progenitor.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Type II = progenitor.

Difficulty: High-Difficulty

**\*\*Q54. The lamellar bodies of Type II cells contain:**

- A) Lysozyme
- B) Melanin
- C) Mucin
- D) Surfactant phospholipid

**Answer: D**

Explanation: Lamellar bodies = surfactant.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: lamellar body = surfactant.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to lamellar body = surfactant.

Difficulty: High-Difficulty

**\*\*Q55. The thinness of the blood air barrier is optimized for:**

- A) Gas diffusion
- B) Cartilage support
- C) Immune defense
- D) Mucous production

**Answer: A**

Explanation: The thin barrier maximizes gas diffusion. Practice Test

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: barrier = diffusion.

Learning Objective mapping: Chapter 14 The Respiratory System LO related to barrier = diffusion.

Difficulty: High-Difficulty

**\*\*Q56. The respiratory epithelium is:**

- A) Transitional
- B) Simple squamous
- C) Pseudostratified ciliated columnar with goblet cells
- D) Stratified squamous

**Answer: C**

Explanation: Respiratory epithelium pseudostratified ciliated columnar with goblet cells, seromucous glands, cartilage in bronchi.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook The Respiratory System section for decisive features.

Key point: Respiratory epithelium pseudostratified ciliated columnar with goblet cells, seromucous glands, cart

Learning Objective mapping: Chapter 14 The Respiratory System LO related to Respiratory epithelium pseudostratified ciliated columnar with goblet cells, ser

Difficulty: High-Difficulty

# Chapter 15 Skin

## Part A Recall

**\*\*Q1. Melanocytes are located in the:**

- A) Stratum granulosum
- B) Stratum corneum
- C) Stratum basale
- D) Dermis

**Answer: C**

Explanation: Melanocytes = stratum basale.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: melanocyte = basale.

Learning Objective mapping: Chapter 15 Skin LO related to melanocyte = basale.

Difficulty: Recall

**\*\*Q2. Merkel cells function as:**

- A) Immune cells
- B) Touch receptors
- C) Melanin producers
- D) Sweat producers

**Answer: B**

Explanation: Merkel cells = touch receptors.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Merkel = touch.

Learning Objective mapping: Chapter 15 Skin LO related to Merkel = touch.

Difficulty: Recall

**\*\*Q3. Thick skin differs from thin skin by:**

- A) Having sebaceous glands
- B) Lacking a stratum corneum
- C) Having hair follicles
- D) Having a stratum lucidum and no hair

**Answer: D**

Explanation: Thick skin = lucidum + no hair.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: thick = lucidum, no hair.

Learning Objective mapping: Chapter 15 Skin LO related to thick = lucidum, no hair.

Difficulty: Recall

**\*\*Q4. The stratum lucidum is found only in:**

- A) Scalp skin
- B) All skin equally
- C) Thin skin
- D) Thick skin (palms/soles)

**Answer: D**

Explanation: Lucidum = thick skin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: lucidum = thick.

Learning Objective mapping: Chapter 15 Skin LO related to lucidum = thick.

Difficulty: Recall

**\*\*Q5. Apocrine sweat glands are most concentrated in the:**

- A) Palms and soles
- B) Axillae and anogenital region
- C) Scalp only
- D) Entire body equally

**Answer: B**

Explanation: Apocrine = axilla/anogenital.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: apocrine = axilla.

Learning Objective mapping: Chapter 15 Skin LO related to apocrine = axilla.

Difficulty: Recall

**\*\*Q6. "Sunburn cells" are:**

- A) Basal cells with mitotic figures
- B) Individual dyskeratotic keratinocytes with pyknotic nuclei
- C) Melanocytes with excess pigment
- D) Merkel cells

**Answer: B**

Explanation: Sunburn cells = dyskeratotic keratinocytes (apoptosis).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: sunburn cell = apoptotic keratinocyte.

Learning Objective mapping: Chapter 15 Skin LO related to sunburn cell = apoptotic keratinocyte.

Difficulty: Recall

**\*\*Q7. Second-degree burns extend into the:**

- A) Subcutaneous fat only
- B) Muscle
- C) Stratum corneum only
- D) Dermis, with blister formation

**Answer: D**

Explanation: Second-degree = into the dermis (blisters).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: 2nd degree = dermis.

Learning Objective mapping: Chapter 15 Skin LO related to 2nd degree = dermis.

Difficulty: Recall

**\*\*Q8. Malignant melanoma arises from:**

- A) Melanocytes
- B) Merkel cells
- C) Langerhans cells
- D) Keratinocytes

**Answer: A**

Explanation: Melanoma = melanocytes.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: melanoma = melanocyte.

Learning Objective mapping: Chapter 15 Skin LO related to melanoma = melanocyte.

Difficulty: Recall

**\*\*Q9. The stratum corneum's cells are:**

- A) Anucleate and keratinized
- B) Ciliated
- C) Pigmented
- D) Nucleated and living

**Answer: A**

**Explanation:** Corneum = anucleate keratinized.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: corneum = anucleate.

Learning Objective mapping: Chapter 15 Skin LO related to corneum = anucleate.

Difficulty: Recall

**\*\*Q10. A pale (clear) layer between granulosum and corneum indicates:**

- A) Melanoma
- B) Thin skin
- C) Thick skin (stratum lucidum)
- D) Psoriasis

**Answer: C**

**Explanation:** Lucidum = thick skin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: lucidum = thick skin.

Learning Objective mapping: Chapter 15 Skin LO related to lucidum = thick skin.

Difficulty: Recall

**\*\*Q11. A biopsy shows a clear (pale) layer between the granular and cornified layers. This is:**

- A) Pemphigus
- B) Thin skin
- C) Thick skin (stratum lucidum)
- D) Psoriasis

**Answer: C**

**Explanation:** Stratum lucidum clear pale layer between granulosum and corneum in thick skin palms soles.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Stratum lucidum clear pale layer between granulosum and corneum in thick skin palms soles.

Learning Objective mapping: Chapter 15 Skin LO related to Stratum lucidum clear pale layer between granulosum and corneum in thick skin pa

Difficulty: Recall

**\*\*Q12. Atypical melanocytes invading the dermis indicate:**

- A) Psoriasis
- B) Vitiligo
- C) Basal cell carcinoma
- D) Melanoma

**Answer: D**

Explanation: Melanoma atypical melanocytes invading dermis, most deadly skin cancer.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Melanoma atypical melanocytes invading dermis, most deadly skin cancer.

Learning Objective mapping: Chapter 15 Skin LO related to Melanoma atypical melanocytes invading dermis, most deadly skin cancer.

Difficulty: Recall

## Part B Understanding

**\*\*Q13. Melanocytes are derived from the:**

- A) Epidermal ectoderm
- B) Neural crest
- C) Mesoderm
- D) Endoderm

**Answer: B**

Explanation: Melanocytes = neural crest.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: melanocyte = neural crest.

Learning Objective mapping: Chapter 15 Skin LO related to melanocyte = neural crest.

Difficulty: Understanding

**\*\*Q14. Melanocytes are derived from:**

- A) Surface ectoderm
- B) Neural crest
- C) Mesoderm
- D) Endoderm

**Answer: B**

Explanation: Melanocytes = neural crest.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: melanocyte = neural crest.

Learning Objective mapping: Chapter 15 Skin LO related to melanocyte = neural crest.

Difficulty: Understanding

**\*\*Q15. The hair follicle bulb encloses the:**

- A) Dermal (hair) papilla
- B) Sebaceous gland
- C) Root sheath only
- D) Arrector pili muscle

**Answer: A**

Explanation: Hair bulb encloses the dermal papilla.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: bulb = dermal papilla.

Learning Objective mapping: Chapter 15 Skin LO related to bulb = dermal papilla.

Difficulty: Understanding

**\*\*Q16. Solar (actinic) elastosis is accumulation of abnormal:**

- A) Reticular fibers
- B) Ground substance
- C) Collagen
- D) Elastic fibers

**Answer: D**

Explanation: Actinic elastosis = degenerated elastic fibers.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: elastosis = elastic.

Learning Objective mapping: Chapter 15 Skin LO related to elastosis = elastic.

Difficulty: Understanding

**\*\*Q17. Skin re-epithelialization after a superficial wound depends on:**

- A) Melanocytes exclusively
- B) The dermis directly
- C) Basal-layer keratinocytes at the wound edge + hair-follicle/gland stem cells
- D) Subcutaneous fat

**Answer: C**

Explanation: Re-epithelialization = basal + appendage stem cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.



D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: healing = basal stem cells.

Learning Objective mapping: Chapter 15    Skin    LO related to healing = basal stem cells.

Difficulty: Understanding

**\*\*Q18. Langerhans cells are derived from:**

- A) Bone marrow (monocyte lineage)
- B) Surface ectoderm
- C) Mesoderm exclusively
- D) Neural crest

**Answer: A**

Explanation: Langerhans cells = bone marrow (monocyte lineage).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Langerhans = bone marrow.

Learning Objective mapping: Chapter 15    Skin    LO related to Langerhans = bone marrow.

Difficulty: Understanding

**\*\*Q19. Birbeck granules are the marker of:**

- A) Langerhans cells
- B) Merkel cells
- C) Keratinocytes
- D) Melanocytes

**Answer: A**

Explanation: Birbeck granules = Langerhans cells.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Birbeck = Langerhans.

Learning Objective mapping: Chapter 15    Skin    LO related to Birbeck = Langerhans.

Difficulty: Understanding

**\*\*Q20. Albinism results from:**

- A) Excess keratin
- B) Langerhans dysfunction
- C) Melanocyte absence
- D) Tyrosinase deficiency (no melanin production)

**Answer: D**

Explanation: Albinism = tyrosinase deficiency (melanocytes present).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: albinism = tyrosinase.

Learning Objective mapping: Chapter 15 Skin LO related to albinism = tyrosinase.

Difficulty: Understanding

**\*\*Q21. The most common skin cancer is:**

- A) Melanoma
- B) Basal cell carcinoma
- C) Squamous cell carcinoma
- D) Merkel cell carcinoma

**Answer: B**

Explanation: Basal cell carcinoma most common skin cancer, least metastatic; melanoma most deadly.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Basal cell carcinoma most common skin cancer, least metastatic; melanoma most deadly.

Learning Objective mapping: Chapter 15 Skin LO related to Basal cell carcinoma most common skin cancer, least metastatic; melanoma most de

Difficulty: Understanding

## **Part C Application**

**\*\*Q22. The correct order of epidermal layers (deep superficial) is:**

- A) Basale granulosum spinosum corneum lucidum
- B) Spinosum basale corneum granulosum lucidum
- C) Corneum lucidum granulosum spinosum basale
- D) Basale spinosum granulosum lucidum corneum

**Answer: D**

Explanation: Basale spinosum granulosum lucidum corneum.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: know the order.

Learning Objective mapping: Chapter 15 Skin LO related to know the order.

Difficulty: Application

**\*\*Q23. Meissner's corpuscles are located in the:**

- A) Dermal papillae (papillary dermis)
- B) Epidermis
- C) Reticular dermis only

D) Hypodermis

**Answer: A**

Explanation: Meissner's = dermal papillae.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Meissner = papillary.

Learning Objective mapping: Chapter 15 Skin LO related to Meissner = papillary.

Difficulty: Application

**\*\*Q24. Eccrine sweat glands secrete by:**

A) Apocrine only

B) Endocrine

C) Holocrine

D) Merocrine

**Answer: D**

Explanation: Eccrine = merocrine.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: eccrine = merocrine.

Learning Objective mapping: Chapter 15 Skin LO related to eccrine = merocrine.

Difficulty: Application

**\*\*Q25. Merkel cells are located in the:**

A) Stratum corneum

B) Stratum basale

C) Stratum granulosum

D) Stratum lucidum

**Answer: B**

Explanation: Merkel cells = basale.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Merkel = basale.

Learning Objective mapping: Chapter 15 Skin LO related to Merkel = basale.

Difficulty: Application

**\*\*Q26. Sebaceous glands secrete sebum via:**

A) Apocrine

- B) Endocrine
- C) Merocrine
- D) Holocrine (entire cell disintegrates)

**Answer: D**

Explanation: Sebaceous = holocrine.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: sebaceous = holocrine.

Learning Objective mapping: Chapter 15    Skin    LO related to sebaceous = holocrine.

Difficulty: Application

**\*\*Q27. Pacinian corpuscles detect:**

- A) Pain
- B) Temperature only
- C) Light touch
- D) Deep pressure and vibration

**Answer: D**

Explanation: Pacinian = deep pressure/vibration.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Pacinian = pressure.

Learning Objective mapping: Chapter 15    Skin    LO related to Pacinian = pressure.

Difficulty: Application

**\*\*Q28. The hypodermis is composed predominantly of:**

- A) Muscle only
- B) Dense collagen
- C) Adipose tissue
- D) Cartilage

**Answer: C**

Explanation: Hypodermis = adipose.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: hypodermis = fat.

Learning Objective mapping: Chapter 15    Skin    LO related to hypodermis = fat.

Difficulty: Application

**\*\*Q29. Basal cell carcinoma arises from:**

- A) Stratum granulosum
- B) The dermis
- C) Stratum corneum
- D) Basal-layer keratinocytes

**Answer: D**

Explanation: BCC = basal keratinocytes.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: BCC = basal keratinocyte.

Learning Objective mapping: Chapter 15 Skin LO related to BCC = basal keratinocyte.

Difficulty: Application

**\*\*Q30. Which gland is associated with hair follicles?**

- A) Eccrine sweat
- B) Sebaceous
- C) Apocrine only
- D) All sweat glands

**Answer: B**

Explanation: Sebaceous glands associate with hair follicles.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: sebaceous = hair.

Learning Objective mapping: Chapter 15 Skin LO related to sebaceous = hair.

Difficulty: Application

**\*\*Q31. Vitiligo results from:**

- A) Excess melanin
- B) Keratinocyte loss
- C) Tyrosinase deficiency
- D) Autoimmune melanocyte destruction

**Answer: D**

Explanation: Vitiligo = autoimmune melanocyte destruction.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: vitiligo = autoimmune.

Learning Objective mapping: Chapter 15 Skin LO related to vitiligo = autoimmune.

Difficulty: Application

**\*\*Q32. Psoriasis shows:**

- A) Acanthosis and parakeratosis
- B) Epidermal thinning
- C) Subepidermal blisters
- D) Melanocyte invasion

**Answer: A**

Explanation: Psoriasis acanthosis thick spinous + parakeratosis nuclei retained in corneum + Munro microabscesses.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Psoriasis acanthosis thick spinous + parakeratosis nuclei retained in corneum + Munro microabscesses

Learning Objective mapping: Chapter 15 Skin LO related to Psoriasis acanthosis thick spinous + parakeratosis nuclei retained in corneum +

Difficulty: Application

## **Part D Integration**

**\*\*Q33. Keratohyalin granules are found in the:**

- A) Stratum basale
- B) Stratum granulosum
- C) Stratum corneum
- D) Stratum lucidum

**Answer: B**

Explanation: Keratohyalin = stratum granulosum.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: keratohyalin = granulosum.

Learning Objective mapping: Chapter 15 Skin LO related to keratohyalin = granulosum.

Difficulty: Integration

**\*\*Q34. The reticular dermis is composed of:**

- A) Loose connective tissue
- B) Dense irregular connective tissue
- C) Adipose tissue
- D) Stratified epithelium

**Answer: B**

Explanation: Reticular dermis = dense irregular CT.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: reticular dermis = dense irregular.

Learning Objective mapping: Chapter 15 Skin LO related to reticular dermis = dense irregular.

Difficulty: Integration

**\*\*Q35. The dermis is organized into a papillary layer and a deeper:**

- A) Basal layer
- B) Hypodermis
- C) Reticular layer (dense irregular CT)
- D) Subcutaneous fat

**Answer: C**

Explanation: Dermis = papillary + reticular (dense irregular).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: reticular = dense irregular.

Learning Objective mapping: Chapter 15 Skin LO related to reticular = dense irregular.

Difficulty: Integration

**\*\*Q36. The arrector pili muscle functions to:**

- A) Elevate the hair (piloerection) and express sebum
- B) Provide sensation
- C) Synthesize keratin
- D) Produce sweat

**Answer: A**

Explanation: Arrector pili = piloerection.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: arrector pili = piloerection.

Learning Objective mapping: Chapter 15 Skin LO related to arrector pili = piloerection.

Difficulty: Integration

**\*\*Q37. Meissner's corpuscles detect:**

- A) Deep pressure
- B) Light touch and low-frequency vibration
- C) Pain
- D) Temperature

**Answer: B**

Explanation: Meissner = light touch.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Meissner = touch.

Learning Objective mapping: Chapter 15 Skin LO related to Meissner = touch.

Difficulty: Integration

**\*\*Q38. Nails are composed of hard keratin produced by the:**

- A) Nail matrix
- B) Eponychium only
- C) Hyponychium only
- D) Dermal papilla

**Answer: A**

Explanation: Nails = nail matrix.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: nail = matrix.

Learning Objective mapping: Chapter 15 Skin LO related to nail = matrix.

Difficulty: Integration

**\*\*Q39. Psoriasis is characterized by:**

- A) Loss of the epidermis
- B) Normal turnover
- C) Epidermal thinning
- D) Acanthosis and parakeratosis

**Answer: D**

Explanation: Psoriasis = acanthosis + parakeratosis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: psoriasis = acanthosis + parakeratosis.

Learning Objective mapping: Chapter 15 Skin LO related to psoriasis = acanthosis + parakeratosis.

Difficulty: Integration

**\*\*Q40. Bullous pemphigoid produces a:**

- A) Intraepidermal blister
- B) Subepidermal blister
- C) Intradermal blister
- D) Subcutaneous blister



**Answer: B**

Explanation: Pemphigoid = subepidermal (anti-hemidesmosome).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: pemphigoid = subepidermal.

Learning Objective mapping: Chapter 15 Skin LO related to pemphigoid = subepidermal.

Difficulty: Integration

**\*\*Q41. The dermal papilla is essential for:**

A) Hair growth (nourishes the bulb)

B) Touch

C) Pigmentation

D) Sweat production

**Answer: A**

Explanation: Dermal papilla = hair growth.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: papilla = hair growth.

Learning Objective mapping: Chapter 15 Skin LO related to papilla = hair growth.

Difficulty: Integration

**\*\*Q42. The receptor for deep pressure and vibration is the:**

A) Merkel cell

B) Free nerve ending

C) Meissner's corpuscle

D) Pacinian corpuscle

**Answer: D**

Explanation: Pacinian deep pressure vibration, Meissner light touch, Merkel touch, Ruffini stretch.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Pacinian deep pressure vibration, Meissner light touch, Merkel touch, Ruffini stretch.

Learning Objective mapping: Chapter 15 Skin LO related to Pacinian deep pressure vibration, Meissner light touch, Merkel touch, Ruffini st

Difficulty: Integration

**\*\*Q43. A third-degree burn requires grafting because it destroys:**

A) The basal/appendage stem-cell compartment

- B) Only the dermis
- C) Only the hypodermis
- D) The stratum corneum only

**Answer: A**

Explanation: Third-degree burn destroys basal stem cells and appendages    needs grafting.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Third-degree burn destroys basal stem cells and appendages    needs grafting.

Learning Objective mapping: Chapter 15    Skin    LO related to Third-degree burn destroys basal stem cells and appendages    needs grafting.

Difficulty: Integration

## Part E    High-Difficulty

**\*\*Q44. Langerhans cells are characterized by:**

- A) Touch reception
- B) Melanin production
- C) Birbeck granules and antigen presentation
- D) Keratin production

**Answer: C**

Explanation: Langerhans = Birbeck granules + antigen presentation.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Langerhans = immune.

Learning Objective mapping: Chapter 15    Skin    LO related to Langerhans = immune.

Difficulty: High-Difficulty

**\*\*Q45. The dermal papillae are a feature of the:**

- A) Epidermis
- B) Papillary dermis
- C) Hypodermis
- D) Reticular dermis

**Answer: B**

Explanation: Dermal papillae = papillary dermis.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: papillae = papillary dermis.

Learning Objective mapping: Chapter 15 Skin LO related to papillae = papillary dermis.

Difficulty: High-Difficulty

**\*\*Q46. Vitamin D synthesis begins with UV-B conversion of a precursor in the:**

- A) Stratum basale and spinosum
- B) Dermis
- C) Subcutaneous fat
- D) Stratum corneum

**Answer: A**

Explanation: Vitamin D precursor = basale/spinosum.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: vitamin D = epidermis.

Learning Objective mapping: Chapter 15 Skin LO related to vitamin D = epidermis.

Difficulty: High-Difficulty

**\*\*Q47. Free nerve endings mediate:**

- A) Fine touch only
- B) Pain and temperature
- C) Vibration only
- D) Deep pressure only

**Answer: B**

Explanation: Free endings = pain + temperature.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: free endings = pain/temperature.

Learning Objective mapping: Chapter 15 Skin LO related to free endings = pain/temperature.

Difficulty: High-Difficulty

**\*\*Q48. Rete ridges function to:**

- A) Detect pain
- B) Produce sweat
- C) Increase the dermal-epidermal junction surface (adhesion)
- D) Synthesize melanin

**Answer: C**

Explanation: Rete ridges = increased adhesion surface.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: rete ridges = adhesion.

Learning Objective mapping: Chapter 15 Skin LO related to rete ridges = adhesion.

Difficulty: High-Difficulty

**\*\*Q49. Which cell type produces melanin?**

- A) Melanocyte
- B) Merkel cell
- C) Keratinocyte
- D) Langerhans cell

**Answer: A**

Explanation: Melanocytes produce melanin.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: melanin = melanocyte.

Learning Objective mapping: Chapter 15 Skin LO related to melanin = melanocyte.

Difficulty: High-Difficulty

**\*\*Q50. Which is correctly paired regarding skin gland secretion modes?**

- A) Apocrine holocrine
- B) Sebaceous apocrine
- C) Sebaceous merocrine
- D) Eccrine merocrine

**Answer: D**

Explanation: Eccrine = merocrine. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: eccrine = merocrine.

Learning Objective mapping: Chapter 15 Skin LO related to eccrine = merocrine.

Difficulty: High-Difficulty

**\*\*Q51. Anti-desmoglein antibodies produce:**

- A) Parakeratosis
- B) Subepidermal blisters
- C) Intraepidermal acantholysis
- D) Acanthosis

**Answer: C**

Explanation: Anti-desmoglein desmosome loss intraepidermal acantholysis pemphigus vulgaris; anti-hemidesmosome subepidermal bullous pemphigoid.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Skin section for decisive features.

Key point: Anti-desmoglein    desmosome loss    intraepidermal acantholysis pemphigus vulgaris;  
anti-hemidesmosom

Learning Objective mapping: Chapter 15    Skin    LO related to Anti-desmoglein    desmosome loss  
intraepidermal acantholysis pemphigus vulgari

Difficulty: High-Difficulty

# Chapter 16 Endocrine Glands

## Part A Recall

**\*\*Q1. GH and prolactin are secreted by which anterior-pituitary cells?**

- A) Pituicytes
- B) Basophils
- C) Acidophils
- D) Chromophobes

**Answer: C**

Explanation: Acidophils = GH + prolactin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: acidophil = GH/PRL.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to acidophil = GH/PRL.

Difficulty: Recall

**\*\*Q2. Colloid-filled follicles are characteristic of the:**

- A) Thyroid
- B) Adrenal
- C) Pituitary
- D) Parathyroid

**Answer: A**

Explanation: Thyroid = colloid follicles.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: colloid = thyroid.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to colloid = thyroid.

Difficulty: Recall

**\*\*Q3. The adrenal medulla contains:**

- A) Follicles
- B) Chief cells
- C) Zona glomerulosa
- D) Chromaffin cells

**Answer: D**

Explanation: Medulla = chromaffin cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: medulla = chromaffin.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to medulla = chromaffin.

Difficulty: Recall

**\*\*Q4. Somatotrophs secrete:**

- A) Prolactin
- B) Growth hormone
- C) ACTH
- D) TSH

**Answer: B**

Explanation: Somatotrophs = GH.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: somatotroph = GH.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to somatotroph = GH.

Difficulty: Recall

**\*\*Q5. The anterior pituitary receives hypothalamic hormones via the:**

- A) Internal carotid directly
- B) Posterior communicating artery
- C) Hypophyseal portal system
- D) Nerve tract

**Answer: C**

Explanation: Anterior = portal system.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: portal = anterior.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to portal = anterior.

Difficulty: Recall

**\*\*Q6. In hyperfunctioning thyroid, follicular cells become:**

- A) Absent
- B) Replaced by fat
- C) Flattened with abundant colloid
- D) Tall/columnar with scalloped colloid

**Answer: D**

Explanation: Hyperthyroid = tall cells + scalloped colloid.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: hyperthyroid = scalloped colloid.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to hyperthyroid = scalloped colloid.

Difficulty: Recall

**\*\*Q7. The zona reticularis secretes:**

- A) Epinephrine
- B) Aldosterone
- C) Androgens (DHEA)
- D) Cortisol exclusively

**Answer: C**

Explanation: Reticularis = androgens.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: reticularis = androgens.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to reticularis = androgens.

Difficulty: Recall

**\*\*Q8. Adrenal zonation (outer inner) secretes, respectively:**

- A) Sex, sugar, salt
- B) Sugar, salt, sex
- C) All identical
- D) Salt, sugar, sex

**Answer: D**

Explanation: GFR = salt (mineralocorticoid), sugar (glucocorticoid), sex (androgen).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: salt/sugar/sex.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to salt/sugar/sex.

Difficulty: Recall

**\*\*Q9. Hashimoto thyroiditis shows:**

- A) Scalloped colloid
- B) Lymphocytic infiltration and Hürthle cells
- C) Chromaffin cells
- D) Brain sand

**Answer: B**



**Explanation:** Hashimoto = lymphocytes + H rthle cells.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Hashimoto = lymphocytes.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Hashimoto = lymphocytes.

Difficulty: Recall

**\*\*Q10. Prolactin excess causes:**

- A) Acromegaly
- B) Cushing syndrome
- C) Diabetes insipidus
- D) Galactorrhea and amenorrhea

**Answer: D**

**Explanation:** Prolactin excess = galactorrhea + amenorrhea.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: prolactin = galactorrhea/amenorrhea.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to prolactin = galactorrhea/amenorrhea.

Difficulty: Recall

**\*\*Q11. Herring bodies are found in the:**

- A) Thyroid
- B) Adrenal medulla
- C) Anterior pituitary
- D) Posterior pituitary

**Answer: D**

**Explanation:** Herring bodies posterior pituitary dilated axon terminals storing ADH oxytocin from hypothalamus.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Herring bodies posterior pituitary dilated axon terminals storing ADH oxytocin from hypothalamus.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Herring bodies posterior pituitary dilated axon terminals storing ADH oxytocin f

Difficulty: Recall

**\*\*Q12. ADH and oxytocin are synthesized in the:**

- A) Hypothalamus
- B) Anterior pituitary
- C) Pineal
- D) Posterior pituitary

**Answer: A**

Explanation: ADH oxytocin synthesized in hypothalamus supraoptic paraventricular nuclei, transported via axons to posterior pituitary stored.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: ADH oxytocin synthesized in hypothalamus supraoptic paraventricular nuclei, transported via axons to

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to ADH oxytocin synthesized in hypothalamus supraoptic paraventricular nuclei, tran

Difficulty: Recall

## Part B Understanding

**\*\*Q13. The "B-FLAT" mnemonic refers to hormones secreted by:**

- A) Basophils
- B) Chromophobes
- C) Chromaffin cells
- D) Acidophils

**Answer: A**

Explanation: Basophils = B-FLAT (FSH, LH, ACTH, TSH).

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: basophil = B-FLAT.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to basophil = B-FLAT.

Difficulty: Understanding

**\*\*Q14. Calcitonin is secreted by:**

- A) Follicular cells
- B) Parafollicular C cells
- C) Chief cells
- D) Oxyphil cells

**Answer: B**

Explanation: C cells = calcitonin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: C cell = calcitonin.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to C cell = calcitonin.

Difficulty: Understanding

**\*\*Q15. Corpora arenacea ("brain sand") are characteristic of the:**

- A) Pituitary
- B) Adrenal
- C) Thyroid
- D) Pineal gland

**Answer: D**

Explanation: Brain sand = pineal.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: brain sand = pineal.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to brain sand = pineal.

Difficulty: Understanding

**\*\*Q16. The zona glomerulosa secretes:**

- A) Catecholamines
- B) Cortisol
- C) Aldosterone
- D) Androgens

**Answer: C**

Explanation: Glomerulosa = aldosterone.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: glomerulosa = aldosterone.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to glomerulosa = aldosterone.

Difficulty: Understanding

**\*\*Q17. Lactotrophs secrete:**

- A) LH/FSH
- B) GH
- C) Prolactin
- D) ACTH

**Answer: C**

Explanation: Lactotrophs = prolactin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: lactotroph = prolactin.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to lactotroph = prolactin.

Difficulty: Understanding

**\*\*Q18. Parathyroid chief cells secrete:**

- A) Thyroxine
- B) Calcitonin
- C) Parathyroid hormone
- D) Cortisol

**Answer: C**

Explanation: Chief cells = PTH.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: chief = PTH.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to chief = PTH.

Difficulty: Understanding

**\*\*Q19. Adrenal cortical cells share which ultrastructural feature?**

- A) Abundant SER and tubular-cristae mitochondria
- B) Dense-core granules
- C) Melanin
- D) Abundant RER

**Answer: A**

Explanation: Steroid cells = SER + tubular-cristae mitochondria.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: steroid = SER.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to steroid = SER.

Difficulty: Understanding

**\*\*Q20. Craniopharyngioma arises from remnants of:**

- A) The adrenal cortex
- B) Neural crest
- C) Rathke's pouch
- D) The pineal

**Answer: C**

**Explanation:** Craniopharyngioma = Rathke's pouch remnant.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: craniopharyngioma = Rathke's.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to craniopharyngioma = Rathke's.

Difficulty: Understanding

**\*\*Q21. Which cell secretes calcitonin?**

- A) Parafollicular C cell
- B) Chief cell
- C) Oxyphil cell
- D) Follicular cell

**Answer: A**

**Explanation:** C cell = calcitonin.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: C cell = calcitonin.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to C cell = calcitonin.

Difficulty: Understanding

**\*\*Q22. Conn syndrome is caused by:**

- A) Catecholamine excess
- B) PTH excess
- C) Cortisol excess
- D) Aldosterone excess (glomerulosa)

**Answer: D**

**Explanation:** Conn = aldosterone (glomerulosa).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Conn = aldosterone.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Conn = aldosterone.

Difficulty: Understanding

**\*\*Q23. A thyroid shows scalloped colloid and follicular hyperplasia. The diagnosis is:**

- A) Medullary carcinoma
- B) Hashimoto thyroiditis

- C) Graves disease
- D) Papillary carcinoma

**Answer: C**

Explanation: Graves disease scalloped colloid follicular hyperplasia hyperthyroidism TSI autoantibodies.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Graves disease scalloped colloid follicular hyperplasia hyperthyroidism TSI autoantibodies.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Graves disease scalloped colloid follicular hyperplasia hyperthyroidism TSI auto

Difficulty: Understanding

## Part C Application

**\*\*Q24. PTH is secreted by which parathyroid cells?**

- A) C cells
- B) Chromaffin cells
- C) Oxyphil cells
- D) Chief (principal) cells

**Answer: D**

Explanation: Chief cells = PTH.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: chief = PTH.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to chief = PTH.

Difficulty: Application

**\*\*Q25. The anterior pituitary is derived from:**

- A) Neural ectoderm
- B) Oral ectoderm (Rathke's pouch)
- C) Mesoderm
- D) Endoderm

**Answer: B**

Explanation: Anterior pituitary = Rathke's pouch (oral ectoderm).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: anterior = Rathke's pouch.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to anterior = Rathke's pouch.

Difficulty: Application

**\*\*Q26. Chromaffin cells share an embryologic origin with:**

- A) Adrenal cortex cells
- B) Postganglionic sympathetic neurons
- C) C cells
- D) Anterior-pituitary cells

**Answer: B**

Explanation: Chromaffin = modified sympathetic neurons.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: chromaffin = sympathetic.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to chromaffin = sympathetic.

Difficulty: Application

**\*\*Q27. Corticotrophs secrete:**

- A) TSH
- B) ACTH
- C) Prolactin
- D) GH

**Answer: B**

Explanation: Corticotrophs = ACTH.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: corticotroph = ACTH.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to corticotroph = ACTH.

Difficulty: Application

**\*\*Q28. PTH raises calcium via bone resorption, renal reabsorption, and:**

- A) Renal phosphate reabsorption
- B) Vitamin D activation (intestinal absorption)
- C) Calcitonin release
- D) Insulin secretion

**Answer: B**

Explanation: PTH activates vitamin D    intestinal Ca    absorption.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: PTH    vitamin D.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to PTH    vitamin D.

Difficulty: Application

**\*\*Q29. The adrenal medulla secretes:**

- A) Aldosterone
- B) Androgens
- C) Cortisol
- D) Epinephrine and norepinephrine

**Answer: D**

Explanation: Medulla = catecholamines.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: medulla = catecholamines.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to medulla = catecholamines.

Difficulty: Application

**\*\*Q30. Diabetes insipidus results from deficiency of:**

- A) Antidiuretic hormone (vasopressin)
- B) Aldosterone
- C) GH
- D) Insulin

**Answer: A**

Explanation: Diabetes insipidus = ADH deficiency.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: DI = ADH.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to DI = ADH.

Difficulty: Application

**\*\*Q31. Which is correctly paired regarding adrenal cortical zones and hormones?**

- A) Glomerulosa    cortisol
- B) Fasciculata    cortisol
- C) Reticularis    aldosterone
- D) Medulla    aldosterone

**Answer: B**

Explanation: Fasciculata = cortisol.



A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: fasciculata = cortisol.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to fasciculata = cortisol.

Difficulty: Application

**\*\*Q32. The posterior pituitary's Herring bodies contain:**

- A) GH and prolactin
- B) ADH and oxytocin
- C) TSH and ACTH
- D) Calcitonin

**Answer: B**

Explanation: Herring bodies = ADH + oxytocin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Herring = ADH/oxytocin.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Herring = ADH/oxytocin.

Difficulty: Application

**\*\*Q33. Corpora arenacea are characteristic of the:**

- A) Pituitary
- B) Parathyroid
- C) Thyroid
- D) Pineal gland

**Answer: D**

Explanation: Corpora arenacea brain sand calcifications pineal gland, pinealocytes melatonin.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Corpora arenacea brain sand calcifications pineal gland, pinealocytes melatonin.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Corpora arenacea brain sand calcifications pineal gland, pinealocytes melatonin.

Difficulty: Application

**\*\*Q34. Pheochromocytoma arises from:**

- A) Zona fasciculata
- B) Chromaffin cells of the medulla

- C) C cells
- D) Chief cells

**Answer: B**

Explanation: Pheochromocytoma adrenal medullary chromaffin cells catecholamines hypertension triad headache palpitations sweating.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Pheochromocytoma adrenal medullary chromaffin cells catecholamines hypertension triad headache palpi

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Pheochromocytoma adrenal medullary chromaffin cells catecholamines hypertension

Difficulty: Application

## Part D Integration

**\*\*Q35. The zona fasciculata is characterized by:**

- A) Anastomosing network
- B) Chromaffin cells
- C) Rounded clusters
- D) Lipid-rich clear/foamy cells in parallel cords

**Answer: D**

Explanation: Fasciculata = clear cells in cords.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: fasciculata = clear cells.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to fasciculata = clear cells.

Difficulty: Integration

**\*\*Q36. Cortisol is produced by the:**

- A) Zona fasciculata
- B) Zona reticularis
- C) Adrenal medulla
- D) Zona glomerulosa

**Answer: A**

Explanation: Fasciculata = cortisol.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: fasciculata = cortisol.

Learning Objective mapping: Chapter 16   Endocrine Glands   LO related to fasciculata = cortisol.  
Difficulty: Integration

**\*\*Q37. The posterior pituitary stores hormones synthesized in the:**

- A) Anterior pituitary
- B) Hypothalamus (supraoptic + paraventricular nuclei)
- C) Thyroid
- D) Adrenal cortex

**Answer: B**

Explanation: Posterior = stores hypothalamic hormones.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: posterior = storage.

Learning Objective mapping: Chapter 16   Endocrine Glands   LO related to posterior = storage.

Difficulty: Integration

**\*\*Q38. The thyroid follicle surrounds a central store of:**

- A) CSF
- B) Blood
- C) Colloid (thyroglobulin)
- D) Lymph

**Answer: C**

Explanation: Follicle = colloid (thyroglobulin).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: colloid = thyroglobulin.

Learning Objective mapping: Chapter 16   Endocrine Glands   LO related to colloid = thyroglobulin.

Difficulty: Integration

**\*\*Q39. Oxyphil cells increase in number with:**

- A) Vitamin D deficiency
- B) Fetal development
- C) Advancing age
- D) Hypocalcemia

**Answer: C**

Explanation: Oxyphil cells   with age.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: oxyphil = age.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to oxyphil = age.

Difficulty: Integration

**\*\*Q40. The pineal's calcified deposits are:**

- A) Corpora arenacea (brain sand)
- B) Psammoma bodies
- C) Colloid
- D) Hassall's corpuscles

**Answer: A**

Explanation: Pineal = corpora arenacea.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: pineal = brain sand.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to pineal = brain sand.

Difficulty: Integration

**\*\*Q41. MEN syndromes show synchronous endocrine tumors due to:**

- A) Diet
- B) Anatomic proximity
- C) An inherited genetic mutation (multiple tissues)
- D) Infection

**Answer: C**

Explanation: MEN = inherited mutation affecting multiple endocrine tissues.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: MEN = genetic.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to MEN = genetic.

Difficulty: Integration

**\*\*Q42. The hypophyseal portal system delivers:**

- A) PTH to the bone
- B) ADH to the posterior pituitary
- C) Releasing hormones to the anterior pituitary
- D) Catecholamines to the medulla

**Answer: C**

Explanation: Portal system anterior pituitary (releasing hormones).

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: portal = anterior.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to portal = anterior.

Difficulty: Integration

**\*\*Q43. Which structure identifies the pineal gland?**

- A) Corpora arenacea
- B) Colloid follicles
- C) Herring bodies
- D) Hassall's corpuscles

**Answer: A**

Explanation: Pineal = corpora arenacea.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: pineal = brain sand.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to pineal = brain sand.

Difficulty: Integration

**\*\*Q44. The hypophyseal portal system delivers hormones to the:**

- A) Adrenal medulla
- B) Pineal
- C) Posterior pituitary
- D) Anterior pituitary

**Answer: D**

Explanation: Hypophyseal portal system delivers hypothalamic releasing hormones to anterior pituitary.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Hypophyseal portal system delivers hypothalamic releasing hormones to anterior pituitary.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to Hypophyseal portal system delivers hypothalamic releasing hormones to anterior p

Difficulty: Integration

## **Part E High-Difficulty**

**\*\*Q45. Pituicytes are found in the:**

- A) Posterior pituitary
- B) Thyroid
- C) Adrenal cortex

D) Anterior pituitary

**Answer: A**

Explanation: Pituicytes = posterior pituitary glia.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: pituicyte = posterior.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to pituicyte = posterior.

Difficulty: High-Difficulty

**\*\*Q46. Aldosterone is produced by the:**

A) Adrenal medulla

B) Zona glomerulosa

C) Zona fasciculata

D) Zona reticularis

**Answer: B**

Explanation: Glomerulosa = aldosterone.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: glomerulosa = aldosterone.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to glomerulosa = aldosterone.

Difficulty: High-Difficulty

**\*\*Q47. ADH and oxytocin reach the posterior pituitary via:**

A) Simple diffusion

B) The systemic circulation

C) The portal blood system

D) Axonal transport (hypothalamo-hypophyseal tract)

**Answer: D**

Explanation: Axonal transport down the stalk.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: axonal transport.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to axonal transport.

Difficulty: High-Difficulty

**\*\*Q48. Iodination of thyroglobulin occurs at the interface between follicular cells and the:**

- A) Basement membrane
- B) Colloid
- C) Adjacent follicle
- D) Capillary lumen

**Answer: B**

Explanation: Iodination = follicular-cell/colloid interface.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: iodination = colloid interface.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to iodination = colloid interface.

Difficulty: High-Difficulty

**\*\*Q49. The zona fasciculata secretes:**

- A) Cortisol
- B) Androgens exclusively
- C) Catecholamines
- D) Aldosterone

**Answer: A**

Explanation: Fasciculata = cortisol.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: fasciculata = cortisol.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to fasciculata = cortisol.

Difficulty: High-Difficulty

**\*\*Q50. Melatonin secretion is predominant during:**

- A) Darkness (night)
- B) Equally all day
- C) REM sleep only
- D) Daylight

**Answer: A**

Explanation: Melatonin = night.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: melatonin = darkness.

Learning Objective mapping: Chapter 16 Endocrine Glands LO related to melatonin = darkness.

Difficulty: High-Difficulty

**\*\*Q51. Graves disease shows:**

- A) Brain sand
- B) Clear cells
- C) Lymphocytic infiltration
- D) Follicular hyperplasia with scalloped colloid

**Answer: D**

Explanation: Graves = hyperplasia + scalloped colloid.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Graves = scalloped colloid.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to Graves = scalloped colloid.

Difficulty: High-Difficulty

**\*\*Q52. Which cell is a modified postganglionic sympathetic neuron?**

- A) Chromaffin cell
- B) Follicular cell
- C) Pituicyte
- D) Chief cell

**Answer: A**

Explanation: Chromaffin = modified sympathetic neuron.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: chromaffin = sympathetic.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to chromaffin = sympathetic.

Difficulty: High-Difficulty

**\*\*Q53. The most common pituitary adenoma is the:**

- A) GH-secreting adenoma
- B) ACTH-secreting adenoma
- C) TSH-secreting adenoma
- D) Prolactinoma

**Answer: D**

Explanation: Prolactinoma = most common pituitary adenoma. Practice Test

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.



Key point: prolactinoma = most common.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to prolactinoma = most common.

Difficulty: High-Difficulty

**\*\*Q54. PTH is secreted by:**

- A) Follicular cells
- B) Oxyphil cells
- C) Chief cells
- D) C cells

**Answer: C**

Explanation: Parathyroid chief cells PTH calcium regulation, oxyphil unknown function.

A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Parathyroid chief cells PTH calcium regulation, oxyphil unknown function.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to Parathyroid chief cells PTH calcium regulation, oxyphil unknown function.

Difficulty: High-Difficulty

**\*\*Q55. Melatonin secretion peaks during:**

- A) Darkness
- B) Dawn
- C) Noon
- D) Daylight

**Answer: A**

Explanation: Melatonin peaks during darkness pineal gland regulates circadian rhythm.

B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Endocrine Glands section for decisive features.

Key point: Melatonin peaks during darkness pineal gland regulates circadian rhythm.

Learning Objective mapping: Chapter 16    Endocrine Glands    LO related to Melatonin peaks during darkness pineal gland regulates circadian rhythm.

Difficulty: High-Difficulty